

Single Stage Transistor Amplifiers

Learning Objectives

- Develop an understanding of single stage transistor amplifier.
- Recognise the practical circuit of transistor amplifier and the functions of various circuit elements.
- Learn phase reversal in common emitter (*CE*) transistor amplifer.
- Explore the d.c. and a.c. equivalent circuits of transistor amplifier.
- Develop a strong understanding of d.c. and a.c. load lines of transistor amplifier.
- Derive an expression for the voltage gain (A_v) of transistor *CE* amplifier in the form : $A_v = \beta \times R_{AC}/R_{in}$.
- Derive an expression for the voltage gain of *CE* amplifier in the form : $A_v = R_{AC}/r'_e$.
- Derive expressions for input impedance and output impedance of *CE* amplifier.
- Learn about the swamped amplifier.
- Develop an understanding of transistor amplifier equivalent circuit using R_{in} (input resistance), R_{out} (output resistance) and no-load voltage gain A_{v0}.

Introduction

In the previous chapter, it was discussed that a properly biased transistor raises the strength of a weak signal and thus acts as an amplifier. Almost all electronic equipments must include means for amplifying electrical signals. For instance, radio receivers amplify very weak signals—sometimes a few millionth of a volt at antenna—until they are strong enough to fill a room with sound. The transducers used in the medical and scientific investigations generate signals in the microvolt (μV) and millivolt (mV) range. These signals must be amplified thousands and millions of times before they will be strong enough to operate indicating instruments. Therefore, electronic amplifiers are a constant and important ingredient of electronic systems.

Our purpose here will be to discuss *single stage transistor amplifier*. By a *stage* we mean a single transistor with its bias and auxiliary equipment. It may be emphasised here that a practical amplifier is always a multistage amplifier *i.e.* it has a number of stages of

amplification. However, it is profitable to consider the multistage amplifier in terms of single stages that are connected together. In this chapter, we shall confine our attention to single stage transistor amplifiers.

8.1 Single Stage Transistor Amplifier

When only one transistor with associated circuitry is used for amplifying a weak signal, the circuit is known as **single stage transistor amplifier.**

A single stage transistor amplifier has one transistor, bias circuit and other auxiliary components. Although a practical amplifier consists of a number of stages, yet such a complex circuit can be conveniently split up into separate single stages. By analysing carefully only a single stage and using this single stage analysis repeatedly, we can effectively analyse the complex circuit. It follows, therefore, that single stage amplifier analysis is of great value in understanding the practical amplifier circuits.

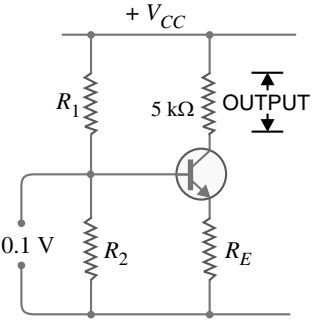


Fig. 8.1

8.2 How Transistor Amplifies ?

Fig. 8.1 shows a single stage transistor amplifier. When a weak a.c. signal is given to the base of transistor, a small base current (which is a.c.) starts flowing. Due to transistor action, a much larger (β times the base current) a.c. current flows through the collector load R_C. As the value of R_C is quite high (usually 4-10 kΩ), therefore, a large voltage appears across R_C. Thus, a weak signal applied in the base circuit appears in amplified form in the collector circuit. It is in this way that a transistor acts as an amplifier.

The action of transistor amplifier can be beautifully explained by referring to Fig. 8.1. Suppose a change of 0.1 V in signal voltage produces a change of 2 mA in the collector current. Obviously, a signal of only 0.1 V applied to the base will give an output voltage = 2 mA × 5 kΩ = 10 V. Thus, the transistor has been able to raise the voltage level of the signal from 0.1 V to 10 V *i.e.* voltage amplification or stage gain is 100.

8.3 Graphical Demonstration of Transistor Amplifier

The function of transistor as an amplifier can also be explained graphically. Fig. 8.2 shows the output characteristics of a transistor in *CE* configuration. Suppose the zero signal base current is 10 μA *i.e.* this is the base current for which the transistor is biased by the biasing network. When an a.c. signal is applied to the base, it makes the base, say positive in the first half-cycle and negative

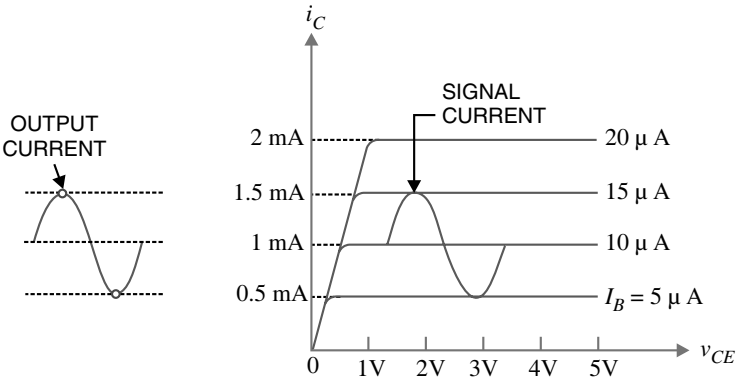


Fig. 8.2

in the second half-cycle. Therefore, the base and collector currents will increase in the first half-cycle when base-emitter junction is more forward-biased. However, they will decrease in the second half-cycle when the base-emitter junction is less forward biased.

For example, consider a sinusoidal signal which increases or decreases the base current by 5 μA in the two half-cycles of the signal. Referring to Fig. 8.2, it is clear that in the absence of signal, the base current is 10μA and the collector current is 1 mA. However, when the signal is applied in the base circuit the base current and hence collector current change continuously. In the first half-cycle peak of the signal, the base current increases to 15 μA and the corresponding collector current is 1.5 mA. In the second half-cycle peak, the base current is reduced to 5 μA and the corresponding collector current is 0.5 mA. For other values of the signal, the collector current is inbetween these values *i.e.* 1.5 mA and 0.5 mA.

It is clear from Fig. 8.2 that 10 μA base current variation results in 1mA (1,000 μA) collector current variation *i.e.* by a factor of 100. This large change in collector current flows through collector resistance R_C. The result is that output signal is much larger than the input signal. Thus, the transistor has done amplification.

8.4 Practical Circuit of Transistor Amplifier

It is important to note that a transistor can accomplish faithful amplification only if proper associated circuitry is used with it. Fig. 8.3 shows a practical single stage transistor amplifier. The various circuit elements and their functions are described below :

(*i*) Biasing circuit. The resistances R_1, R_2 and R_E form the biasing and stabilisation circuit. The biasing circuit must establish a proper operating point otherwise a part of the negative half-cycle of the signal may be cut-off in the output.

(*ii*) Input capacitor C_{in}. An electrolytic capacitor C_{in} (≃ 10 μF) is used to couple the signal to the base of the transistor. If it is not used, the signal source resistance will come across R_2 and thus change the bias. The capacitor C_{in} allows only a.c. signal to flow but isolates the signal source from R_2.*

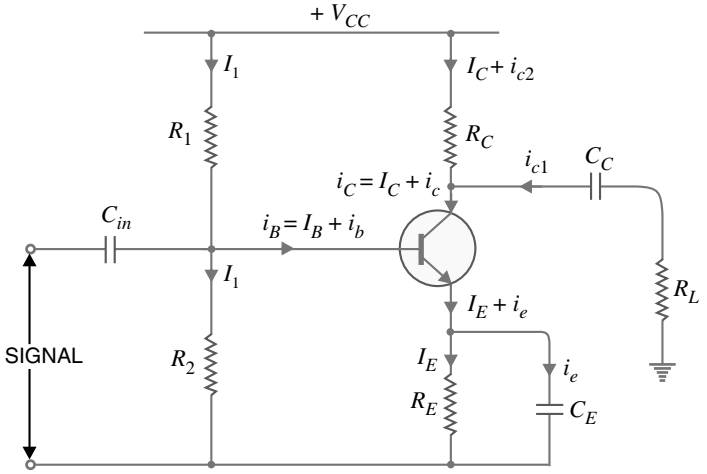


Fig. 8.3

* It may be noted that a capacitor offers infinite reactance to d.c. and blocks it completely whereas it allows a.c. to pass through it.

(*iii*) Emitter bypass capacitor C_E. An emitter bypass capacitor C_E ($\simeq$ 100 μF) is used in parallel with R_E to provide a low reactance path to the amplified a.c. signal. If it is not used, then amplified a.c. signal flowing through R_E will cause a voltage drop across it, thereby reducing the output voltage.

(*iv*) Coupling capacitor C_C. The coupling capacitor C_C ($\simeq$ 10 μF) couples one stage of amplification to the next stage. If it is not used, the bias conditions of the next stage will be drastically changed due to the shunting effect of R_C. This is because R_C will come in parallel with the upper resistance R_1 of the biasing network of the next stage, thereby altering the biasing conditions of the latter. In short, the coupling capacitor C_C isolates the d.c. of one stage from the next stage, but allows the passage of a.c. signal.

Various circuit currents. It is useful to mention the various currents in the complete amplifier circuit. These are shown in the circuit of Fig. 8.3.

(*i*) Base current. When no signal is applied in the base circuit, d.c. base current I_B flows due to biasing circuit. When a.c. signal is applied, a.c. base current i_b also flows. Therefore, with the application of signal, total base current i_B is given by ;

$$i_B = I_B + i_b$$

(*ii*) Collector current. When no signal is applied, a d.c. collector current I_C flows due to biasing circuit. When a.c. signal is applied, a.c. collector current i_c also flows. Therefore, the total collector current i_C is given by ;

$$i_C = I_C + i_c$$

where $I_C = \beta I_B$ = Zero signal collector current

$i_c = \beta i_b$ = Collector current due to signal

(*iii*) Emitter current. When no signal is applied, a d.c. emitter current I_E flows. With the application of signal, total emitter current i_E is given by ;

$$i_E = I_E + i_e$$

It is useful to keep in mind that :

$$I_E = I_B + I_C$$

$$i_e = i_b + i_c$$

Now base current is usually very small, therefore, as a reasonable approximation,

$$I_E \simeq I_C \quad \text{and} \quad i_e \simeq i_c$$

Example 8.1. *What is the role of emitter bypass capacitor C_E in CE amplifier circuit shown in Fig. 8.3? Illustrate with a numerical example.*

Solution. The emitter bypass capacitor C_E (See Fig. 8.3) connected in parallel with R_E plays an important role in the circuit. **If it is not used, the amplified a.c. signal flowing through R_E will cause a voltage drop across it, thereby reducing the a.c. output voltage and hence the voltage gain of the amplifier.**

Let us illustrate the effect of C_E with a numerical example. Suppose $R_E = 1000\,\Omega$ and capacitive reactance of C_E at the signal frequency is $100\,\Omega$ (*i.e.* $X_{C_E} = 100\,\Omega$). Then 10/11 of a.c emitter current will flow through C_E and only 1/11 through R_E. The signal voltage developed across R_E is, therefore, only 1/11 of the voltage which would have been developed if C_E were not present. In practical circuits, the value of C_E is so selected that it almost

entirely bypasses the a.c. signal (the name for C_E is obvious). *For all practical purposes, we consider C_E to be a short for a.c. signals.*

Example 8.2. *Select a suitable value for the emitter bypass capacitor in Fig. 8.4 if the amplifier is to operate over a frequency range from 2 kHz to 10 kHz.*

Solution. An amplifier usually handles more than one frequency. Therefore, the value of C_E is so selected that it provides adequate bypassing for the *lowest* of all the frequencies. Then it will also be a good bypass ($X_C \propto 1/f$) for all the higher frequencies. Suppose the minimum frequency to be handled by C_E is f_{min}. Then C_E is considered a good bypass if at f_{min},

$$X_{C_E} = \frac{R_E}{10}$$

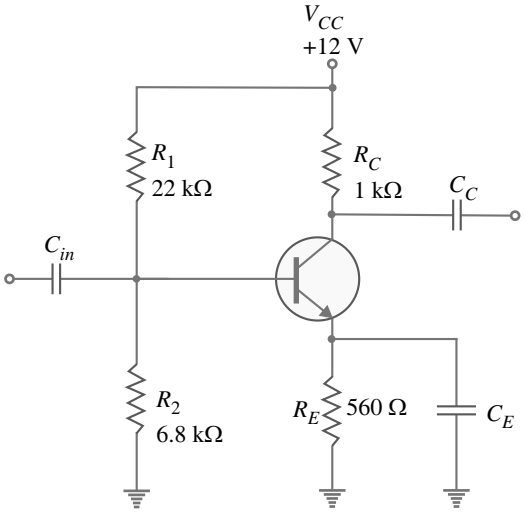


Fig. 8.4

In the given problem, f_{min} = 2kHz ; R_E = 560Ω.

$$\therefore \quad 10\, X_{C_E} = 560$$

or $$X_{C_E} = 560/10 = 56\Omega$$

or $$\frac{1}{2\pi f_{min} C_E} = 56$$

$$\therefore \quad C_E = \frac{1}{2\pi f_{min}\, 56} = \frac{1}{2\pi \times (2 \times 10^3) \times 56} = 1.42 \times 10^{-6}\ \text{F} = \mathbf{1.42\ \mu F}$$

Note. While discussing *CE* amplifier, the reader should be very particular about the role of C_E.

8.5 Phase Reversal

In common emitter connection, when the input signal voltage increases in the positive sense, the output voltage increases in the negative direction and *vice-versa*. In other words, there is a phase difference of 180° between the input and output voltage in *CE* connection. This is called phase reversal.*

The phase difference of 180° between the signal voltage and output voltage in a common emitter amplifier is known as **phase reversal.**

Consider a common emitter amplifier circuit shown in Fig. 8.5. The signal is fed at the input terminals (*i.e.* between base and emitter) and output is taken from collector and emitter end of supply. The total instantaneous output voltage v_{CE} is given by;

$$**v_{CE} = V_{CC} - i_C R_C \qquad ...(i)$$

* This is so if output is taken from collector and emitter end of supply as is always done. However, if the output is taken across R_C, it will be in phase with the input.

** Reactance of C_C (= 10μF) is negligible at ordinary signal frequencies. Therefore, it can be considered a short for the signal.

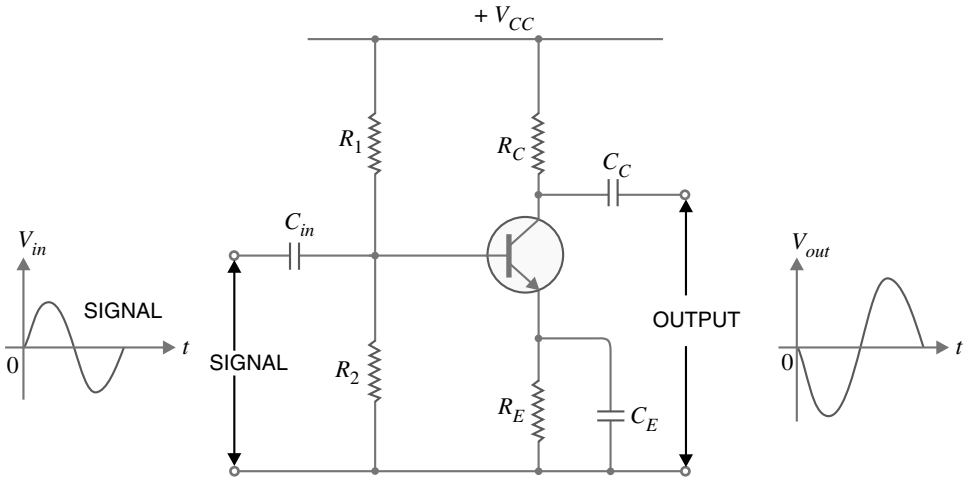


Fig. 8.5

When the signal voltage increases in the positive half-cycle, the base current also increases. The result is that collector current and hence voltage drop $i_C R_C$ increases. As V_{CC} is constant, therefore, output voltage v_{CE} decreases. In other words, as the signal voltage is increasing in the positive half-cycle, the output voltage is increasing in the negative sense *i.e.* output is 180° out of phase with the input. It follows, therefore, that in a common emitter amplifier, the positive half-cycle of the signal appears as amplified negative half-cycle in the output and *vice-versa*. It may be noted that amplification is not affected by this phase reversal.

The fact of phase reversal can be readily proved mathematically. Thus differentiating exp. (*i*), we get,

$$dv_{CE} = 0 - di_c\, R_C$$

or $$dv_{CE} = - di_c\, R_C$$

The negative sign shows that output voltage is 180° out of phase with the input signal voltage.

Graphical demonstration. The fact of phase reversal in *CE* connection can be shown graphically with the help of output characteristics and load line (See Fig. 8.6).

In Fig. 8.6, *AB* is the load line. The base current fluctuates between, say ± 5μA with 10μA as the zero signal base current. From the figure, it is clear that when the base current is maximum in the positive direction, v_{CE} becomes

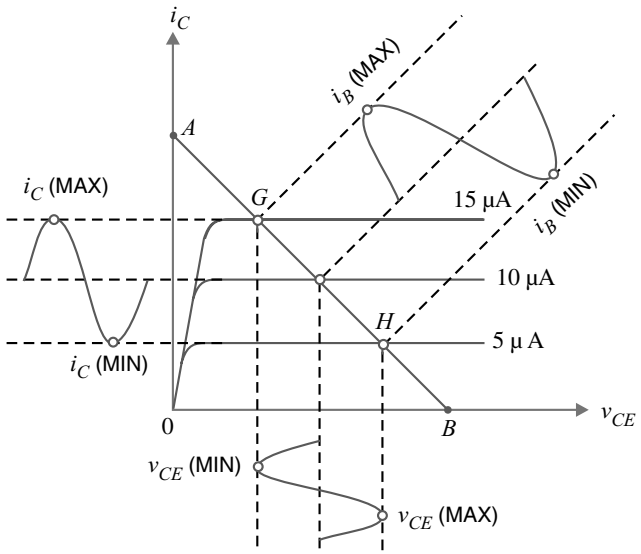


Fig. 8.6

maximum in the negative direction (*point G* in Fig. 8.6). On the other hand, when the base current is maximum in the negative direction, v_{CE} is maximum in the positive sense (*point H* in Fig. 8.6). Thus, the input and output voltages are in *phase opposition* or equivalently, the transistor is said to produce a 180° phase reversal of output voltage *w.r.t.* signal voltage.

Note. No phase reversal of voltage occurs in common base and common collector amplifiers. The a.c. output voltage is in phase with the a.c. input signal. For all three amplifier configurations ; input and output currents are in phase.

Example 8.3. *Illustrate the phenomenon of phase reversal in CE amplifier assuming typical circuit values.*

Solution. In every type of amplifier, the input and output currents are in phase. However, common emitter amplifier has the unique property that input and output voltages are 180° out of phase, even though the input and output currents are in phase. This point is illustrated in Fig. 8.7. Here it is assumed that *Q*-point value of I_B = 10 μA, a.c. signal peak value is 5 μA and β = 100. This means that input current varies by 5 μA both above and below a 10 μA d.c. level. At any instant, the output current will be 100 times the input current at that instant. Thus when the input current is 10 μA, output current is $i_C = 100 \times 10$ μA = 1 mA. However, when the input current is 15 μA, then output current is $i_C = 100 \times 15$ μA = 1.5 mA and so on. Note that input and output currents are in phase.

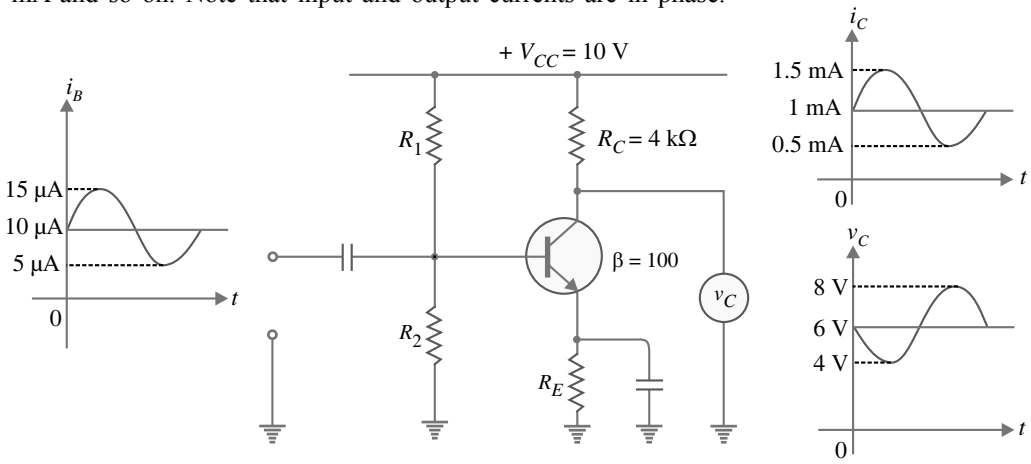


Fig. 8.7

The output voltage, $v_C = V_{CC} - i_C R_C$

(*i*) When signal current is zero (*i.e.*, in the absence of signal), i_C = 1 mA.

$\therefore \quad v_C = V_{CC} - i_C R_C = 10 \text{ V} - 1 \text{ mA} \times 4 \text{ k}\Omega = 6\text{V}$

(*ii*) When signal reaches positive peak value, i_C = 1.5 mA.

$\therefore \quad v_C = V_{CC} - i_C R_C = 10 \text{ V} - 1.5 \text{ mA} \times 4 \text{ k}\Omega = 4 \text{ V}$

Note that as i_C increases from 1mA to 1.5 mA, v_C decreases from 6V to 4V. Clearly, output voltage is 180° out of phase from the input voltage as shown in Fig. 8.7.

(*iii*) When signal reaches negative peak, i_C = 0.5 mA.

$\therefore \quad v_C = V_{CC} - i_C R_C = 10 \text{ V} - 0.5 \text{ mA} \times 4 \text{ k}\Omega = 8 \text{ V}$

Note that as i_C decreases from 1.5 mA to 0.5 mA, v_C increases from 4 V to 8 V. Clearly, output voltage is 180° out of phase from the input voltage. The following points may be noted carefully about *CE* amplifier :

(*a*) The input voltage and input current are in phase.

(*b*) Since the input current and output current are in phase, input voltage and output current are in phase.

(*c*) Output current is 180° out of phase with the output voltage (v_C). *Therefore, input voltage and output voltage are 180° out of phase.*

8.6 Input/Output Phase Relationships

The following points regarding the input / output phase relationships between currents and voltages for the various transistor configurations may be noted :

(*i*) *For every amplifier type (CE, CB and CC), the input and output currents are in phase.* When the input current decreases, the output current also decreases and *vice-versa*.

(*ii*) *Remember that common emitter (CE) circuit is the only configuration that has input and output voltages 180° out of phase.*

(*iii*) *For both common base (CB) and common collector (CC) circuits, the input and output voltages are in phase.* If the input voltage decreases, the output voltage also decreases and vice-versa.

8.7 Important Parameters of an Amplifier

There are many types of amplifiers *e.g.,* voltage amplifiers, power amplifiers, current amplifiers *etc.* A voltage amplifier is designed to raise the voltage level of the signal. Likewise, a power amplifier is designed to raise the power level of the signal. Regardless of their type, all amplifiers have the following five important circuit parameters :

(*i*) Input resistance (R_{in})
(*ii*) Output resistance (R_{out})
(*iii*) Current gain (A_i)
(*iv*) Voltage gain (A_v)
(*v*) Power gain (A_p)

The above circuit parameters play an important role in the design and analysis of an amplifier. Fig. 8.8 shows the block diagram of a transistor amplifier. There are two input terminals and two output terminals. The input side (the side to which the signal V_S is normally applied) is to the left and the output side (where the load R_L is connected) is to the right. It may be added here that we *generally use r.m.s. values of voltages and currents to determine the above parameters of the amplifier.

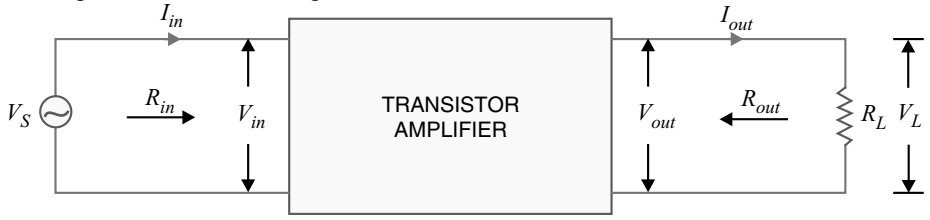


Fig. 8.8

(*i*) Input resistance, $R_{in} = \dfrac{V_{in}}{I_{in}}$

* Except for power calculations (where r.m.s. values must be used), we can also use peak or peak-to-peak values.

Note that input *resistance is the resistance "seen" by the input signal source V_S. It is a measure of current (I_{in}) drawn by the amplifier. The input resistance depends upon whether the transistor is in the common base (*CB*), common emitter (*CE*) or common collector (*CC*) arrangement.

(*ii*) Output resistance, $R_{out} = \dfrac{V_{out}}{I_{out}}$

Note that output resistance (R_{out}) is the resistance "seen" by the load resistance (R_L) when "looking back" into the output terminals of the amplifier. In fact, R_{out} is the internal resistance seen from the output terminals of the amplifier. *It may be thought as Thevenin resistance at the output terminals of the amplifier.*

Note. If reactances of coupling capacitors and bypass capacitors of an amplifier are large and cannot be neglected, then, the terms *input impedance* (Z_{in}) instead of input resistance (R_{in}) and *output impedance* (Z_{out}) instead of output resistance (R_{out}) should be used.

(*iii*) Current gain, $A_i = \dfrac{I_{out}}{I_{in}}$

Although this parameter receives less attention than voltage gain, it is an important quantity that can have significant impact on the overall performance of an amplifier. The current gain of a transistor amplifier may range from less than 1 to a value that may exceed 100.

(*iv*) Voltage gain, $A_v = \dfrac{V_{out}}{V_{in}}$

When load resistance R_L is not connected to the amplifier (unloaded condition) as shown in Fig. 8.9 (*i*), then V_{out}/V_{in} is called *no-load voltage gain* A_{v0} of the amplifier *i.e.*

No-load voltage gain, $A_{v0} = \dfrac{V_{out}}{V_{in}}$ ... in Fig. 8.9 (*i*)

When load resistance R_L is connected to the amplifier (loaded condition) as shown in Fig. 8.9 (*ii*), then V_L/V_{in} is called *loaded voltage gain* A_v of the amplifier *i.e.*

Loaded voltage gain, $A_v = \dfrac{V_L}{V_{in}}$ ... in Fig. 8.9 (*ii*)

Note that **$V_L < V_{out}$ (open-circuited or unloaded output voltage) so that *the loaded voltage gain of an amplifier is always less than its unloaded voltage gain.*

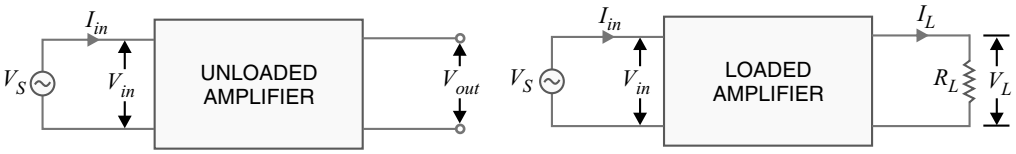


Fig. 8.9

* For signal frequencies < 100 kHz, the input impedance of the amplifier is purely resistive in nature. It is because for signal frequencies less than 100 kHz, the reactances of coupling and bypass capacitors have negligible effects.

** When R_L is not connected to the amplifier, the output terminals are open so that R_{out} is ineffective. However, when R_L is connected, R_{out} comes into picture so that $V_L < V_{out}$.

Voltage gain formula. We can easily derive the loaded voltage gain formula in terms of current gain (A_i).

Input voltage, $V_{in} = I_{in} R_{in}$

Output voltage, $V_L = I_L R_L = I_{out} R_L$ $(\because I_L = I_{out})$

$\therefore$ Loaded voltage gain, $A_v = \dfrac{V_L}{V_{in}} = \dfrac{I_{out} R_L}{I_{in} R_{in}} = \dfrac{I_{out}}{I_{in}} \times \dfrac{R_L}{R_{in}}$

$\therefore \quad A_v = A_i \dfrac{R_L}{R_{in}}$ ... (*i*) $\left(\because \dfrac{I_{out}}{I_{in}} = A_i\right)$

Eq. (*i*) is a general expression for the loaded voltage gain of a transistor amplifier and will always hold true regardless of the type of the amplifier.

(*v*) Power gain, $A_p = \dfrac{P_{out}}{P_{in}}$

Since $P_{in} = V_{in} I_{in}$ and $P_{out} = V_{out} I_{out}$

$\therefore \quad A_p = \dfrac{V_{out} I_{out}}{V_{in} I_{in}} = \dfrac{I_{out}}{I_{in}} \times \dfrac{V_{out}}{V_{in}} = A_i \times A_v$

$\therefore \quad A_p = A_i A_v$ = Current gain × Voltage gain

It may be noted that while carrying out power calculations, the r.m.s. values of currents and voltages must be used.

Example 8.4. *Fig. 8.10 shows the block diagram of a transistor amplifier. The signal source V_S = 12 mV (r.m.s.) and signal source resistance R_S = 1 kΩ. Determine (i) input current (ii) input resistance (iii) output voltage (iv) output current and (v) current gain.*

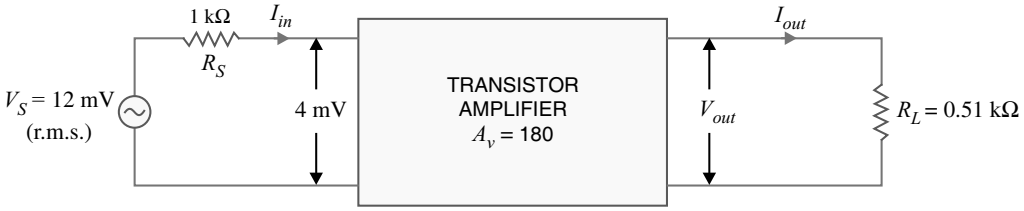


Fig. 8.10

Solution. V_S = 12 mV (r.m.s.) ; R_S = 1 kΩ ; V_{in} = 4 mV (r.m.s.) ; R_L = 0.51 kΩ ; A_v = 180

(*i*) Input current, $I_{in} = \dfrac{V_S - V_{in}}{R_S} = \dfrac{12\text{mV} - 4\text{mV}}{1\text{k}\Omega} = \dfrac{8\text{mV}}{1\text{k}\Omega} = \mathbf{8\ \mu A}$

(*ii*) Input resistance, $R_{in} = \dfrac{V_{in}}{I_{in}} = \dfrac{4\text{ mV}}{8\,\mu\text{A}} = \mathbf{500\ \Omega}$

(*iii*) Output voltage, $V_{out} = A_v V_{in} = 180 \times 4$ mV $= \mathbf{720\ mV}$

(*iv*) Output current, $I_{out} = \dfrac{V_{out}}{R_L} = \dfrac{720\text{ mV}}{0.51\text{k}\Omega} = \mathbf{1.41\ mA}$

(*v*) Current gain, $A_i = \dfrac{I_{out}}{I_{in}} = \dfrac{1.41\text{ mA}}{8\,\mu\text{A}} = \mathbf{176.25}$

Example 8.5. *Fig. 8.11 shows the block diagram of a transistor amplifier. Determine (i) input voltage (ii) input current (iii) input resistance and (iv) voltage gain between source voltage and amplifier output.*

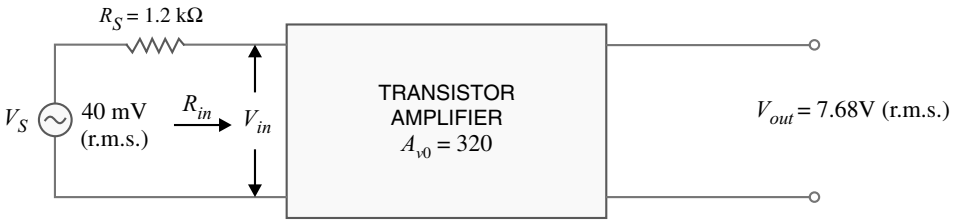


Fig. 8.11

Solution. V_S = 40 mV (r.m.s.) ; R_S = 1.2 kΩ ; A_{v0} = 320 ; V_{out} = 7.68V (r.m.s.)

(*i*) Input voltage, $V_{in} = \dfrac{V_{out}}{A_{v0}} = \dfrac{7.68\,\text{V}}{320}$ = **24 mV**

(*ii*) Input current, $I_{in} = \dfrac{V_S - V_{in}}{R_S} = \dfrac{40\text{ mV} - 24\text{ mV}}{1.2\text{ k}\Omega}$ = **13.33 μA**

(*iii*) Input resistance, $R_{in} = \dfrac{V_{in}}{I_{in}} = \dfrac{24\text{ mV}}{13.33\,\mu\text{A}}$ = **1.8 kΩ**

(*iv*) The source resistance R_S is in series with the input resistance R_{in} of the amplifier. Therefore, by voltage-divider rule,

$$V_{in} = \frac{V_S}{R_{in} + R_S} \times R_{in}$$

or
$$\frac{V_{in}}{V_S} = \frac{R_{in}}{R_{in} + R_S}$$

∴ Voltage gain between source voltage (V_S) and amplifier output voltage (V_{out}) is

$$A_{vS} = \frac{V_{out}}{V_S} = \frac{V_{in}}{V_S} \times \frac{V_{out}}{V_{in}} = \frac{R_{in}}{R_{in} + R_S} \times A_{v0}$$

where $A_{v0} = V_{out}/V_{in}$ = No-load voltage gain = 320

∴
$$A_{vS} = \frac{1.8\,\text{k}\Omega}{1.8\,\text{k}\Omega + 1.2\,\text{k}\Omega} \times 320 = \mathbf{192}$$

Example 8.6. *The signal source connected to the input of an amplifier has an internal resistance of 1.2 kΩ. The voltage gain of the amplifier from its input to its output is 140. What minimum value of input resistance should the amplifier have if the voltage gain from signal source to amplifier output is to be atleast 100 ?*

Solution. As shown in the above example, the voltage gain A_{vS} between source voltage (V_S) and amplifier output voltage (V_{out}) is

$$A_{vS} = \frac{R_{in}}{R_{in} + R_S} \times A_{v0}$$

Here, A_{vS} = 100 ; A_{v0} = No-load voltage gain = 140 ; R_S = 1.2 kΩ

$$\therefore \qquad 100 = \frac{R_{in}}{R_{in} + 1.2} \times 140$$

or $\qquad 100\ R_{in} + 120 = 140\ R_{in} \qquad \therefore\ R_{in} = 3\ \text{k}\Omega$

Therefore, the amplifier should have minimum input resistance of 3 kΩ.

8.8 Transistor is an Energy Converter

We have seen that a transistor amplifier provides a power gain *i.e.* output a.c. power is greater than the input a.c. signal power. It does not mean that a transistor is a power generating device because it is simply a two *pn* junction device and cannot be a source of power. It appears that law of conservation of energy is violated but that is not so. A transistor receives d.c. power from the d.c. supply (V_{CC}), converts it into a.c. power and adds it to the signal. Therefore, law of conservation of energy holds good. In fact, a transistor is an energy converter *i.e.* it converts d.c. power into a.c. power and the addition of this a.c. power to the signal results in the greater a.c. output power.

8.9 D.C. and A.C. Equivalent Circuits of Transistor Amplifier

In a transistor amplifier, both d.c. and a.c. conditions prevail. The d.c. sources set up d.c. currents and voltages whereas the a.c. source (*i.e.* signal) produces fluctuations in the transistor currents and voltages. Therefore, a simple way to analyse the action of a transistor is to split the analysis into two parts *viz.* a d.c. analysis and an a.c. analysis. In the d.c. analysis, we consider all the d.c. sources at the same time and work out the d.c. currents and voltages in the circuit. On the other hand, for a.c. analysis, we consider all the a.c. sources at the same time and work out the a.c. currents and voltages. By adding the d.c. and a.c. currents and voltages, we get the total currents and voltages in the circuit. For example, consider the amplifier circuit shown in Fig. 8.12. This circuit can be easily analysed by splitting it into *d.c. equivalent circuit* and *a.c. equivalent circuit.*

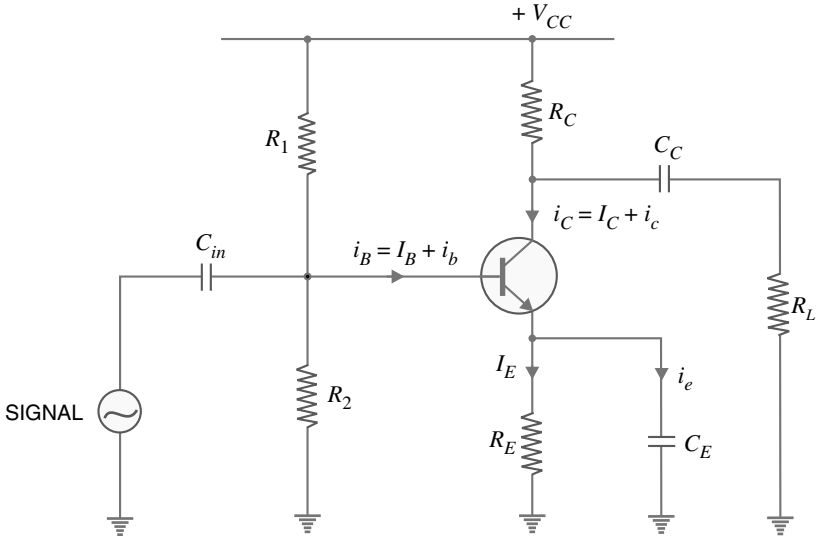


Fig. 8.12

5. Show that the output voltage of a single stage common emitter transistor amplifier is 180º out of phase with the input voltage.
6. What do you understand by d.c. and a.c. load lines? How will you construct them on the output characteristics?
7. Draw the d.c. and a.c. equivalent circuits of a transistor amplifier.
8. Derive an expression for the voltage gain of a transistor amplifier from its a.c. equivalent circuit.
9. Write short notes on the following :
   (*i*) phase reversal (*ii*) d.c. and a.c. load lines
   (*iii*) operating point (*iv*) classification of amplifiers.

Problems

1. In transistor amplifier, the collector current swings from 2 mA to 5 mA as the base current is changed from 5 μA to 15 μA. Find the current gain. **[300]**
2. A transistor amplifier employs a 4 kΩ as collector load. If the input resistance is 1 kΩ, determine the voltage gain. Given β = 100, g_m =10 mA/volt and signal voltage = 50 mV. **[1.04]**
3. Fig. 8.60 shows the transistor amplifier. If R_C = 4 kΩ, R_E = 5 kΩ and V_{CC} = 30 V, draw the d.c. load line.

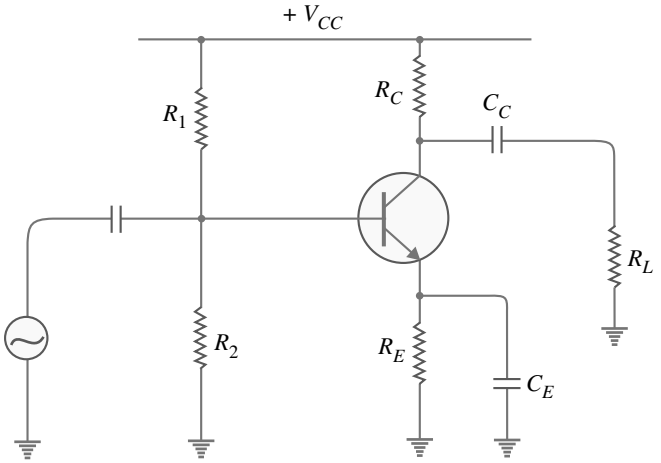


Fig. 8.60

4. Find the operating point for Fig. 8.60, V_{CC} = 30 V, R_1 = 20 kΩ, R_2 = 20 kΩ, R_C = 4 kΩ, R_E =5 kΩ. **[13.2 V, 1.85 mA]**
5. For the circuit shown in Fig. 8.60, find the voltage gain if β = 100, R_C = 3 kΩ, R_L = 6 kΩ and R_{in} = 2 kΩ. **[100]**
6. In the circuit shown in Fig. 8.60, V_{CC} = 30 V, R_1 = 2 kΩ, R_2 = 1 kΩ, R_C = 2 kΩ, R_L = 2 kΩ, R_E = 1 kΩ. Draw the d.c. and a.c. load lines.
7. A voltage-divider biased circuit has an emitter voltage of 2 V and an emitter resistor of 4.7 kΩ. What is the ac resistance of emitter diode? **[58.7 Ω]**
8. A transistor amplifier has a d.c. collector current of 5 mA. What is the a.c. resistance of the base if β = 200? **[1000 Ω]**

9. Determine the voltage gain for the amplifier circuit shown in Fig. 8.61.

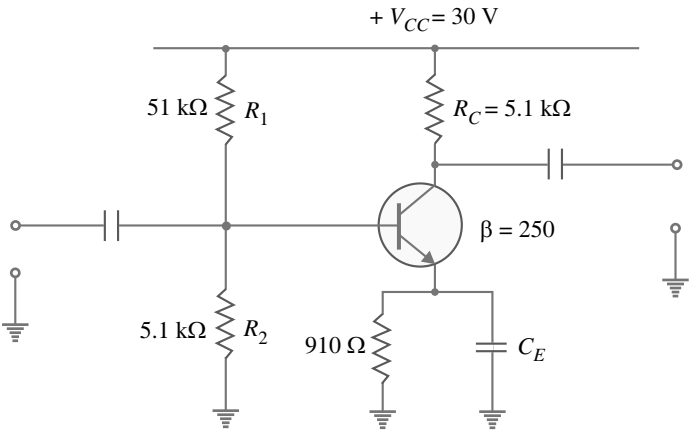


Fig. 8.61

10. What is the input impedance of the amplifier circuit shown in Fig. 8.61? **[1.75 kΩ]**

11. A voltage-divider biased amplifier has the values of R_1 = 40 kΩ, R_2 = 10 kΩ, R_C = 6 kΩ, R_E = 2 kΩ, V_{CC} = +10 V and β = 80. Determine the ac emitter resistance of the transistor. **[38.46Ω]**

12. A standard *CE* amplifier has the following values: V_{CC} = 30V, R_1 = 51 kΩ, R_2 = 5.1 kΩ, R_C = 5.1 kΩ, R_E = 910 Ω and β = 250. Determine the voltage gain of the amplifier. **[455.4]**

13. A *CE* amplifier has a voltage gain A_v = 59.1 and β = 200. Determine the power gain and output power of the amplifier when input power is 80 μW. **[11820; 945.6 mW]**

14. Determine the voltage gain for the first stage in Fig. 8.62. **[53.03]**

15. If the value of β for the second stage in Fig. 8.62 is increased to 280, determine the voltage gain of the first amplifier stage. **[58.08]**

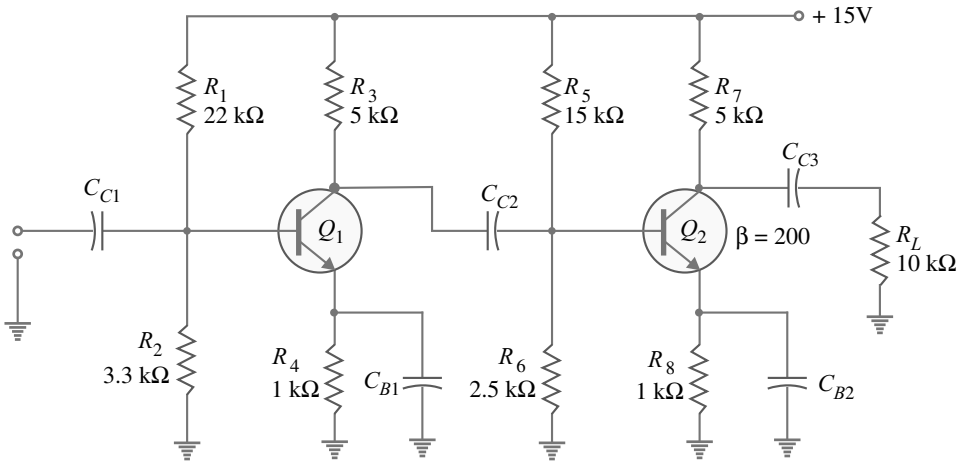


Fig. 8.62



Multistage Transistor Amplifiers

Learning Objectives

- Understand a multistage transistor amplifier and its need.
- Learn the effect of source resistance, load resistance and interstage loading on the voltage gain of multistage transistor amplifier.
- Explore the characteristics of *RC* coupling, transformer coupling and direct coupling in a multistage amplifier.
- Develop an understanding of expressing amplifier gain in decibels (dB).
- Recognise the importance of frequency response and bandwidth of an amplifier.
- Learn the characteristics of *RC* coupled transistor amplifier.
- Understand the important aspects of transformer coupled transistor amplifier.
- Recognise the need of direct coupled transistor amplifier.

Introduction

The output from a single stage amplifier is usually insufficient to drive an output device. In other words, the gain of a single amplifier is inadequate for practical purposes. Consequently, additional amplification over two or three stages is necessary. To achieve this, the output of each amplifier stage is *coupled* in some way to the input of the next stage. The resulting system is referred to as multistage amplifier. It may be emphasised here that a practical amplifier is always a multistage amplifier. For example, in a transistor radio receiver, the number of amplification stages may be six or more. In this chapter, we shall focus our attention on the various multistage transistor amplifiers and their practical applications.

9.1 Multistage Transistor Amplifier

A transistor circuit containing more than one stage of amplification is known as **multistage transistor amplifier.**

In a multistage transistor amplifier, a number of single amplifier stages are connected in **cascade arrangement i.e.* output of first stage is connected to the input of the second stage, the output of second stage is connected to the input of the third stage and so on.

* The term *cascade* means *connected in series*.

Generally, a single amplifier stage cannot provide the required gain (voltage gain or power gain or current gain). The primary need of a multistage amplifier is to provide the additional gain. For example, in an audio amplifier system, we require a large amount of power (of the order of watts) to operate the loudspeaker. However, the signal originating at the microphone has very small power (of the order of mW or less). This requires the use of a multistage amplifier to "boost" the signal to provide the necessary power.

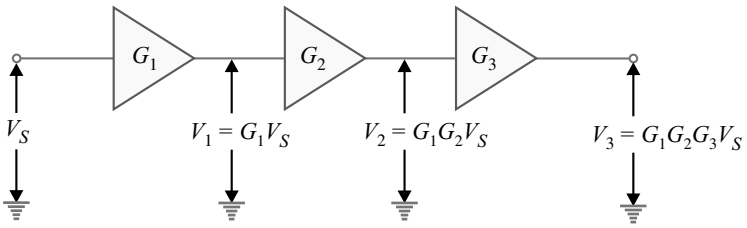


Fig. 9.1

Gain. *The total *gain (G) of a multistage amplifier is equal to the product of the gains of the individual stages (not the sum!).* Consider a 3-stage amplifier shown in Fig. 9.1. If G_1, G_2 and G_3 are the voltage gains of the first stage, second stage and third stage respectively, then total voltage gain G is given by ;

$$G = G_1 \times G_2 \times G_3 \quad \ldots (i)$$

The above relation can be easily proved. Suppose the input voltage to the first stage is V_S. Then,

$$\text{Output voltage of first stage} = G_1V_S$$

$$\text{Output voltage of second stage} = (G_1V_S)G_2 = G_1G_2V_S$$

$$\text{Output voltage of third stage} = (G_1G_2V_S)G_3 = G_1G_2G_3V_S$$

$$\therefore \quad \text{Total voltage gain, } G = \frac{\text{Output voltage of third stage}}{V_S}$$

or
$$G = \frac{G_1G_2G_3V_S}{V_S} = G_1G_2G_3$$

Thus the total voltage gain is equal to the *product* of individual stage voltage gains. It may be noted that actually the total voltage gain $G < G_1G_2G_3$. It is because while deriving eq. (*i*) above, we have made the following simplyfying assumptions :

(*i*) The resistance R_S of the voltage source connected to the first stage is zero (*i.e.* $R_S = 0$).

(*ii*) The load R_L is not connected to the last stage.

(*iii*) The *interstage loading* effect is neglected. It means that we have ignored the loading effect of the input impedance of the next stage.

In a practical multistage ampifier, all the above factors are present so that $G < G_1G_2G_3$. Nevertheless, the total voltage gain (G) of the multistage amplifier is very large compared to that of the single amplifier stage. This will become quite apparent from the following examples.

* It may be voltage gain (A_v) or power gain (A_p) or current gain (A_i).

Example 9.1. *Fig. 9.2 shows a 3-stage transistor amplifier and the a.c. r.m.s. voltages at several points in the amplifier. Note that V_1 is the input voltage delivered by a signal source having zero resistance (i.e. $R_S = 0$) and that V_3 is the output voltage with no load connected. Find (i) voltage gain of each stage (ii) overall voltage gain V_3/V_1.*

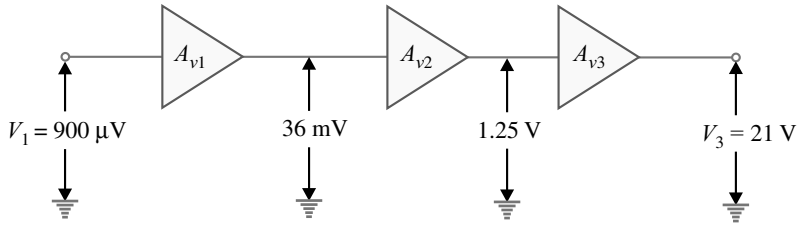


Fig. 9.2

Solution. We neglect the loading effect of the input impedance of next stage in this multistage amplifier.

(*i*) Voltage gain of first stage, $A_{v1} = \dfrac{36\,\text{mV}}{900\,\mu\text{V}} = \mathbf{40}$

Voltage gain of second stage, $A_{v2} = \dfrac{1.25\,\text{V}}{36\,\text{mV}} = \mathbf{34.72}$

Voltage gain of third stage, $A_{v3} = \dfrac{21\,\text{V}}{1.25\,\text{V}} = \mathbf{16.8}$

(*ii*) Overall voltage gain, $A_v = A_{v1} \times A_{v2} \times A_{v3}$

$= 40 \times 34.72 \times 16.8 = \mathbf{23{,}333}$

Alternatively. $A_v = \dfrac{V_3}{V_1} = \dfrac{21\,\text{V}}{900\,\mu\text{V}} = 23{,}333$

Example 9.2. *The voltage gains of the stages in a 4-stage transistor amplifier are shown in Fig. 9.3. Find (i) the overall voltage gain, V_{out}/V_{in} (ii) the voltage gain that would be necessary in a fifth stage, which, if added to the cascade, would make the overall voltage gain 10^5.*

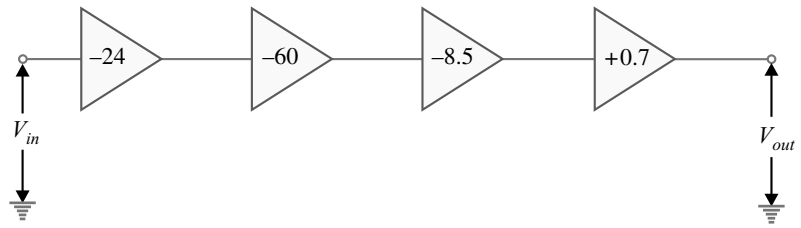


Fig. 9.3

Solution. $A_{v1} = -24$; $A_{v2} = -60$; $A_{v3} = -8.5$; $A_{v4} = +0.7$

(*i*) The overall voltage gain A_v $(= V_{out}/V_{in})$ of the 4-stage amplifier is equal to the product of voltage gains of the individual stages *i.e.*

$A_v = A_{v1} \times A_{v2} \times A_{v3} \times A_{v4}$

$= (-24) \times (-60) \times (-8.5) \times (+0.7) = \mathbf{-8568}$

Recall that a *CE* amplifier causes a phase inversion of 180° of the input signal. A multistage amplifier may contain one or more *CE* amplifier stages. If the number of *CE* amplifier stages is *odd*, the final output voltage (V_{out}) will be out of phase with the input voltage (V_{in}). On the other hand, if the number of inverting stages is *even* (or zero) V_{out} will be in phase with V_{in}.

(*ii*) Suppose the voltage gain of the fifth stage is A'_v. Then as per the conditions of the problem,

$$(-8568) \times A'_v = 10^5$$

$$\therefore \quad A'_v = \frac{10^5}{-8568} = \mathbf{-11.67}$$

Example 9.3. *It is desired to construct a 3-stage amplifier whose overall voltage gain is 500. The *in-circuit voltage gains of the first two stages are to be equal and the voltage gain of the third stage is to be one-half that of each of the first two. What should be the voltage gain of each stage ?*

Solution. Suppose the voltage gain of the first stage is A_{v1}. As per the conditions of the problem,

Voltage gain of second stage, $A_{v2} = A_{v1}$

Voltage gain of third stage, $A_{v3} = \dfrac{A_{v1}}{2}$

It is given that overall voltage gain of 3-stage amplifier is $A_v = 500$.

Now, $A_v = A_{v1} \times A_{v2} \times A_{v3}$

or $500 = A_{v1} \times A_{v1} \times \dfrac{A_{v1}}{2}$

$$\therefore \quad A_{v1} = \sqrt[3]{1000} = 10$$

$\therefore$ Voltage gain of first stage $= A_{v1} = \mathbf{10}$

Voltage gain of second stage $= A_{v1} = \mathbf{10}$

Voltage gain of third stage $= A_{v1}/2 = 10/2 = \mathbf{5}$

9.2 Effect of R_S, R_L and Interstage Loading

Fig. 9.4 shows a 3-stage amplifier. The signal source V_S having internal resistance R_S (also called *source resistance*) is connected to the input of the first stage. The load R_L is connected to the output terminals of the last stage. The load may be a loudspeaker, antenna, an indicating instrument or any other useful device. Further, there is interstage loading *i.e.* output impedance of one stage interacts with the input impedance of the next stage. The presence of R_S, R_L and interstage loading reduces the overall voltage gain of the multistage amplifier.

(*i*) Effect of source resistance R_S. As shown in Fig. 9.4, the source resistance R_S is in series with the input impedance Z_{in} of the first stage. Therefore, the a.c. voltage V_1 at the input terminals of the first stage is

$$V_1 = I_1 Z_{in} = \frac{V_S}{R_S + Z_{in}} \times Z_{in}$$

* The in-circuit gains mean the gains when all the stages are connected *i.e.* we take into account interstage loading.

or $$V_1 = \left(\frac{Z_{in}}{R_S + Z_{in}}\right) V_S \quad \ldots (i)$$

It may be seen from eq. (*i*) above that $V_1 < V_S$. Therefore, source resistance R_S reduces the overall voltage gain of the multistage amplifier.

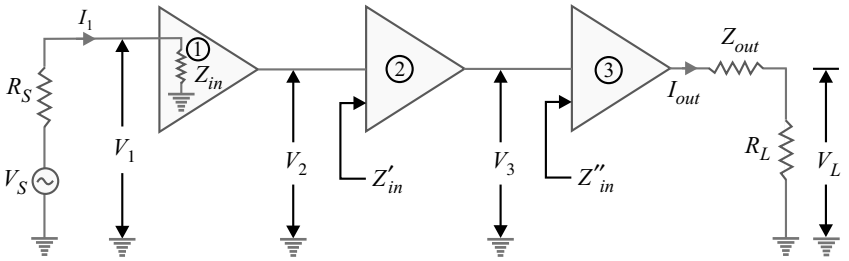


Fig. 9.4

(*ii*) Effect of load R_L. As shown in Fig. 9.4, the output impedance Z_{out} of the last stage is in series with the load R_L. Therefore, a.c. voltage V_L across the load R_L is

$$V_L = I_{out} R_L = \frac{A_{v0} V_1}{Z_{out} + R_L} \times R_L$$

or $$V_L = A_{v0} V_1 \left(\frac{R_L}{Z_{out} + R_L}\right) \quad \ldots (ii)$$

where A_{v0} = No-load voltage gain of multistage amplifier

It may be seen from eq. (*ii*) that $V_L < A_{v0} V_1$ (= unloaded output voltage). Therefore, application of R_L to the last stage reduces the overall voltage gain of the multistage amplifier.

(*iii*) Effect of interstage loading. As shown in Fig. 9.4, the output impedance of the first stage comes in parallel with the input impedance Z'_{in} of the second stage. This is called interstage loading and *reduces the voltage gain of the first stage. Likewise, the output impedance of the second stage comes in parallel with the input impedance Z''_{in} of the third stage and as a result, the voltage gain of the second stage is reduced. Therefore, the effect of interstage loading is to reduce the overall voltage gain of the multistage amplifier.

It may be noted that inspite of the fact that the presence of R_S, R_L and interstage loading reduces the voltage gain of the individual stages significantly, the overall voltage gain of a multistage amplifier is still quite large compared to that of a single amplifier stage.

Example 9.4. *Fig. 9.5 shows a 3-stage amplifier where source resistance R_S = 0 and load R_L is not connected to the last stage. Now, suppose that source resistance R_S = 2000 Ω and the load R_L = 25 Ω. Further, stage 1 has input resistance of 1 kΩ and stage 3 has output resistance of 50 Ω. Find :*

(*i*) *the overall voltage gain from signal source to load (i.e. V_L/V_S).*

(*ii*) *power gain between input to the first stage and the load.*

(*iii*) *current gain between input to the first stage and the load.*

* Because a.c. load of first stage is reduced.

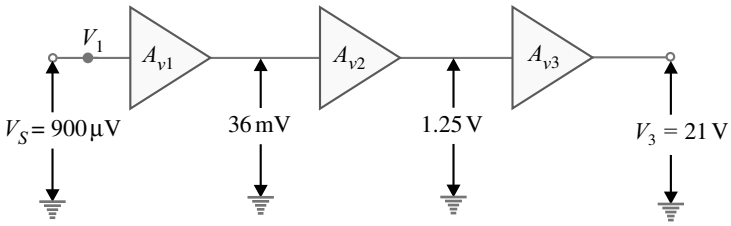


Fig. 9.5

Solution.

(*i*) The no-load voltage gain A_v of the multistage amplifier is given by ;

$$A_{v0} = \frac{V_3}{V_S} = \frac{21\,\text{V}}{900\,\mu\text{V}} = 23{,}333$$

The overall voltage gain of the 3-stage amplifier from source to load is (See Art. 8.24)

$$A_{vs} = \frac{V_L}{V_S} = A_{v0}\left(\frac{R_{in}}{R_S + R_{in}}\right)\left(\frac{R_L}{R_{out} + R_L}\right)$$

Here, A_{v0} = 23,333 ; R_{in} = 1 kΩ = 1000 Ω ; R_S = 2000 Ω ; R_L = 25 Ω ; R_{out} = 50 Ω

$$\therefore \quad A_{vs} = 23{,}333\left(\frac{1000}{2000+1000}\right)\left(\frac{25}{50+25}\right) = \mathbf{2592.5}$$

(*ii*) The source resistance R_S(= 2000 Ω) is in series with the input resistance R_{in} (= 1000 Ω) of the first stage. Therefore, a.c. voltage V_1 at the input terminals of the first stage is

$$V_1 = \frac{R_{in}}{R_S + R_{in}} \times V_S = \frac{1000}{2000+1000} \times 900\ \mu\text{V} = 300\ \mu\text{V}$$

∴ Input power to the first stage is

$$P_{in} = \frac{V_1^2}{R_{in}} = \frac{(300\times10^{-6})^2}{1000} = 90 \times 10^{-12}\ \text{W}$$

The output resistance R_{out} (= 50 Ω) is in series with the load R_L (= 25 Ω). Therefore, a.c. voltage V_L across the load is

$$V_L = A_{v0}V_1\left(\frac{R_L}{R_{out} + R_L}\right)$$

$$= 23{,}333 \times 300\ \mu\text{V}\left(\frac{25}{50+25}\right) = 2.33\ \text{V}$$

∴ Output power developed across the load is

$$P_{out} = P_L = \frac{V_L^2}{R_L} = \frac{(2.33)^2}{25} = 0.217\ \text{W}$$

$$\therefore \quad \text{Power gain, } A_p = \frac{P_{out}}{P_{in}} = \frac{0.217\,\text{W}}{90\times10^{-12}\,\text{W}} = \mathbf{2.41 \times 10^9}$$

(*iii*) The voltage gain between the input to the first stage and the load is

$$A_{vL} = \frac{V_L}{V_1} = \frac{2.33\,\text{V}}{300\,\mu\text{V}} = 7766$$

Now, $A_p = A_{vL} \times A_i$

$\therefore$ Current gain, $A_i = \dfrac{A_p}{A_{vL}} = \dfrac{2.41 \times 10^9}{7766} = \mathbf{3.1 \times 10^5}$

9.3 Coupling of Transistor Amplifiers

In a multistage transistor amplifier, the output of one amplifier stage is connected to the input of the next amplifier stage through a suitable *coupling device* and so on. The purpose of coupling device (*e.g.* a capacitor, transformer *etc.*) is (*i*) to transfer a.c. output of one amplifier stage to the input of the next amplifier stage and (*ii*) to isolate the d.c. conditions of one stage from the next stage. Fig. 9.6 shows the block diagram of a 3-stage amplifier. Each stage consists of one transistor and associated circuitry and is coupled to the next stage through a coupling device. The name of the amplifier is usually given after the type of coupling used *e.g.*

Name of coupling	*Name of multistage amplifier*
Capacitor coupling	Capacitively coupled amplifier
Transformer coupling	Transformer coupled amplifier
Direct coupling	Direct coupled amplifier



Fig. 9.6

Let us briefly examine the characteristics of each of these three types of coupling.

(*i*) Capacitor coupling. In this type of coupling, the output of one amplifier stage is connected to the input of the next amplifier stage through a capacitor C (called *coupling capacitor*) as shown in Fig. 9.7. We know that a capacitor blocks d.c. but allows the a.c. to pass through it. Therefore, the capacitor C allows the a.c. signal to pass from the output of one amplifier stage to the input of the next amplifier stage but prevents direct current (d.c.) from flowing between the output of one stage and the input of the next stage. In this way, d.c. bias conditions of one stage are isolated from the d.c. bias conditions of the next stage.

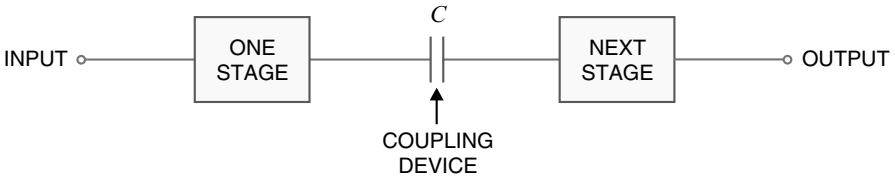


Fig. 9.7

The advantage of capacitor coupling is that it is simple and inexpensive. The disadvantage is that at low signal frequencies, the capacitor poses problems. At *low signal frequencies, the reactance X_C (= $1/2\pi fC$) of the coupling capacitor is quite large and considerable amount

* At high signal frequencies, the capacitor behaves as a short.

of a.c. signal is dropped across it. This reduces the a.c. signal at the input of the next stage and hence output voltage from this stage.

(*ii*) Transformer coupling. In this type of coupling, the output of one amplifier stage is connected to the input of the next amplifier stage through a transformer as shown in Fig. 9.8. We know that a transformer blocks d.c. while it transfers a.c. from primary to the secondary. Therefore, the transformer allows the a.c. signal to pass from the output of one amplifier stage to the input of the next amplifier stage but prevents direct current (d.c.) from flowing between the output of one stage and the input of the next stage. In this way, the transformer provides d.c. isolation between the two stages.

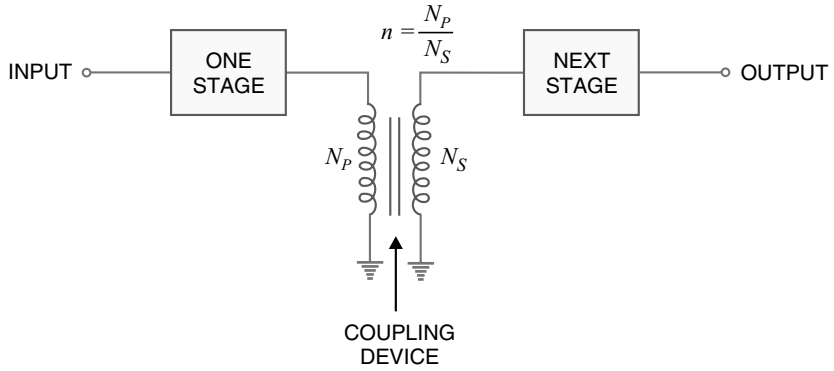


Fig. 9.8

An important advantage of transformer coupling is that it permits impedance matching for maximum power transfer. If a step-down transformer of turn ratio n (= N_P/N_S) is used, then value of primary impedance is $^*Z_P = n^2 Z_L$ where Z_L (load) is the input impedance of the second stage. If turn ratio n is so selected to make Z_P equal to the output impedance of the first stage, then maximum power will be transferred to the second stage. The disadvantage of transformer coupling is that there are losses in a transformer.

(*iii*) Direct coupling. In this type of coupling, the output of one amplifier stage is connected *directly* to the input of the next amplifier stage as shown in Fig. 9.9. The major advantage of direct coupling is that there are no losses in the coupling, either at low or high frequencies.

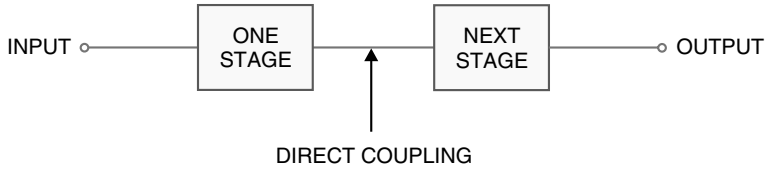


Fig. 9.9

The disadvantage of direct coupling is that the input impedance of the second stage affects the voltage gain as well as *Q*-point of the first stage.

9.4 Role of Capacitors in Transistor Amplifiers

Regardless of the manner in which a capacitor is connected in a transistor amplifier, its behaviour towards d.c. and a.c. is as follows. *A capacitor blocks d.c. i.e. a capacitor behaves*

* The d.c. resistance of primary is very small and is neglected.

as an "open" to d.c.* Therefore, for d.c. analysis, we can remove the capacitors from the transistor amplifier circuit. A capacitor offers reactance (= $1/2\pi fC$) to a.c. depending upon the values of f and C. In practical transistor circuits, the size of capacitors is so selected that they offer negligible (ideally zero) reactance to the range of frequencies handled by the circuits. Therefore, *for a.c. analysis, we can replace the capacitors by a short i.e. by a wire.* The capacitors serve the following two roles in transistor amplifiers :

1. As coupling capacitors
2. As bypass capacitors

1. As coupling capacitors. In most applications, you will not see a single transistor amplifier. Rather we use a multistage amplifier *i.e.* a number of transistor amplifiers are connected in series or cascaded. The capacitors are commonly used to connect one amplifier stage to another. When a capacitor is used for this purpose, it is called a *coupling capacitor.* Fig. 9.10 shows the coupling capacitors (C_{C1}, C_{C2} , C_{C3} and C_{C4}) in a multistage amplifier. A coupling capacitor performs the following two functions :

(*i*) It blocks d.c. *i.e.* it provides d.c. isolation between the two stages of a multistage amplifier.

(*ii*) It passes the a.c. signal from one stage to the next with little or no distortion.

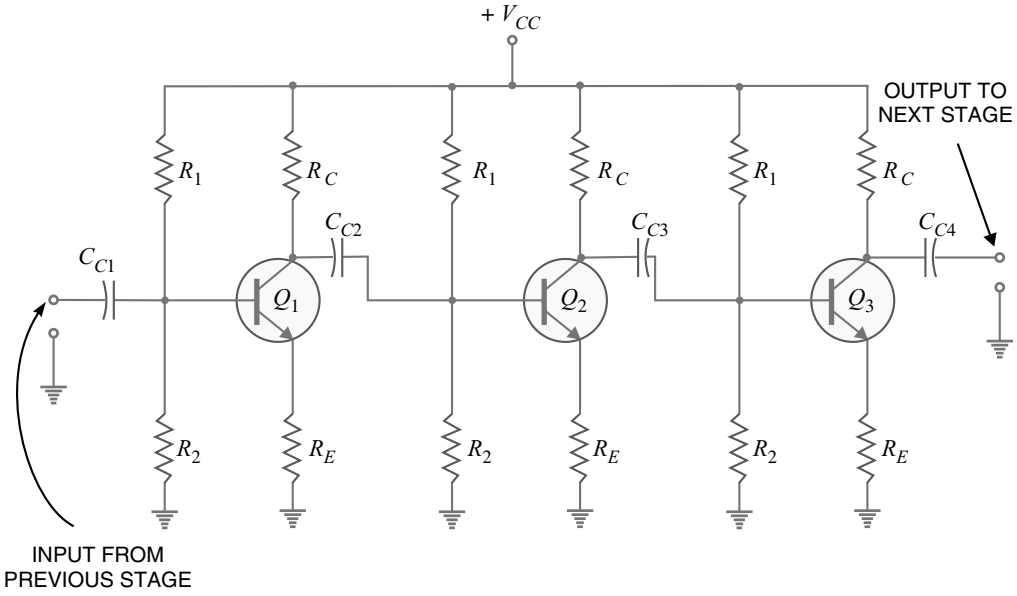


Fig. 9.10

2. As bypass capacitors. Like a coupling capacitor, a bypass capacitor also blocks d.c. and behaves as a short or wire (due to proper selection of capacitor size) to an a.c. signal. But it is used for a different purpose. A bypass capacitor is connected in parallel with a circuit component (*e.g.* resistor) to bypass the a.c. signal and hence the name. Fig. 9.11 shows a bypass capacitor C_E connected across the emitter resistance R_E. Since C_E behaves as a short to the a.c. signal, the whole of a.c. signal (i_e) passes through it. Note that C_E keeps

* $X_C = \dfrac{1}{2\pi fC}$. For d.c., $f = 0$ so that $X_C \to \infty$. Therefore, a capacitor behaves as an open to d.c.

the emitter at a.c. ground. Thus for a.c. purposes, R_E does not exist. We have already seen in the previous chapter that C_E plays an important role in determining the voltage gain of the amplifier circuit. If C_E is removed, the voltage gain of the amplifier is greatly reduced. Note that C_{in} is the coupling capacitor in this circuit.

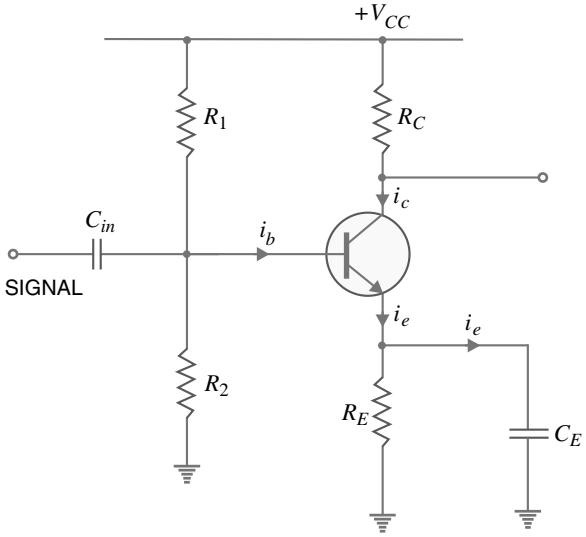


Fig. 9.11

9.5 Amplifier Gain in Decibels (dB)

While designing and analysing a multistage amplifier, the most important consideration is the gain of the individual amplifier stages. Although the gain of an amplifier (voltage gain or power gain or current gain) can be expressed as a number, yet it is of great practical importance to assign it a unit. The unit assigned is *bel or decibel* (*dB*).

The common logarithm (*log to the base 10*) *of power gain is known as* **bel power gain** *i.e.*

$$\text{Power gain} = \log_{10} \frac{P_{out}}{P_{in}} \text{ bel}$$

$$1 \text{ bel} = 10 \text{ dB}$$



Fig. 9.12

$$\therefore \qquad \text{Power gain} = 10 \log_{10} \frac{P_{out}}{P_{in}} \text{ dB}$$

If $P_{out} > P_{in}$, then power gain in dB is *positive*, indicating increase in power. If $P_{out} < P_{in}$, then power given in dB is *negative*, indicating reduction in power. However, if $P_{out} = P_{in}$, then power given is 0 dB since $\log_{10} 1 = 0$.

If the two powers are developed in the same resistance or equal resistances, then,

$$P_1 = \frac{V_{in}^2}{R} = I_{in}^2 R$$

$$P_2 = \frac{V_{out}^2}{R} = I_{out}^2 R$$

$$\therefore \quad \text{Voltage gain in dB} = 10 \log_{10} \frac{V_{out}^2 / R}{V_{in}^2 / R} = 20 \log_{10} \frac{V_{out}}{V_{in}}$$

$$\text{Current gain in dB} = 10 \log_{10} \frac{I_{out}^2 R}{I_{in}^2 R} = 20 \log_{10} \frac{I_{out}}{I_{in}}$$

Advantages. The following are the advantages of expressing the gain in dB :

(*i*) The unit dB is a logarithmic unit. Our ear response is also logarithmic *i.e.* loudness of sound heard by ear is not according to the intensity of sound but according to the log of intensity of sound. Thus if the intensity of sound given by speaker (*i.e.* power) is increased 100 times, our ears hear a doubling effect ($\log_{10}$ 100 = 2) *i.e.* as if loudness were doubled instead of made 100 times. Hence, this unit tallies with the natural response of our ears.

(*ii*) When the gains are expressed in dB, the overall gain of a multistage amplifier is the sum of gains of individual stages in dB. Thus referring to Fig. 9.13,

$$\text{Gain as number} = \frac{V_2}{V_1} \times \frac{V_3}{V_2}$$

$$\text{Gain in dB} = 20 \log_{10} \frac{V_2}{V_1} \times \frac{V_3}{V_2}$$

$$= 20 \log_{10} \frac{V_2}{V_1} + 20 \log_{10} \frac{V_3}{V_2}$$

$$= \text{1st stage gain in dB + 2nd stage gain in dB}$$

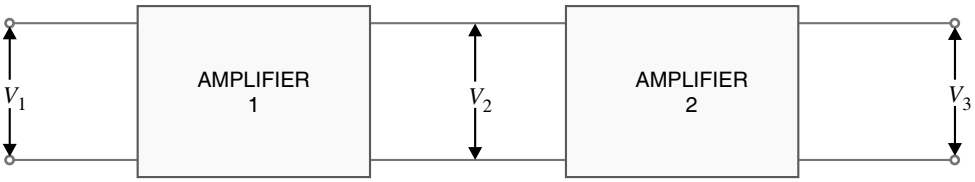


Fig. 9.13

However, absolute gain is obtained by multiplying the gains of individual stages. Obviously, it is easier to add than to multiply.

9.6 Properties of dB Gain

The power gain expressed as a number is called ordinary power gain. Similarly, the voltage gain expressed as a number is called ordinary voltage gain.

1. **Properties of dB power gain.** The following are the useful rules for dB power gain:

(*i*) *Each time the ordinary power gain increases (decreases) by a factor of 10, the dB power gain increases (decreases) by 10 dB.*

For example, suppose the ordinary power gain increases from 100 to 1000 (*i.e.* by a factor of 10).

$$\therefore \quad \text{Increase in dB power gain} = 10 \log_{10} 1000 - 10 \log_{10} 100$$
$$= 30 - 20 = 10 \text{ dB}$$

This property also applies for the decrease in power gain.

(*ii*) *Each time the ordinary power gain increases (decreases) by a factor of 2, the dB power gain increases (decreases) by 3 dB.*

For example, suppose the power gain increases from 100 to 200 (*i.e.* by a factor of 2).

$$\therefore \quad \text{Increase in dB power gain} = 10 \log_{10} 200 - 10 \log_{10} 100$$
$$= 23 - 20 = 3 \text{ dB}$$

2. Properties of dB voltage gain. The following are the useful rules for dB voltage gain:

(*i*) *Each time the ordinary voltage gain increases (decreases) by a factor of 10, the dB voltage gain increases (decreases) by 20 dB.*

For example, suppose the voltage gain increases from 100 to 1000 *(i.e.* by a factor of 10).

$$\therefore \quad \text{Increase in dB voltage gain} = 20 \log_{10} 1000 - 20 \log_{10} 100$$
$$= 60 - 40 = 20 \text{ dB}$$

(*ii*) *Each time the ordinary voltage gain increases (decreases) by a factor of 2, the dB voltage gain increases (decreases) by 6 dB.*

For example, suppose the voltage gain increases from 100 to 200 (*i.e.* by a factor of 2).

$$\therefore \quad \text{Increase in dB voltage gain} = 20 \log_{10} 200 - 20 \log_{10} 100$$
$$= 46 - 40 = 6 \text{ dB}$$

Example 9.5. *Find the gain in dB in the following cases :*

(i) Voltage gain of 30 *(ii) Power gain of 100*

Solution. (*i*) Voltage gain $= 20 \log_{10} 30$ dB = **29.54 dB**

(*ii*) Power gain $= 10 \log_{10} 100$ dB = **20 dB**

Example 9.6. *Express the following gains as a number :*

(i) Power gain of 40 dB *(ii) Power gain of 43 dB*

Solution. (*i*) Power gain = 40 dB = 4 bel

If we want to find the gain as a number, we should work from logarithm back to the original number.

$$\therefore \quad \text{Gain} = \text{Antilog } 4 = 10^4 = \mathbf{10{,}000}$$

(*ii*) Power gain = 43 dB = 4.3 bel

$$\therefore \quad \text{Power gain} = \text{Antilog } 4.3 = 2 \times 10^4 = \mathbf{20{,}000}$$

Alternatively. $10 \log_{10} \dfrac{P_2}{P_1} = 43 \text{ dB}$

or $\log_{10} \dfrac{P_2}{P_1} = 43/10 = 4.3$

$$\therefore \quad \frac{P_2}{P_1} = (10)^{4.3} = 20{,}000$$

In general, we have,

$$\frac{V_2}{V_1} = (10)^{gain\ in\ dB/20}$$

$$\frac{P_2}{P_1} = (10)^{gain\ in\ dB/10}$$

Example 9.7. *A three-stage amplifier has a first stage voltage gain of 100, second stage voltage gain of 200 and third stage voltage gain of 400. Find the total voltage gain in dB.*

Solution.

First-stage voltage gain in dB $= 20 \log_{10} 100 = 20 \times 2 = 40$

Second-stage voltage gain in dB $= 20 \log_{10} 200 = 20 \times 2.3 = 46$

Third-stage voltage gain in dB $= 20 \log_{10} 400 = 20 \times 2.6 = 52$

$\therefore$ Total voltage gain $= 40 + 46 + 52 =$ **138 dB**

Example 9.8. (*i*) *A multistage amplifier employs five stages each of which has a power gain of 30. What is the total gain of the amplifier in dB?*

(*ii*) *If a negative feedback of 10 dB is employed, find the resultant gain.*

Solution. Absolute gain of each stage $= 30$

No. of stages $= 5$

(*i*) Power gain of one stage in dB $= 10 \log_{10} 30 = 14.77$

$\therefore$ Total power gain $= 5 \times 14.77 =$ **73.85 dB**

(*ii*) Resultant power gain with negative feedback

$= 73.85 - 10 =$ **63.85 dB**

It is clear from the above example that by expressing the gain in dB, calculations have become very simple.

Example 9.9. *In an amplifier, the output power is 1.5 watts at 2 kHz and 0.3 watt at 20 Hz, while the input power is constant at 10 mW. Calculate by how many decibels gain at 20 Hz is below that at 2 kHz?*

Solution.

dB power gain at 2 kHz. At 2 kHz, the output power is 1.5 W and input power is 10 mW.

$$\therefore \quad \text{Power gain in dB} = 10 \log_{10} \frac{1.5\ \text{W}}{10\ \text{mW}} = 21.76$$

dB power gain at 20 Hz. At 20 Hz, the output power is 0.3 W and input power is 10 mW.

$$\therefore \quad \text{Power gain in dB} = 10 \log_{10} \frac{0.3\ \text{W}}{10\ \text{mW}} = 14.77$$

Fall in gain from 2 kHz to 20 Hz $= 21.76 - 14.77 =$ **6.99 dB**

Example 9.10. *A certain amplifier has voltage gain of 15 dB. If the input signal voltage is 0.8 V, what is the output voltage?*

Solution. dB voltage gain $= 20 \log_{10} V_2/V_1$

or $15 = 20 \log_{10} V_2/V_1$

or $15/20 = \log_{10} V_2/V_1$

or $0.75 = \log_{10} V_2/0.8$

Taking antilogs, we get,

$$\text{Antilog } 0.75 = \text{Antilog } (\log_{10} V_2/0.8)$$

or $10^{0.75} = V_2/0.8$

$\therefore$ $V_2 = 10^{0.75} \times 0.8 = \mathbf{4.5\ V}$

Example 9.11. *An amplifier has an open-circuit voltage gain of 70 dB and an output resistance of 1.5 kΩ. Determine the minimum value of load resistance so that voltage gain is not more than 67 dB.*

Solution. $A_{v0} = 70\text{ dB}$; $A_v = 67\text{ dB}$

$$A_{v0} \text{ in dB} - A_v \text{ in dB} = 70 - 67 = 3\text{ dB}$$

or $20 \log_{10} A_{v0} - 20 \log_{10} A_v = 3$

or $20 \log_{10} \dfrac{A_{v0}}{A_v} = 3$

or $\dfrac{A_{v0}}{A_v} = (10)^{3/20} = 1.41$

But $\dfrac{A_v}{A_{v0}} = \dfrac{R_L}{R_{out} + R_L}$ [See Art. 8.24]

$\therefore$ $\dfrac{1}{1.41} = \dfrac{R_L}{1.5 + R_L}$

or $R_L = \mathbf{3.65\ k\Omega}$

Example 9.12. *An amplifier feeding a resistive load of 1 kΩ has a voltage gain of 40 dB. If the input signal is 10 mV, find (i) output voltage (ii) load power.*

Solution.

(i) $\dfrac{V_{out}}{V_{in}} = (10)^{dB\ gain/20} = (10)^{40/20} = 100$

$\therefore$ $V_{out} = 100 \times V_{in} = 100 \times 10\text{ mV} = 1000\text{ mV} = \mathbf{1\ V}$

(ii) $\text{Load power} = \dfrac{V_{out}^2}{R_L} = \dfrac{(1)^2}{1000} = 10^{-3}\text{ W} = \mathbf{1\ mW}$

Example 9.13. *An amplifier rated at 40 W output is connected to a 10 Ω speaker.*

(i) Calculate the input power required for full power output if the power gain is 25 dB.

(ii) Calculate the input voltage for rated output if the amplifier voltage gain is 40 dB.

Solution.

(i) $\text{Power gain in dB} = 10 \log_{10} \dfrac{P_2}{P_1}$ or $25 = 10 \log_{10} \dfrac{40\text{ W}}{P_1}$

$\therefore$ $P_1 = \dfrac{40\text{ W}}{\text{Antilog } 2.5} = \dfrac{40\text{ W}}{3.16 \times 10^2} = \dfrac{40\text{ W}}{316} = \mathbf{126.5\ mW}$

(*ii*) Voltage gain in dB $= 20 \log_{10} \frac{V_2}{V_1}$ or $40 = 20 \log_{10} \frac{V_2}{V_1}$

$\therefore \quad \frac{V_2}{V_1} = \text{Antilog } 2 = 100$

Now, $V_2 = \sqrt{P_2 R} = \sqrt{40\text{W} \times 10\,\Omega} = 20\text{ V}$

$\therefore \quad V_1 = \frac{V_2}{100} = \frac{20\text{V}}{100} = \mathbf{200\text{ mV}}$

Example 9.14. *The input voltage to an amplifier is 4 mV (r.m.s.). At point 1 in the amplifier, the voltage gain w.r.t. the input is – 4.2 dB and at point 2, the voltage gain w.r.t. point 1 is 18.5 dB. Find the (i) voltage at point 1 (ii) voltage at point 2 (iii) voltage gain in dB at point 2 w.r.t. the input.*

Solution. Input voltage, $V_i = 4\text{ mV}$ (r.m.s.) $= 4 \times 10^{-3}$ V (r.m.s.)

(*i*) Suppose at point 1, the voltage is V_1 (r.m.s.). Then,

$$20 \log_{10} \frac{V_1}{V_i} = -4.2$$

or $$20 \log_{10} \frac{V_1}{4 \times 10^{-3}} = -4.2$$

or $$\log_{10} \frac{V_1}{4 \times 10^{-3}} = -4.2/20 = -0.21$$

Taking antilog of both sides, we get,

$$\frac{V_1}{4 \times 10^{-3}} = 0.617$$

$\therefore \quad V_1 = 2.47 \times 10^{-3}\text{ V} = \mathbf{2.47\text{ mV}}$

(*ii*) Suppose the voltage at point 2 is V_2 (r.m.s.). Then,

$$20 \log_{10} \frac{V_2}{V_1} = 18.5$$

or $$20 \log_{10} \frac{V_2}{2.47 \times 10^{-3}} = 18.5$$

or $$\log_{10} \frac{V_2}{2.47 \times 10^{-3}} = \frac{18.5}{20} = 0.925$$

$\therefore \quad V_2 = 2.47 \times 10^{-3}$ antilog (0.925)

$= 20.75 \times 10^{-3}\text{ V} = \mathbf{20.75\text{ mV}}$

(*iii*) Voltage gain in dB at point 2 *w.r.t.* the input is

$$\text{Voltage gain} = 20 \log_{10} \frac{V_2}{V_i} = 20 \log_{10} \frac{20.75}{4} = \mathbf{14.3\text{ dB}}$$

Alternatively. Voltage gain $= -4.2\text{ dB} + 18.5\text{ dB} = 14.3\text{ dB}$

Note that overall voltage gain (*i.e.* from the input to point 2) in dB is the sum of intermediate gains in dB.

9.7 Frequency Response and Bandwidth of an Amplifier

In the design of an amplifier, the *frequency response* and the *bandwidth* of the amplifier are very important considerations. It is because the performance of an amplifier in a circuit largely depends upon the shape of the frequency response and value of the bandwidth.

1. Frequency response. *The curve between voltage gain (or power gain) and the signal frequency of an amplifier is called* **frequency response** *of the amplifier.*

The frequency response of an amplifier indicates how voltage gain (or power gain) of the amplifier *responds* to the changes in the signal frequency. The voltage gain of an amplifier varies with signal frequency. It is because reactance of the capacitors in the circuit changes with signal frequency and hence affects the output voltage. Fig. 9.14 shows the frequency response of a typical amplifier. The gain of the amplifier increases as the frequency increases from zero till it becomes maximum at f_r, called *resonant frequency*. If the frequency of signal increases beyond f_r, the gain decreases. Note that voltage gain of the amplifier is small at both low and high frequencies.

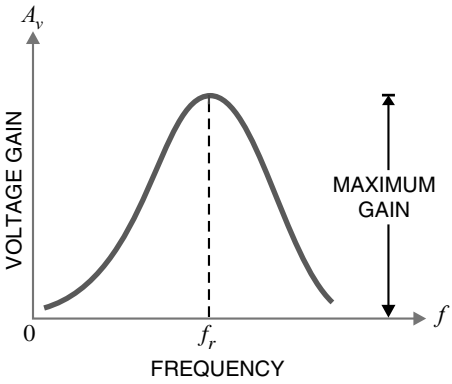


Fig. 9.14

The performance of an amplifier depends to a considerable extent upon its frequency response. While designing an amplifier, appropriate steps must be taken to ensure that gain is essentially uniform over some specified frequency range. For instance, in case of an audio amplifier, which is used to amplify speech or music, it is necessary that all the frequencies in the sound spectrum (*i.e.* 20 Hz to 20 kHz) should be uniformly amplified otherwise speaker will give a distorted sound output.

The following points may be noted :

(*i*) The shape of the frequency response depends upon the type of the amplifier. For example, Fig. 9.15 shows the frequency response of an audio voltage amplifier. Note that the voltage gain of the amplifier is *nearly* constant over the audio frequency range (20 Hz to 20 kHz) but falls off at low and high frequencies. The frequency range over which the gain is more or less constant (*flat*) is called *midband range* (or *mid-frequency region*). You may see that when the signal frequency starts to go outside the mid-frequency region (or midband range), the voltage gain of the amplifier begins to decrease rapidly.

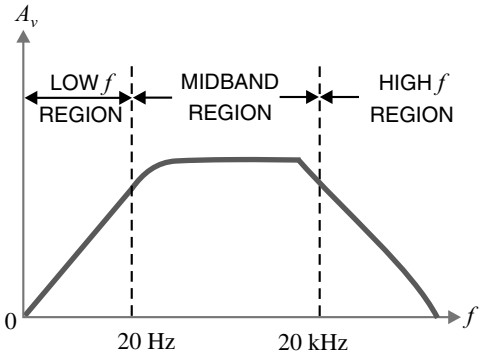


Fig. 9.15

(*ii*) We generally divide the frequency response of an amplifier into three regions (See Fig. 9.15) *viz. low frequency region, high-frequency region* **and** *mid-frequency region*. The

D.C. emitter voltage, $V_{E2} = V_{C1} - V_{BE} = 5.18\,\text{V} - 0.7\,\text{V} = \mathbf{4.48\,V}$

D.C. emitter current, $I_{E2} = \dfrac{V_{E2}}{R_6} = \dfrac{4.48\,\text{V}}{10\,\text{k}\Omega} = 0.448\,\text{mA}$

D.C. voltage at collector, $V_{C2} = V_{CC} - I_{C2}\,R_5 \qquad (\because I_{E2} \simeq I_{C2})$

$= 12\,\text{V} - 0.448\,\text{mA} \times 10\,\text{k}\Omega = \mathbf{7.52\,V}$

(*ii*) **Voltage gain.** To find voltage gain, we shall use the standard formula : total a.c. collector load divided by total ac emitter resistance.

First stage

$$r'_{e1} = \frac{25\,\text{mV}}{I_{E1}} = \frac{25\,\text{mV}}{0.31\,\text{mA}} = 80.6\,\Omega$$

Input impedance Z_{in} of the second stage is given by ;

$$Z_{in} = \beta\, r'_{e2}$$

Here, $r'_{e2} = \dfrac{25\,\text{mV}}{I_{E2}} = \dfrac{25\,\text{mV}}{0.448\,\text{mA}} = 55.8\,\Omega$

$\therefore \quad Z_{in} = \beta\, r'_{e2} = 125 \times (55.8\,\Omega) \simeq 7000\,\Omega = 7\,\text{k}\Omega$

Total a.c. collector load, $R_{AC} = R_3 \parallel Z_{in} = 22\,\text{k}\Omega \parallel 7\,\text{k}\Omega = 5.31\,\text{k}\Omega$

$\therefore$ Voltage gain, $A_{v1} = \dfrac{R_{AC}}{r'_{e1}} = \dfrac{5.31\,\text{k}\Omega}{80.6\,\Omega} = \mathbf{66}$

Second stage. There is no loading effect of any subsequent stage. Therefore, total a.c. collector load, $R_{AC} = R_5 = 10\,\text{k}\Omega$.

Voltage gain, $A_{v2} = \dfrac{R_5}{r'_{e2}} = \dfrac{10\,\text{k}\Omega}{55.8\,\Omega} = \mathbf{179}$

Overall voltage gain $= A_{v1} \times A_{v2} = 66 \times 179 = \mathbf{11{,}814}$

Example 9.31. *For the two-stage direct coupled amplifier shown in Fig. 9.30, find (i) the approximate I_C of the transistor Q_1 (ii) the potential of collector of Q_1 w.r.t. ground (iii) the approximate I_C of transistor Q_2 and (iv) the approximate overall voltage gain of the circuit. The transistors are of silicon.*

Solution. For a silicon transistor, $V_{BE} = 0.7$ V.

(*i*) Input voltage $= V_{BE}$ + Voltage across $680\,\Omega$

or $1.4 = 0.7 + I_E \times 680$

or $I_E = \dfrac{(1.4 - 0.7)\,\text{V}}{680\,\Omega} \simeq 1\,\text{mA}$

$\therefore \quad I_C \simeq I_E = \mathbf{1\,mA}$

(*ii*) Drop across $27\,\text{k}\Omega = 1\,\text{mA} \times 27\,\text{k}\Omega = 27\,\text{V}$

$\therefore$ Potential of collector of Q_1 *w.r.t.* ground $= 30 - 27 = \mathbf{3\,V}$

(*iii*) Now, the input to the transistor Q_2 is 3 V.

Now, $3 = V_{BE} + I_E \times 2400$

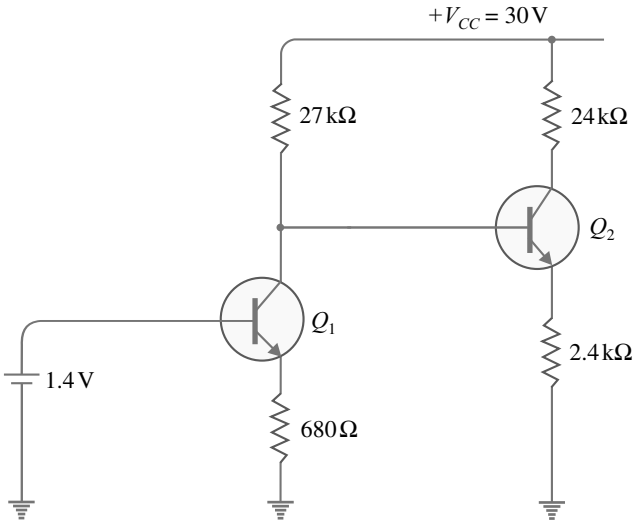


Fig. 9.30

or $$I_E = \frac{(3-0.7)\,\text{V}}{2400\ \Omega} = 1\ \text{mA}$$

$\therefore$ $$I_C \simeq I_E = \mathbf{1\ mA}$$

(*iv*) The approximate voltage gain of the two stages is given by ;

$$\text{First stage gain} = \frac{R_C}{R_E} = \frac{27\ \text{k}\Omega}{680\ \Omega} = 40$$

$$\text{Second stage gain} = \frac{R_C}{R_E} = \frac{24\ \text{k}\Omega}{2.4\ \text{k}\Omega} = 10$$

Approximate overall voltage gain = 40 × 10 = **400**

9.11 Comparison of Different Types of Coupling

S. No.	Particular	*RC* coupling	Transformer coupling	Direct coupling
1.	*Frequency response*	Excellent in the audio frequency range	Poor	Best
2.	*Cost*	Less	More	Least
3.	*Space and weight*	Less	More	Least
4.	*Impedance matching*	Not good	Excellent	Good
5.	*Use*	For voltage amplification	For power amplification	For amplifiying extremely low frequencies

Short Answer Questions

Q.1. *Why is RC coupling called so?*

Ans. Because in this arrangement, the voltage developed across the collector load (R_C) in one stage is coupled through a coupling capacitor (C_C) to the next stage.

Q.2. *Why is RC coupling very popular?*

Ans. This circuit is very cheap because it employs inexpensive resistors and capacitors. In addition, it provides constant voltage gain over audio-frequency range.

Q.3. *Why is RC coupling employed in the initial stages of a multistage amplifier and not in the final stages?*

Ans. The initial stages of a multistage amplifier are required to do the job of voltage amplification and this is beautifully done by *RC* coupling due to least frequency distortion. However, the final stage is expected to deliver maximum power to the output device *e.g.* speaker. For this purpose, impedance matching is required – a quality which *RC* coupled amplifier does not possess because its output impedance is several hundred ohms compared to a few ohms impedance of the speaker.

Q.4. *Why is transformer coupling not used in the initial stages of a multi-stage amplifier?*

Ans. A transformer coupling has a high frequency distortion *i.e.* output varies considerably with signal frequency. If it is employed in the initial stages, the output may have intolerable frequency distortion due to amplification of distortion by various stages.

Q.5. *Why is transformer coupling invariably employed in the final stage of a multistage amplifier?*

Ans. Transformer coupling provides excellent impedance matching. Usually, the impedance of output device (*e.g.* speaker) is a few ohms whereas the output impedance of a transistor (*CE* arrangement) is several hundred times this value. In order to match impedance, a step-down transformer of proper turn ratio (N_P/N_S) is used. Impedance matching permits the transfer of maximum power to the output device which is really the goal in the final stage.

Q.6. *Why is the total gain of a multistage amplifier less than the product of individual stages?*

Ans. Because of loading effect of next stages. For example, consider two amplifiers each having a stage gain of 80.When these are coupled, the total gain will be less than $80 \times 80 = 6400$. It is because the gain of first stage will be less than 80 due to the loading effect of input impedance of second stage. However, the gain of second stage will remain 80 because it has no loading effect of any subsequent stage.

Q.7. *Why should the coupling capacitors in transistor amplifiers possess high values?*

Ans. A transistor amplifier (*CE* arrangement) has low input impedance of the order of 500 Ω or so. It is desired that reactance offered by the coupling capacitor should not be more than 20 Ω. To achieve this, large values of coupling capacitances must be used. Generally, the capacitance of coupling capacitors ranges from 1 μF to 10 μF.

Q.8. *Why can more gain be obtained by using transformer coupling instead of RC coupling between the stages?*



Amplifiers with Negative Feedback

Learning Objectives

- Understand the importance of negative feedback in amplifiers.
- Derive an expression for voltage gain of negative voltage feedback amplifier.
- Explore the advantages of negative voltage feedback in amplifiers.
- Develop an understanding of negative current feedback in amplifiers.
- Understand the characteristics of emitter follower e.g. voltage gain, d.c. analysis, input impedance and output impedance.
- Explore the importance of Darlington Amplifier.

Introduction

A practical amplifier has a gain of nearly one million *i.e.* its output is one million times the input. Consequently, even a casual disturbance at the input will appear in the amplified form in the output. There is a strong tendency in amplifiers to introduce *hum* due to sudden temperature changes or stray electric and magnetic fields. Therefore, every high gain amplifier tends to give noise along with signal in its output. The noise in the output of an amplifier is undesirable and must be kept to as small a level as possible.

The noise level in amplifiers can be reduced considerably by the use of *negative feedback i.e.* by injecting a fraction of output in phase opposition to the input signal. The object of this chapter is to consider the effects and methods of providing negative feedback in transistor amplifiers.

11.1 Feedback

The process of injecting a fraction of output energy of some device back to the input is known as **feedback**.

The principle of feedback is probably as old as the invention of first machine but it is only some 75 years ago that feedback has come into use in connection with electronic circuits. It has been found very useful in reducing noise in amplifiers and making amplifier operation stable. Depending upon whether the feedback energy aids or opposes the input

signal, there are two basic types of feedback in amplifiers *viz.* *positive feedback* and *negative feedback*.

(*i*) Positive feedback. When the feedback energy (voltage or current) is in phase with the input signal and thus aids it, it is called *positive feedback*. This is illustrated in Fig. 11.1. Both amplifier and feedback network introduce a phase shift of 180°. The result is a 360° phase shift around the loop, causing the *feedback voltage* V_f to be in phase with the input signal V_{in}.

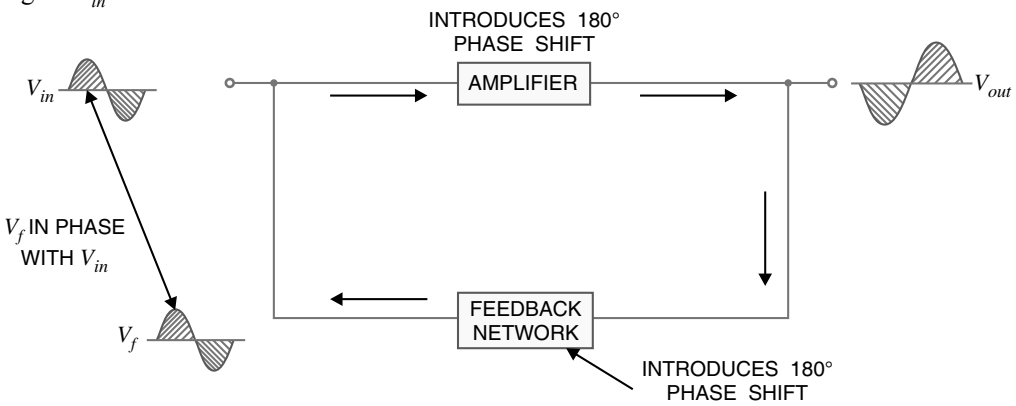


Fig. 11.1

The positive feedback increases the gain of the amplifier. However, it has the disadvantages of increased distortion and instability. Therefore, positive feedback is seldom employed in amplifiers. One important use of positive feedback is in oscillators. As we shall see in the next chapter, if positive feedback is sufficiently large, it leads to oscillations. As a matter of fact, an oscillator is a device that converts d.c. power into a.c. power of any desired frequency.

(*ii*) Negative feedback. When the feedback energy (voltage or current) is out of phase with the input signal and thus opposes it, it is called *negative feedback*. This is illustrated in Fig. 11.2. As you can see, the amplifier introduces a phase shift of 180° into the circuit while the feedback network is so designed that it introduces no phase shift (*i.e.* 0° phase shift). The result is that the *feedback voltage* V_f is 180° out of phase with the input signal V_{in}.

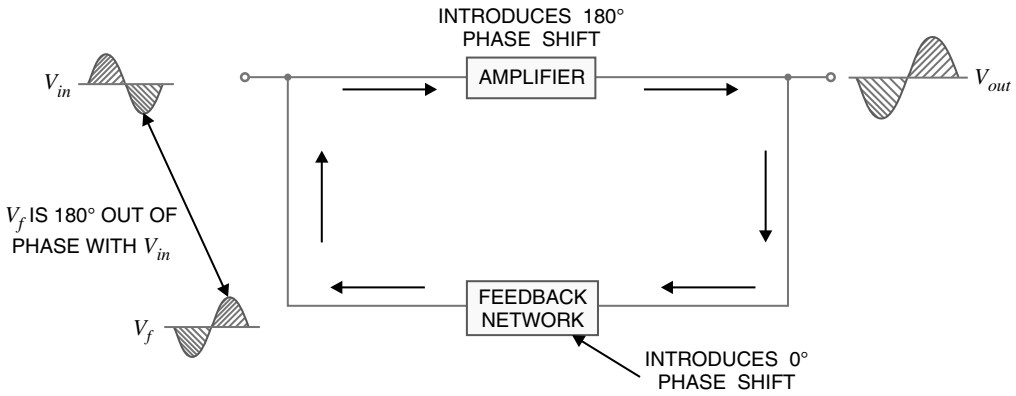


Fig. 11.2

Negative feedback reduces the gain of the amplifier. However, the advantages of negative feedback are : reduction in distortion, stability in gain, increased bandwidth and improved input and output impedances. It is due to these advantages that negative feedback is frequently employed in amplifiers.

11.2 Types of Negative Feedback in Amplifiers

There are two basic types of negative feedback that are used in amplifiers *viz.* (***i***) negative voltage feedback (***ii***) negative current feedback. In negative feedback (voltage or current), a fraction of the output signal is continuously fed back (called *feedback signal*) to the input side and is subtracted from the input signal to create a signal (= input signal–feedback signal) which is then applied to the input of the amplifier. It is clear that negative voltage feedback would decrease input voltage to the amplifier and hence its voltage gain will be reduced. Likewise, negative current feedback reduces the current gain of the amplifier because input current to the amplifier is decreased. However, at the cost of reduced gain (voltage or current), negative feedback in amplifiers offers such advantages as gain stability, modification (increase or decrease) of input and output impedances, reduced distortion, increased bandwidth *etc*. Thus negative feedback improves the performance of the amplifier. It may be mentioned here that loss of gain due to negative feedback can be compensated by adding extra amplifier stage/stages.

(*i*) Negative voltage feedback. In negative voltage feedback, a fraction of the output voltage (called *feedback voltage* v_f) of the amplifier is fed back to the signal voltage (v_s) in such a way that the difference of v_s and v_f (*i.e.* $v_s - v_f$) is applied to the input of the amplifier as shown in Fig. 11.3. The output voltage v_0 of the amplifier is applied to the feedback circuit which reduces v_0 to produce the feedback voltage v_f. The feedback circuit is simply a resistive voltage divider. The feedback voltage v_f is applied in series with the signal v_s. As v_f is *180° out of phase with v_s, the input voltage v_i to the amplifier is equal to the *difference* of the two *i.e.*

$$\text{Input voltage, } v_i = v_s - v_f$$

Thus negative voltage feedback has reduced the input voltage to the amplifier. Therefore, the output voltage is lower than that produced when negative feedback is not used. This means that voltage gain of the amplifier is reduced with negative voltage feedback.

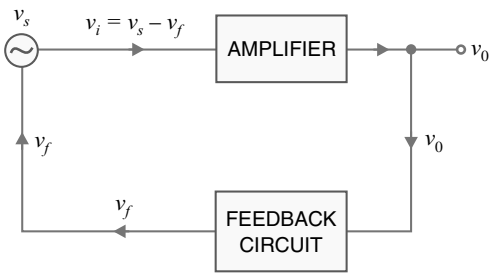


Fig. 11.3

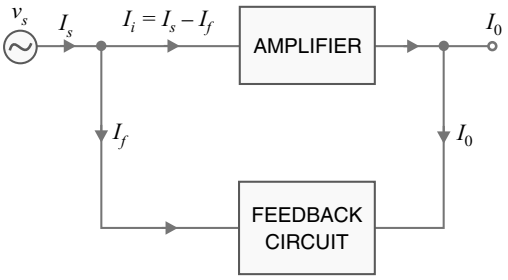


Fig. 11.4

* v_0 is 180° out of phase with v_s. Since feedback circuit is resistive, it produces zero degree phase shift. Therefore, v_f is 180° out of phase with v_s.

(*ii*) Negative current feedback. In negative current feedback, a fraction of the output current (called *feedback current* I_f) of the amplifier is fed back to the signal current (I_s) in such a manner that the difference of I_s and I_f (*i.e.* $I_s - I_f$) is fed to the input of the amplifier as shown in Fig. 11.4. The fact that the arrowhead of the feedback current (I_f) being fed forward is because it is *negative feedback*. It is clear that input current I_i to the amplifier is equal to the *difference* of the two *i.e.*

$$\text{Input current, } I_i = I_s - I_f$$

Again, the negative current feedback has reduced the input current to the amplifier. Therefore, the output current is lower than that produced when negative feedback is not used. This means that current gain of the amplifier is reduced with negative current feedback.

Discussion. The following points may be noted about negative feedback :

(*i*) *The negative voltage feedback decreases the voltage gain* (A_v) *and output impedance* (Z_{out}) *of the amplifier*. It does not affect the current gain of the amplifier. However, negative voltage feedback will either increase or decrease the input impedance (Z_{in}) of the amplifier, depending on the circuit arrangement.

(*ii*) *The negative current feedback decreases current gain* (A_i) *and increases output impedance* (Z_{out}) *of the amplifier*. It does not affect the voltage gain of the amplifier. However, negative current feedback will either increase or decrease the input impedance (Z_{in}) of the amplifier, depending on the circuit arrangement.

(*iii*) Since power gain $A_p = A_v A_i$, both types of negative feedback will decrease the power gain of the amplifier. However, *the drawback of reduced power gain is offset by the advantage of increased bandwidth.*

11.3 Principles of Negative Voltage Feedback in Amplifiers

A feedback amplifier has two parts *viz.* an amplifier and a feedback circuit. The feedback circuit usually consists of resistors and returns a fraction of output energy back to the input. Fig. 11.5 *shows the principles of negative voltage feedback in an amplifier. Typical values

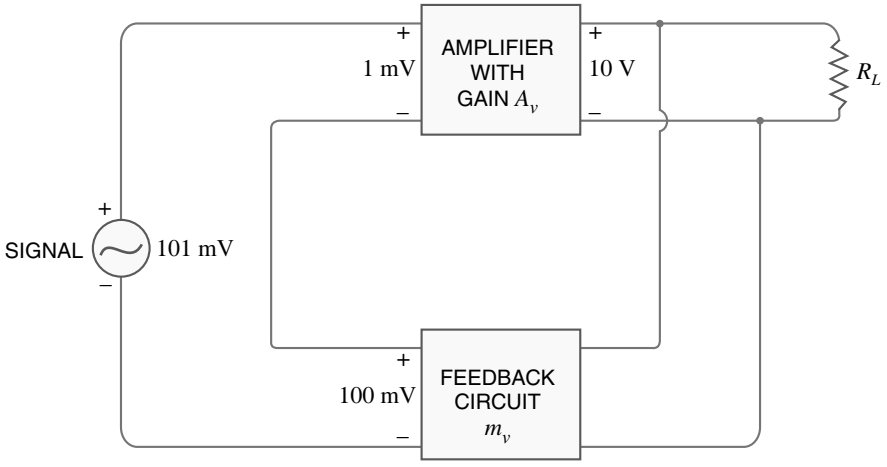


Fig. 11.5

* Note that amplifier and feedback circuits are connected in *series-parallel*. The inputs of amplifier and feedback circuits are in *series* but the outputs are in *parallel*. In practice, this circuit is widely used.

have been assumed to make the treatment more illustrative. The output of the amplifier is 10 V. The fraction m_v of this output *i.e.* 100 mV is fed back to the input where it is applied in series with the input signal of 101 mV. As the feedback is negative, therefore, only 1 mV appears at the input terminals of the amplifier.

Referring to Fig. 11.5, we have,

$$\text{Gain of amplifier without feedback, } A_v = \frac{10 \text{ V}}{1 \text{ mV}} = 10{,}000$$

$$\text{Fraction of output voltage fed back, } m_v = \frac{100 \text{ mV}}{10 \text{ V}} = 0.01$$

$$\text{Gain of amplifier with negative feedback, } A_{vf} = \frac{10 \text{ V}}{101 \text{ mV}} = 100$$

The following points are worth noting :

(*i*) When negative voltage feedback is applied, the gain of the amplifier is *reduced. Thus, the gain of above amplifier without feedback is 10,000 whereas with negative feedback, it is only 100.

(*ii*) When negative voltage feedback is employed, the voltage *actually* applied to the amplifier is extremely small. In this case, the signal voltage is 101 mV and the negative feedback is 100 mV so that voltage applied at the input of the amplifier is only 1 mV.

(*iii*) In a negative voltage feedback circuit, the feedback fraction m_v is always between 0 and 1.

(*iv*) The gain with feedback is sometimes called *closed-loop gain* while the gain without feedback is called *open-loop gain*. These terms come from the fact that amplifier and feedback circuits form a "loop". When the loop is "opened" by disconnecting the feedback circuit from the input, the amplifier's gain is A_v, the "open-loop" gain. When the loop is "closed" by connecting the feedback circuit, the gain decreases to A_{vf}, the "closed-loop" gain.

11.4 Gain of Negative Voltage Feedback Amplifier

Consider the negative voltage feedback amplifier shown in Fig. 11.6. The gain of the amplifier without feedback is A_v. Negative feedback is then applied by feeding a fraction m_v of the output voltage v_0 back to amplifier input. Therefore, the actual input to the amplifier is the signal voltage v_s *minus* feedback voltage $m_v v_0$ *i.e.*

$$\text{Actual input to amplifier, } v_i = v_s - m_v v_0$$

The output voltage v_0 must be equal to the input voltage $v_s - m_v v_0$ multiplied by gain A_v of the amplifier *i.e.*

$$(v_s - m_v v_0) A_v = v_0$$

or $$A_v v_s - A_v m_v v_0 = v_0$$

or $$v_0 (1 + A_v m_v) = A_v v_s$$

or $$\frac{v_0}{v_s} = \frac{A_v}{1 + A_v m_v}$$

But v_0/v_s is the voltage gain of the amplifier with negative voltage feedback.

$\therefore$ Voltage gain with negative voltage feedback is

---

* Since with negative voltage feedback the voltage gain is decreased and current gain remains unaffected, the power gain A_p ($= A_v \times A_i$) will decrease. However, the drawback of reduced power gain is offset by the advantage of increased bandwidth.

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

It may be seen that the gain of the amplifier without feedback is A_v. However, when negative voltage feedback is applied, the gain is reduced by a factor $1 + A_v\, m_v$. It may be noted that negative voltage feedback does not affect the current gain of the circuit.

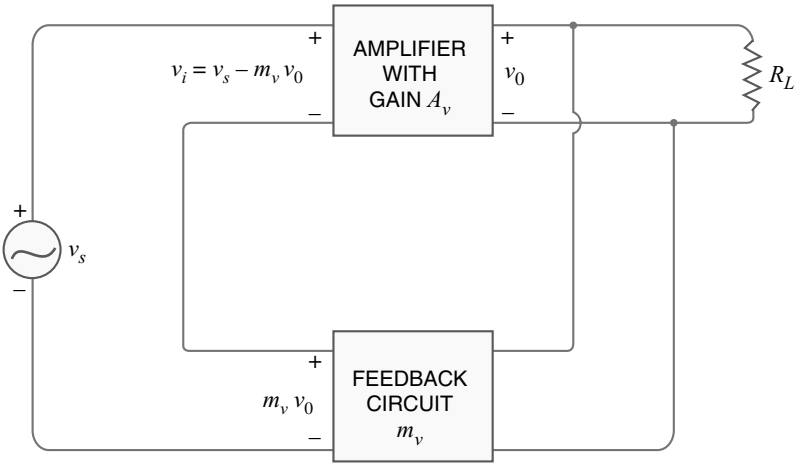


Fig. 11.6

Discussion. The following points are worth noting :

(*i*) The various relations in negative voltage feedback amplifier are :

$$\text{Open-loop voltage gain, } A_v = \frac{v_0}{v_i}$$

$$\text{Feedback fraction, } m_v = \frac{v_f}{v_0}$$

$$\text{Closed-loop voltage gain, } A_{vf} = \frac{A_v}{1 + A_v m_v}$$

(*ii*) The quantity $(1 + A_v m_v)$ is called **feedback factor** of the negative voltage feedback amplifier. When negative voltage feedback is applied to an amplifier the voltage gain, input/output impedance and frequency response are controlled by this factor.

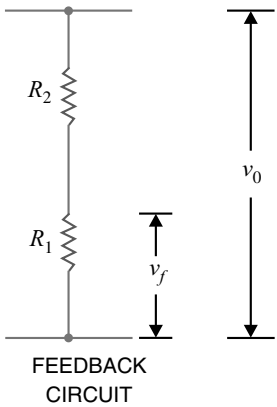


Fig. 11.7

(*iii*) The feedback circuit provides a path from the amplifier output back to the input. The purpose of this path is to "feed" a portion of the output signal back to the amplifier input. The feedback circuit may be as simple as the resistive voltage divider as shown in Fig. 11.7. The output voltage v_0 of the amplifier is applied to the input of the feedback circuit which reduces v_0 by a fraction m_v to produce a feedback voltage v_f $(= m_v v_0)$.

Clearly, $$m_v = \frac{v_f}{v_0}$$

Since feedback circuit is resistive, it produces no phase shift in the output voltage.

Example 11.1. *The voltage gain of an amplifier without feedback is 3000. Calculate the voltage gain of the amplifier if negative voltage feedback is introduced in the circuit. Given that feedback fraction $m_v = 0.01$.*

Solution. $A_v = 3000 \; ; \quad m_v = 0.01$

$\therefore$ Voltage gain with negative voltage feedback is

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v} = \frac{3000}{1 + 3000 \times 0.01} = \frac{3000}{31} = \mathbf{97}$$

Example 11.2. *The overall gain of a multistage amplifier is 140. When negative voltage feedback is applied, the gain is reduced to 17.5. Find the fraction of the output that is fed back to the input.*

Solution. $A_v = 140 \; ; \quad A_{vf} = 17.5$

Let m_v be the feedback fraction. Voltage gain with negative voltage feedback is

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

or $$17.5 = \frac{140}{1 + 140\, m_v}$$

or $$17.5 + 2450\, m_v = 140$$

$\therefore$ $$m_v = \frac{140 - 17.5}{2450} = \mathbf{\frac{1}{20}}$$

Example 11.3. *When negative voltage feedback is applied to an amplifier of gain 100, the overall gain falls to 50.*

(*i*) *Calculate the fraction of the output voltage fed back.*

(*ii*) *If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain is to be 75.*

Solution.

(***i***) Gain without feedback, $A_v = 100$

Gain with feedback, $A_{vf} = 50$

Let m_v be the fraction of the output voltage fed back.

Now, $$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

or $$50 = \frac{100}{1 + 100\, m_v}$$

or $$50 + 5000\, m_v = 100$$

or $$m_v = \frac{100 - 50}{5000} = \mathbf{0.01}$$

(***ii***) $A_{vf} = 75 \; ; \quad m_v = 0.01 \; ; \quad A_v = ?$

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

or $$75 = \frac{A_v}{1 + 0.01\, A_v}$$

or $75 + 0.75\ A_v = A_v$

$\therefore \quad A_v = \dfrac{75}{1 - 0.75} = \mathbf{300}$

Example 11.4. *With a negative voltage feedback, an amplifier gives an output of 10 V with an input of 0.5 V. When feedback is removed, it requires 0.25 V input for the same output. Calculate (i) gain without feedback (ii) feedback fraction m_v.*

Solution.

(*i*) Gain without feedback, $A_v = 10/0.25 = \mathbf{40}$

(*ii*) Gain with feedback, $A_{vf} = 10/0.5 = 20$

Now, $A_{vf} = \dfrac{A_v}{1 + A_v\ m_v}$

or $20 = \dfrac{40}{1 + 40\ m_v}$

or $20 + 800\ m_v = 40$

or $m_v = \dfrac{40 - 20}{800} = \mathbf{\dfrac{1}{40}}$

Example 11.5. *The gain of an amplifier without feedback is 50 whereas with negative voltage feedback, it falls to 25. If due to ageing, the amplifier gain falls to 40, find the percentage reduction in stage gain (i) without feedback and (ii) with negative feedback.*

Solution. $A_{vf} = \dfrac{A_v}{1 + A_v m_v}$

or $25 = \dfrac{50}{1 + 50\ m_v}$

or $m_v = 1/50$

(*i*) Without feedback. The gain of the amplifier without feedback is 50. However, due to ageing, it falls to 40.

$\therefore$ % age reduction in stage gain $= \dfrac{50 - 40}{50} \times 100 = \mathbf{20\%}$

(*ii*) With negative feedback. When the gain without feedback was 50, the gain with negative feedback was 25. Now the gain without feedback falls to 40.

$\therefore$ New gain with negative feedback $= \dfrac{A_v}{1 + A_v\ m_v} = \dfrac{40}{1 + (40 \times 1/50)} = 22.2$

$\therefore$ % age reduction in stage gain $= \dfrac{25 - 22.2}{25} \times 100 = \mathbf{11.2\%}$

Example 11.6. *An amplifier has a voltage amplification A_v and a fraction m_v of its output is fed back in opposition to the input. If $m_v = 0.1$ and $A_v = 100$, calculate the percentage change in the gain of the system if A_v falls 6 dB due to ageing.*

Solution. $A_v = 100\ ; \quad m_v = 0.1\ ; \quad A_{vf} = ?$

$A_{vf} = \dfrac{A_v}{1 + A_v\ m_v} = \dfrac{100}{1 + 100 \times 0.1} = 9.09$

Fall in gain = 6 dB

Let A_{v1} be the new absolute voltage gain without feedback.

Then, $20 \log_{10} A_v/A_{v1} = 6$

or $\log_{10} A_v/A_{v1} = 6/20 = 0.3$

or $\frac{A_v}{A_{v1}} = \text{Antilog } 0.3 = 2$

or $A_{v1} = A_v/2 = 100/2 = 50$

$\therefore$ New $A_{vf} = \frac{A_{v1}}{1 + A_{v1}\, m_v} = \frac{50}{1 + 50 \times 0.1} = 8.33$

% age change in system gain $= \frac{9.09 - 8.33}{9.09} \times 100 =$ **8.36%**

Example 11.7. *An amplifier has a voltage gain of 500 without feedback. If a negative feedback is applied, the gain is reduced to 100. Calculate the fraction of the output fed back. If, due to ageing of components, the gain without feedback falls by 20%, calculate the percentage fall in gain with feedback.*

Solution. $A_v = 500$; $A_{vf} = 100$; $m_v = ?$

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

or $100 = \frac{500}{1 + 500\, m_v}$

$\therefore$ $m_v =$ **0.008**

Now, $A_v = \frac{80}{100} \times 500 = 400$; $m_v = 0.008$; $A_{vf} = ?$

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v} = \frac{400}{1 + 400 \times 0.008} = \frac{400}{4.2} = 95.3$$

$\therefore$ % age fall in $A_{vf} = \frac{100 - 95.3}{100} \times 100 =$ **4.7%**

Note that without negative feedback, the change in gain is 20%. However, when negative feedback is applied, the change in gain (4.7%) is much less. This shows that negative feedback provides voltage gain stability.

Example 11.8. *An amplifier has an open-loop gain A_v = 100,000. A negative feedback of 10 dB is applied. Find (i) voltage gain with feedback (ii) value of feedback fraction m_v.*

Solution.

(*i*) dB voltage gain without feedback

$= 20 \log_{10} 100{,}000 = 20 \log_{10} 10^5 = 100$ dB

Voltage gain with feedback = 100 – 10 = 90 dB

Now, $20 \log_{10} (A_{vf}) = 90$

or $\log_{10} (A_{vf}) = 90/20 = 4.5$

$\therefore \quad A_{vf} = \text{Antilog } 4.5 = \mathbf{31622}$

(*ii*) $\quad A_{vf} = \dfrac{A_v}{1 + A_v\, m_v}$

or $\quad 31622 = \dfrac{100{,}000}{1 + 100{,}000 \times m_v}$

$\therefore \quad m_v = \mathbf{2.17 \times 10^{-5}}$

Example 11.9. *An amplifier with an open-circuit voltage gain of 1000 has an output resistance of 100 Ω and feeds a resistive load of 900 Ω. Negative voltage feedback is provided by connecting a resistive voltage divider across the output and one-fiftieth of the output voltage is fed back in series with the input signal. Determine the voltage gain with negative feedback.*

Solution. Fig. 11.8 shows the equivalent circuit of an amplifier along with the feedback circuit.

Voltage gain of the amplifier without feedback is

$$A_v = \frac{A_{v0}\, R_L}{R_{out} + R_L} \qquad \text{...See Art. 8.24}$$

$$= \frac{1000 \times 900}{100 + 900} = 900$$

$\therefore \quad A_{vf} = \dfrac{A_v}{1 + A_v\, m_v} = \dfrac{900}{1 + 900 \times (1/50)} = \mathbf{47.4}$

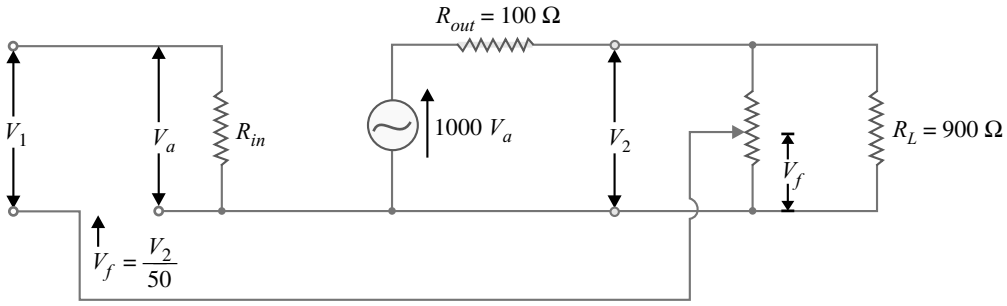


Fig. 11.8

Example 11.10. *An amplifier is required with a voltage gain of 100 which does not vary by more than 1%. If it is to use negative feedback with a basic amplifier the voltage gain of which can vary by 20%, determine the minimum voltage gain required and the feedback factor.*

Solution.

$$100 = \frac{A_v}{1 + A_v\, m_v}$$

or $\quad 100 + 100\, A_v\, m_v = A_v \qquad \ldots (i)$

Also, $\quad 99 = \dfrac{0.8\, A_v}{1 + 0.8\, A_v\, m_v}$

or $\quad 99 + 79.2\, A_v\, m_v = 0.8\, A_v \qquad \ldots (ii)$

Multiplying eq. (*i*) by 0.792, we have,

$$79.2 + 79.2\ A_v\ m_v = 0.792\ A_v \qquad \ldots (iii)$$

Subtracting [(*ii*) − (*iii*)], we have,

$$19.8 = 0.008\ A_v \quad \therefore \quad A_v = \frac{19.8}{0.008} = \mathbf{2475}$$

Putting the value of A_v (= 2475) in eq. (*i*), we have,

$$100 + 100 \times 2475 \times m_v = 2475$$

$$\therefore \quad m_v = \frac{2475 - 100}{100 \times 2475} = \mathbf{0.0096}$$

11.5 Feedback Circuit

The function of the feedback circuit is to return a fraction of the output voltage to the input of the amplifier. Fig. 11.9 shows the feedback circuit of negative voltage feedback amplifier. It is essentially a potential divider consisting of resistances R_1 and R_2. The output voltage of the amplifier is fed to this potential divider which gives the feedback voltage to the input of the amplifier.

Referring to Fig. 11.9, it is clear that :

$$\text{Voltage across } R_1,\ v_f = \left(\frac{R_1}{R_1 + R_2}\right) v_0$$

$$\text{Feedback fraction, } m_v = \frac{v_f}{v_0} = \frac{R_1}{R_1 + R_2}$$

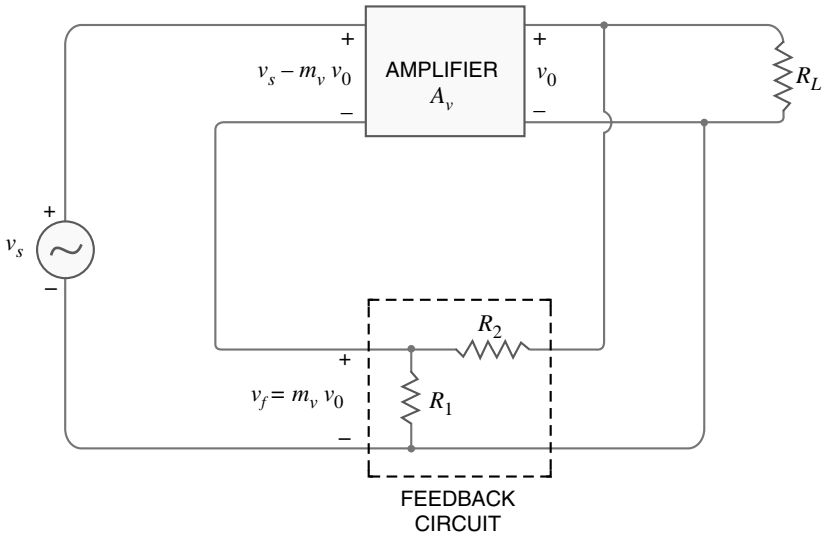


Fig. 11.9

Approximate voltage gain. It has been shown in Art. 11.4 that the voltage gain (A_{vf}) of the negative voltage feedback amplifier is given by ;

$$A_{vf} = \frac{A_v}{1 + A_v\ m_v}$$

If $A_v m_v >> 1$ (as is generally the case), then 1 can be neglected as compared to $A_v m_v$.

$$\therefore \quad A_{vf} = \frac{A_v}{A_v m_v} = \frac{1}{m_v} \quad \text{... } A_v m_v >> 1$$

Now, $$m_v = \frac{R_1}{R_1 + R_2}$$

$$\therefore \quad A_{vf} = \frac{1}{m_v} = \frac{R_1 + R_2}{R_1} \quad \text{... } (i)$$

Eq. (*i*) gives the expression for the voltage gain of negative voltage feedback amplifier provided $A_v m_v >> 1$.

(*i*) While designing a negative voltage feedback amplifier, we ensure that $A_v m_v >> 1$ so that :

$$A_{vf} = \frac{1}{m_v} = \frac{R_1 + R_2}{R_1}$$

This is a very interesting result. It shows that *the voltage gain of a negative voltage feedback amplifier is almost entirely determined by the feedback circuit*. Since feedback circuit is a resistive network, the voltage gain with negative voltage feedback (A_{vf}) is unaffected by changes in temperature, variations in transistor parameters and signal frequency. Hence, voltage gain with negative feedback is extremely stable.

(*ii*) The condition $A_v m_v >> 1$ means that $A_v >> 1/m_v$ *i.e.*

$$A_v >> A_{vf} \qquad (\because 1/m_v = A_{vf})$$

Therefore, *for satisfactory operation of negative voltage feedback amplifier, the open-loop voltage gain (A_v) must be much greater than the required closed-loop voltage gain (A_{vf})*. In general, for stable A_{vf}, the A_v should be equal to or greater than 100 A_{vf} (*i.e.* $A_v \geq 100\ A_{vf}$). For example, if it is desired that $A_{vf} = 100$, then the values of R_1 and R_2 of the feedback circuit should be so adjusted that $m_v = 1/A_{vf} = 1/100 = 0.01$.

Example 11.11. *For the negative voltage feedback amplifier shown in Fig. 11.10, the open-loop voltage gain is 100,000 and feedback fraction = 1/100. Determine the closed-loop voltage gain of the amplifier. What will be the values of the closed-loop gain when the open-loop gain is changed by ± 50% ?*

Solution. Here, $A_v = 100{,}000$; $m_v = 1/100 = 0.01$

$$\therefore \quad \text{Closed-loop voltage gain, } A_{vf} = \frac{A_v}{1 + A_v m_v} = \frac{100{,}000}{1 + 100{,}000 \times 0.01} = \mathbf{99.9}$$

When A_v = 150,000. This is the case when A_v (open-loop voltage gain) increases by 50%.

$$\therefore \quad A_{vf} = \frac{A_v}{1 + A_v m_v} = \frac{150{,}000}{1 + 150{,}000 \times 0.01} = \mathbf{99.93}$$

When A_v = 50,000. This is the case when A_v decreases by 50%.

$$\therefore \quad A_{vf} = \frac{A_v}{1 + A_v m_v} = \frac{50{,}000}{1 + 50{,}000 \times 0.01} = \mathbf{99.8}$$

This example shows that when the open-loop voltage gain (A_v) changes by ±50%, the closed-loop voltage gain (A_{vf}) of the negative voltage feedback amplifier changes within ±0.1%.Therefore, the voltage gain of the amplifier with negative voltage feedback is extremely stable.

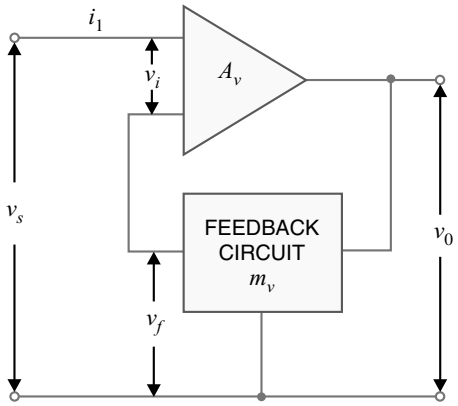


Fig. 11.10

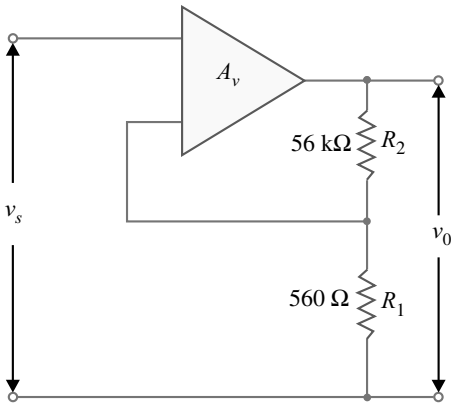


Fig. 11.11

Example 11.12. *For the negative voltage feedback amplifier shown in Fig. 11.11, the open-loop voltage gain is 100,000. Calculate (i) feedback fraction and (ii) closed-loop voltage gain.*

Solution. Here, $A_v = 100{,}000$; $R_1 = 560\,\Omega$; $R_2 = 56\,\text{k}\Omega$

(*i*) Feedback fraction, $m_v = \dfrac{R_1}{R_1 + R_2} = \dfrac{560\ \Omega}{560\ \Omega + 56\ \text{k}\Omega} = \dfrac{1}{101}$

(*ii*) Closed-loop voltage gain of the amplifier with negative feedback is

$$A_{vf} = \frac{A_v}{1 + A_v m_v} = \frac{100,000}{1 + 100,000 \times (1/101)} = 100.9$$

Also, $$A_{vf} \simeq \frac{1}{m_v} = \frac{1}{1/101} = 101$$

Note that error is about 0.1%.

11.6 Advantages of Negative Voltage Feedback

The following are the advantages of negative voltage feedback in amplifiers :

(*i*) Gain stability. An important advantage of negative voltage feedback is that the resultant gain of the amplifier can be made independent of transistor parameters or the supply voltage variations.

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

For negative voltage feedback in an amplifier to be effective, the designer deliberately makes the product $A_v\, m_v$ much greater than unity. Therefore, in the above relation, 1 can be neglected as compared to $A_v\, m_v$ and the expression becomes :

$$A_{vf} = \frac{A_v}{A_v\, m_v} = \frac{1}{m_v}$$

It may be seen that the gain now depends only upon feedback fraction m_v *i.e.* on the characteristics of feedback circuit. As feedback circuit is usually a voltage divider (a resistive network), therefore, it is unaffected by changes in temperature, variations in transistor parameters and frequency. Hence, the gain of the amplifier is extremely stable.

(*ii*) Reduces non-linear distortion. A large signal stage has non-linear distortion because its voltage gain changes at various points in the cycle. The negative voltage feedback reduces the non-linear distortion in large signal amplifiers. It can be proved mathematically that :

$$D_{vf} = \frac{D}{1 + A_v\, m_v}$$

where D = Distortion in amplifier without feedback

D_{vf} = Distortion in amplifier with negative feedback

It is clear that by applying negative voltage feedback to an amplifier, distortion is reduced by a factor $1 + A_v\, m_v$.

(*iii*) Improves frequency response. As feedback is usually obtained through a resistive network, therefore, voltage gain of the amplifier is *independent of signal frequency. The result is that voltage gain of the amplifier will be substantially constant over a wide range of signal frequencies. The negative voltage feedback, therefore, improves the frequency response of the amplifier.

Fig. 11.12 shows the effect of negative voltage feedback on the bandwidth of the amplifier. Here, BW is the bandwidth of the amplifier without feedback and $BW_{(f)}$ is the amplifier bandwidth with negative voltage feedback. Note that with negative voltage feedback, the bandwidth ($BW_{(f)}$) is increased, the lower cut-off frequency [$f_{1(f)}$] is decreased and upper cut-off frequency [$f_{2(f)}$] is increased.

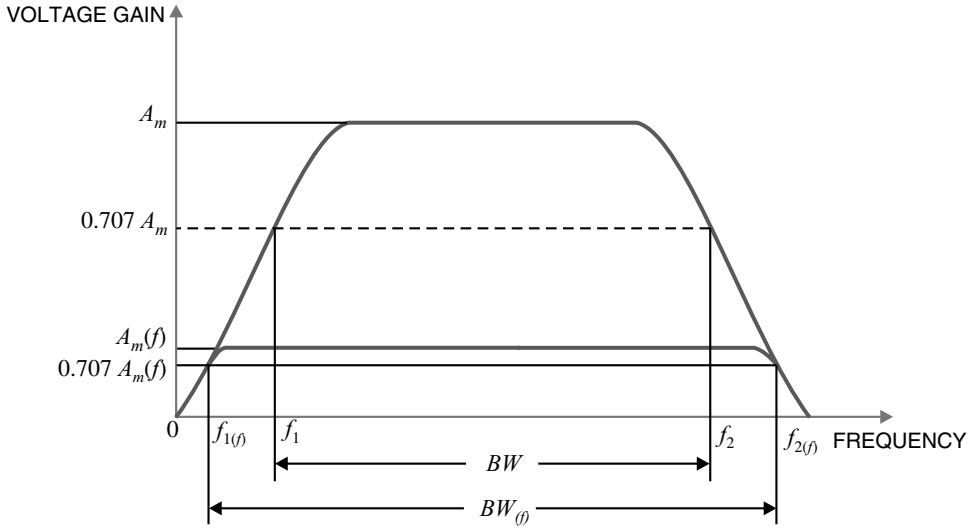


Fig. 11.12

It can be proved that with negative voltage feedback, we have,

* $A_{vf} = 1/m_v$. Now m_v depends upon feedback circuit. As feedback circuit consists of resistive network, therefore, value of m_v is unaffected by change in signal frequency.

$$\text{Lower cut-off frequency, } f_{1(f)} = \frac{f_1}{1 + A_v m_v}$$

$$\text{Upper cut-off frequency, } f_{2(f)} = f_2(1 + A_v m_v)$$

$$\text{Bandwidth, } BW_{(f)} = BW\ (1 + A_v m_v)$$

Here, f_1 and f_2 are the lower and upper cut-off frequencies respectively without feedback. Note that with negative voltage feedback, the lower cut-off frequency is *decreased* by $(1 + A_v m_v)$ while upper cut-off frequency is *increased* by the same factor. As a result, the bandwidth of the amplifier is *increased* by $(1 + A_v m_v)$.

(*iv*) Increases circuit stability. The output of an ordinary amplifier is easily changed due to variations in ambient temperature, frequency and signal amplitude. This changes the gain of the amplifier, resulting in distortion. However, by applying negative voltage feedback, voltage gain of the amplifier is stabilised or accurately fixed in value. This can be easily explained. Suppose the output of a negative voltage feedback amplifier has increased because of temperature change or due to some other reason. This means more negative feedback since feedback is being given from the output. This tends to oppose the increase in amplification and maintains it stable. The same is true should the output voltage decrease. Consequently, the circuit stability is considerably increased.

(*v*) Increases input impedance and decreases output impedance. The negative voltage feedback increases the input impedance and decreases the output impedance of amplifier. Such a change is profitable in practice as the amplifier can then serve the purpose of impedance matching.

(*a*) Input impedance. The increase in input impedance with negative voltage feedback can be explained by referring to Fig. 11.13. Suppose the input impedance of the amplifier is Z_{in} without feedback and $Z_{in(f)}$ with negative feedback. Let us further assume that input current is i_1.

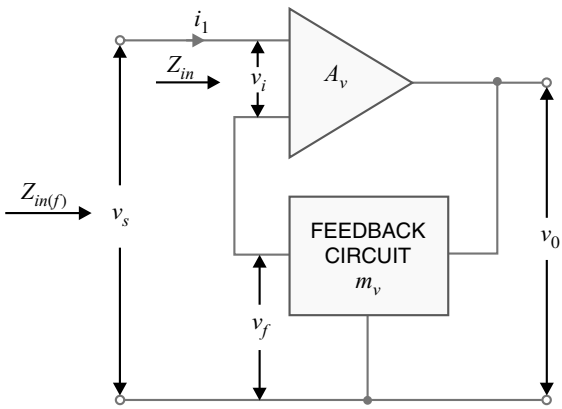


Fig. 11.13

The input impedance of a transistor amplifier without negative feedback is the impedance *looking into* the base of the transistor. If no negative feedback is present in the amplifier shown in Fig. 11.13, the input impedance Z_{in} is

$$Z_{in} = \frac{v_i}{i_1}$$

$$\therefore \qquad i_1 = \frac{v_i}{Z_{in}}$$

With negative voltage feedback, the input impedance $Z_{in(f)}$ of the amplifier is

$$Z_{in(f)} = \frac{v_s}{i_1} = \frac{v_s}{v_i} \times Z_{in} \qquad \text{... } (i) \qquad \left(\because \; i_1 = \frac{v_i}{Z_{in}} \right)$$

Now, $v_i = v_s - v_f = v_s - m_v\, v_0 \qquad (\because\; v_f = m_v\, v_0)$

or $v_i = v_s - m_v\, A_v\, v_i \qquad (\because\; v_0 = A_v\, v_i)$

or $v_i(1 + A_v\, m_v) = v_s$

or $\frac{v_s}{v_i} = (1 + A_v\, m_v)$

From eq. (*i*), $Z_{in(f)} = Z_{in}\,(1 + A_v\, m_v)$

It is clear that by applying negative voltage feedback, the input impedance of the amplifier is increased by a factor $(1 + A_v m_v)$. As $A_v m_v$ is much greater than unity, therefore, input impedance is increased considerably. This is an advantage, since the amplifier will now present less of a load to its source circuit.

(*b*) Output impedance. Recall that output impedance of an amplifier is the Thevenin impedance at its output terminals. Therefore, in order to find the output impedance $[Z_{out(f)}]$ of a negative voltage feedback amplifier, short-circuit the input signal source and apply voltage v_0 at the output terminals as shown in Fig. 11.14. Since the input terminals are shorted, the feedback voltage v_f is the only voltage at the amplifier input. Referring to Fig. 11.14, the voltage v_0 produces the feedback voltage v_f and this generates a voltage $A_v v_f$ in series with Z_{out}. Here, Z_{out} is the output impedance of the amplifier without feedback. Assuming an instantaneous positive polarity for v_0, the output current i_0 flows in the direction shown. Applying Kirchhoff's voltage law to the output circuit, we have,

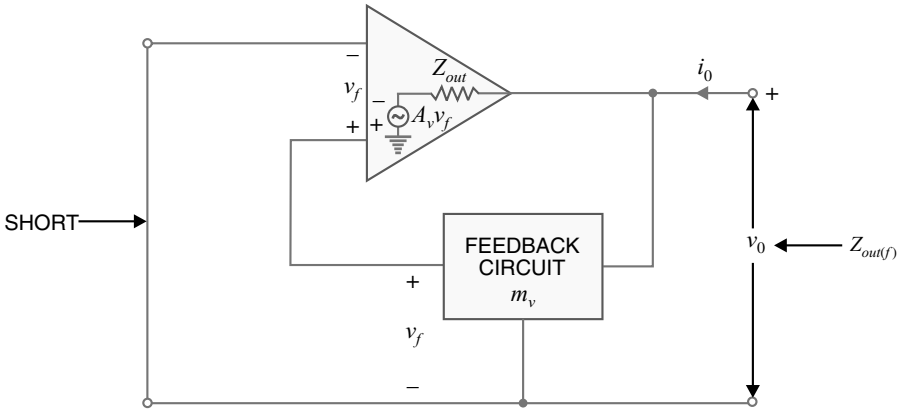


Fig. 11.14

$$v_0 = i_0 Z_{out} - A_v v_f$$

or $i_0 Z_{out} = v_0 + A_v v_f$

or $i_0 Z_{out} = v_0 + A_v\, m_v\, v_0 \qquad (\because\; v_f = m_v v_0)$

or $$i_0 Z_{out} = v_0(1 + A_v\, m_v) \quad \text{... } (i)$$

With negative voltage feedback, the output impedance of the amplifier is

$$Z_{out(f)} = \frac{v_0}{i_0} = \frac{Z_{out}}{1 + A_v m_v} \quad \text{... From eq. } (i)$$

$$\therefore \quad Z_{out(f)} = \frac{Z_{out}}{1 + A_v m_v}$$

It is clear that by applying negative feedback, the output impedance of the amplifier is decreased by a factor $(1 + A_v\, m_v)$. This is an added benefit of using negative voltage feedback. With lower value of output impedance, the amplifier is much better suited to drive low impedance loads.

Example 11.13. *Fig. 11.15 shows the negative voltage feedback amplifier. If the gain of the amplifier without feedback is 10,000, find :*

(i) feedback fraction (ii) overall voltage gain (iii) output voltage if input voltage is 1 mV.

Solution. $A_v = 10{,}000\ ;\quad R_1 = 2\,\text{k}\Omega\ ;\quad R_2 = 18\,\text{k}\Omega$

(*i*) Feedback fraction, $m_v = \dfrac{R_1}{R_1 + R_2} = \dfrac{2}{2 + 18} = \mathbf{0.1}$

(*ii*) Voltage gain with negative feedback is

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v} = \frac{10{,}000}{1 + 10{,}000 \times 0.1} = \mathbf{10}$$

(*iii*) Output voltage = A_{vf} × Input voltage

= 10 × 1 mV = **10 mV**

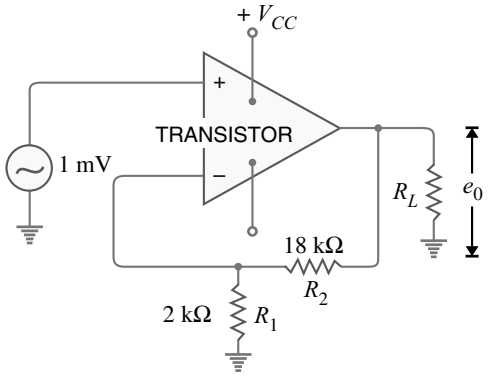


Fig. 11.15

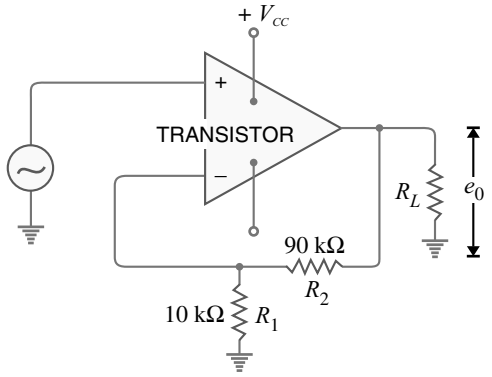


Fig. 11.16

Example 11.14. *Fig. 11.16 shows the circuit of a negative voltage feedback amplifier. If without feedback, A_v = 10,000, Z_{in} = 10 kΩ, Z_{out} = 100 Ω, find :*

(i) feedback fraction *(ii) gain with feedback*

(iii) input impedance with feedback *(iv) output impedance with feedback.*

Solution.

(*i*) Feedback fraction, $m_v = \dfrac{R_1}{R_1 + R_2} = \dfrac{10}{10 + 90} = \mathbf{0.1}$

(*ii*) Gain with negative feedback is

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v} = \frac{10{,}000}{1 + 10{,}000 \times 0.1} = \mathbf{10}$$

(*iii*) With negative voltage feedback, input impedance is increased and is given by ;

$$\begin{aligned} Z_{in(f)} &= (1 + A_v\, m_v)\, Z_{in} \\ &= (1 + 10{,}000 \times 0.1)\ 10\,\text{k}\Omega \\ &= 1001 \times 10\,\text{k}\Omega \\ &= \mathbf{10\ M\Omega} \end{aligned}$$

(*iv*) With negative voltage feedback, output impedance is decreased and is given by ;

$$Z_{out(f)} = \frac{Z_{out}}{1 + A_v\, m_v} = \frac{100\ \Omega}{1 + 10{,}000 \times 0.1} = \frac{100\ \Omega}{1001} = \mathbf{0.1\ \Omega}$$

Example 11.15. *The gain and distortion of an amplifier are 150 and 5% respectively without feedback. If the stage has 10% of its output voltage applied as negative feedback, find the distortion of the amplifier with feedback.*

Solution.

Gain without feedback, $A_v = 150$

Distortion without feedback, $D = 5\% = 0.05$

Feedback fraction, $m_v = 10\% = 0.1$

If D_{vf} is the distortion with negative voltage feedback, then,

$$D_{vf} = \frac{D}{1 + A_v\, m_v} = \frac{0.05}{1 + 150 \times 0.1} = 0.00313 = \mathbf{0.313\%}$$

It may be seen that by the application of negative voltage feedback, the amplifier distortion is reduced from 5% to 0.313%.

Example 11.16. *An amplifier has a gain of 1000 without feedback and cut-off frequencies are f_1 = 1.5 kHz and f_2 = 501.5 kHz. If 1% of output voltage of the amplifier is applied as negative feedback, what are the new cut-off frequencies ?*

Solution. $A_v = 1000$; $m_v = 0.01$

The new lower cut-off frequency with negative voltage feedback is

$$f_{1(f)} = \frac{f_1}{1 + A_v\, m_v} = \frac{1.5\ \text{kHz}}{1 + 1000 \times 0.01} = \mathbf{136.4\ Hz}$$

The new upper cut-off frequency with negative voltage feedback is

$$\begin{aligned} f_{2(f)} &= f_2\ (1 + m_v\, A_v) \\ &= (501.5\ \text{kHz})\ (1 + 1000 \times 0.01) = \mathbf{5.52\ MHz} \end{aligned}$$

Note the effect of negative voltage feedback on the bandwidth of the amplifier. The lower cut-off frequency is decreased by a factor $(1 + m_v\, A_v)$ while upper cut-off frequency is increased by a factor $(1 + m_v\, A_v)$. In other words, the bandwidth of the amplifier is increased approximately by a factor $(1 + m_v\, A_v)$.

$$BW_{(f)} \simeq BW\,(1 + m_v\,A_v)$$

where BW = Bandwidth of the amplifier without feedback

$BW_{(f)}$ = Bandwidth of the amplifier with negative voltage feedback

Example 11.17. *Fig. 11.17 shows negative voltage feedback amplifier having open-loop voltage gain of 100,000. When negative feedback is not used, the input impedance of the amplifier is 1 kΩ. Find the input impedance of the circuit with negative voltage feedback.*

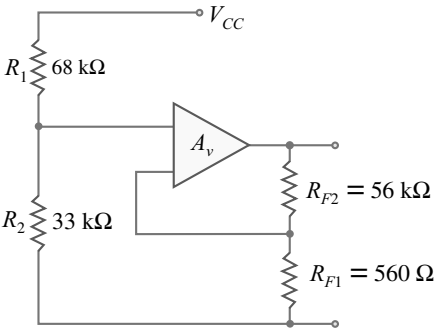


Fig. 11.17

Solution. The input impedance of the circuit means input impedance with biasing resistors (R_1 and R_2). The biasing resistors are not affected by the negative feedback because these are outside the feedback loop.

∴ Input impedance of the circuit is

$$Z_{\substack{in\\(circuit)}} = R_1 \parallel R_2 \parallel Z_{in(f)}$$

$$\text{Feedback fraction, } m_v = \frac{R_{F1}}{R_{F1} + R_{F2}} = \frac{560\ \Omega}{560\ \Omega + 56\ \text{k}\Omega} = \frac{1}{101}$$

Input impedance of the amplifier without negative feedback is Z_{in} = 1 kΩ. Therefore, with negative voltage feedback, the input impedance of the amplifier is

$$Z_{in(f)} = Z_{in}\,(1 + A_v m_v) = 1\ \text{k}\Omega\ (1 + 100{,}000 \times 1/101) = 991\ \text{k}\Omega$$

$$\therefore \quad Z_{\substack{in\\(circuit)}} = 68\ \text{k}\Omega \parallel 33\ \text{k}\Omega \parallel 991\ \text{k}\Omega = \mathbf{21.7\ k\Omega}$$

11.7 Two-stage Amplifier with Negative Voltage Feedback

Fig. 11.18 shows two-stage *CE* amplifier with negative voltage feedback. The feedback path is from the collector of transistor Q_2 to the emitter circuit of transistor Q_1 through the components C_{F1}, R_{F1} and R_{F2}. Here, resistors R_{F1} and R_{F2} form the *feedback circuit* because C_{F1} behaves as a short for a.c. The capacitor C_{F1} is a d.c. blocking capacitor and prevents the d.c. voltage at the collector of Q_2 from affecting the bias conditions of Q_1.

Fig. 11.19 shows the *a.c. equivalent circuit of the feedback circuit.* The capacitor C_2 behaves as a short for a.c. so that C_2 and R_4 (shorted by C_2) are absent from the a.c. equivalent circuit. Referring to Fig. 11.19, the signal voltage (v_s) is applied between the base of Q_1 and ground and the output voltage (v_0) is developed between collector of Q_2 and ground. The feedback voltage (v_f) is developed across resistor R_{F2}. The input voltage (v_i) across the base-emitter of Q_1 is $v_i = v_s - v_f$. This can be easily proved. The output voltage (v_0) of two-stage amplifier is in phase with the input signal (v_s). When v_s is positive going, v_0 is also positive going and consequently v_f is in phase *opposition with v_s.

$$\therefore \quad v_i = v_s - v_f$$

* Since v_s and v_f work in opposition in forcing current through the base emitter junction of Q_1.

26. The basic purpose of applying negative voltage feedback is to .....
(*i*) increase voltage gain
(*ii*) reduce distortion
(*iii*) keep the temperature within limits
(*iv*) none of the above

27. If the voltage gain of an amplifier without feedback is 20 and with negative voltage feedback it is 12, then feedback fraction is .....
(*i*) 5/3 (*ii*) 3/5
(*iii*) 1/5 (*iv*) 0.033

28. In an emitter follower, we employ ..... negative current feedback.
(*i*) 50% (*ii*) 25%
(*iii*) 100% (*iv*) 75%

29. An amplifier has an open loop voltage gain of 1,00,000. With negative voltage feedback, the voltage gain is reduced to 100. What is the sacrifice factor?
(*i*) 1000 (*ii*) 100
(*iii*) 5000
(*iv*) None of the above

30. In the above question, what will happen to circuit performance?
(*i*) Distortion is increased 1000 times
(*ii*) Input impedance is increased 1000 times
(*iii*) Output impedance is increased 1000 times
(*iv*) None of the above

31. The non-linear distortion of an amplifier is D without feedback. The amplifier has an open-loop voltage gain of A_v and feedback fraction is m_v. With negative voltage feedback, the non-linear distortion will be .....
(*i*) $D\ (1 + A_v\ m_v)$ (*ii*) $D\ (1 - A_v\ m_v)$
(*iii*) $\dfrac{1 + A_v m_v}{D}$ (*iv*) $\dfrac{D}{1 + A_v m_v}$

32. The output and input voltages of an emitter follower have a phase difference of .....
(*i*) 180º (*ii*) 90º
(*iii*) 0º (*iv*) 270º

33. It is most necessary to control signal-to-noise ratio at .....
(*i*) initial stage (*ii*) driver stage
(*iii*) output stage (*iv*) detector stage

34. In order to obtain good gain stability in a negative voltage feedback amplifier (A_v = voltage gain without feedback ; m_v = feedback fraction), ...
(*i*) $A_v\ m_v = 1$ (*ii*) $A_v\ m_v >> 1$
(*iii*) $A_v\ m_v < 1$
(*iv*) none of the above

35. Emitter follower is also known as .....
(*i*) grounded emitter circuit
(*ii*) grounded base circuit
(*iii*) grounded collector circuit
(*iv*) none of the above

Answers to Multiple-Choice Questions

1. (*ii*)	**2.** (*i*)	**3.** (*iii*)	**4.** (*iv*)	**5.** (*ii*)
6. (*i*)	**7.** (*iii*)	**8.** (*ii*)	**9.** (*i*)	**10.** (*iii*)
11. (*ii*)	**12.** (*ii*)	**13.** (*i*)	**14.** (*ii*)	**15.** (*iii*)
16. (*i*)	**17.** (*iii*)	**18.** (*ii*)	**19.** (*iv*)	**20.** (*i*)
21. (*iv*)	**22.** (*iii*)	**23.** (*i*)	**24.** (*ii*)	**25.** (*i*)
26. (*ii*)	**27.** (*iv*)	**28.** (*iii*)	**29.** (*i*)	**30.** (*ii*)
31. (*iv*)	**32.** (*iii*)	**33.** (*i*)	**34.** (*ii*)	**35.** (*iii*)

Chapter Review Topics

1. What do you understand by feedback? Why is negative feedback applied in high gain amplifiers?
2. Discuss the principles of negative voltage feedback in amplifiers with a neat diagram.

3. Derive an expression for the gain of negative voltage feedback amplifier.
4. What is a feedback circuit? Explain how it provides feedback in amplifiers.
5. Describe the action of emitter follower with a neat diagram.
6. Derive the expressions for (*i*) voltage gain (*ii*) input impedance and (*iii*) output impedance of an emitter follower.

Problems

1. An amplifier has a gain of 2×10^5 without feedback. Determine the gain if negative voltage feedback is applied. Take feedback fraction $m_v = 0.02$. **[50]**
2. An amplifier has a gain of 10,000 without feedback. With negative voltage feedback, the gain is reduced to 50. Find the feedback fraction. **[m_v = 0.02]**
3. A feedback amplifier has an internal gain A_v = 40 dB and feedback fraction m_v = 0.05. If the input impedance of this circuit is 12 kΩ, what would have been the input impedance if feedback were not present? **[2 kΩ]**
4. Calculate the gain of a negative voltage feedback amplifier with an internal gain A_v = 75 and feedback fraction m_v = 1/15. What will be the gain if A_v doubles? **[12.5 ; 13.64]**
5. An amplifier with negative feedback has a voltage gain of 100. It is found that without feedback, an input signal of 50 mV is required to produce a given output, whereas with feedback, the input signal must be 0.6 V for the same output. Calculate (*i*) gain without feedback (*ii*) feedback fraction. **[(*i*) 1200 (*ii*) 0.009]**

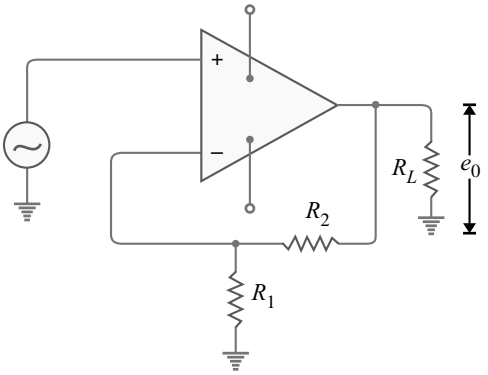


Fig. 11.44

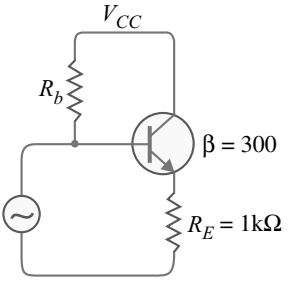


Fig. 11.45

6. Fig. 11.44 shows the negative feedback amplifier. If the gain of the amplifier without feedback is 10^5 and R_1 = 100 Ω, R_2 = 100 kΩ, find (*i*) feedback fraction (*ii*) gain with feedback. **[(*i*) 0.001 (*ii*) 1000]**
7. In Fig. 11.45, if input and output impedances without feedback are 2 MΩ and 500 Ω respectively, find their values after negative voltage feedback. **[302 MΩ ; 1.6 Ω]**
8. An amplifier has a current gain of 240 without feedback. When negative current feedback is applied, determine the effective current gain of the amplifier. Given that current attenuation m_i = 0.015. **[52.7]**
9. An amplifier has an open-loop gain and input impedance of 200 and 15 kΩ respectively. If negative current feedback is applied, what is the effective input impedance of the amplifier? Given that current attenuation m_i = 0.012. **[4.41 kΩ]**
10. An amplifier has A_i = 200 and m_i = 0.012. The open-loop output impedance of the amplifier is 2 kΩ. If negative current feedback is applied, what is the effective output impedance of the amplifier? **[6.8 kΩ]**



Sinusoidal Oscillators

Learning Objectives

- Understand the underlying principle for producing sustained oscillations.
- Develop mathematical condition for proper operation of an oscillator.
- Explore the operation of *LC* oscillators and *RC* oscillators.
- Understand the operation of various types of *LC* oscillators.
- Develop an understanding of *RC* Phase shift oscillator and Wien Bridge oscillator.
- Derive the expression for operating frequency (f_o) for the *RC* Phase Shift oscillator.
- Derive the expression for the balance frequency (f_o) of Wien Bridge oscillator.
- Learn the limitations of *LC* and *RC* oscillators.
- Understand the working and characteristics of Quartz crystal.
- Understand the operation of Transistor Crystal oscillator.

Introduction

In the previous chapter, we have seen that negative feedback in an amplifier improves the circuit performance of the amplifier. However, a proper amount of positive feedback in an amplifier enables it to work as an oscillator *i.e.* it produces continuous oscillations of any desired frequency without any external signal source. The only input power to an oscillator is the d.c. power supply. Therefore, an oscillator is simply a positive feedback amplifier. The frequency of oscillations is determined by the components of the circuit.

Many electronic devices require a source of energy at a specific frequency which may range from a few Hz to several MHz. This is achieved by an electronic device called an *oscillator*. Oscillators are extensively used in electronic equipment. For example, in radio and television receivers, oscillators are used to generate high frequency wave (called *carrier wave*) in the tuning stages. Audio frequency and radio-frequency signals are required for the repair of radio, television and other electronic equipment. Oscillators are also widely used in radar, electronic computers and other electronic devices. Oscillators can produce sinusoidal or non-sinusoidal (*e.g.* square wave) waves. In this chapter, we shall confine our attention to sinusoidal oscillators *i.e.* those which produce sine-wave signals.

12.1 Sinusoidal Oscillator

An electronic device that generates sinusoidal oscillations of desired frequency without any external signal source is known as a **sinusoidal oscillator.**

Note that an oscillator generates oscillations of any desired frequency without any external signal source. The only input power to an oscillator is the d.c. power supply. As we shall see, an oscillator is a *positive feedback amplifier* which provides its own input signal *via* the feedback circuit. Therefore, the oscillator generates or produces oscillations without any externally applied input signal. The frequency of oscillations is determined by the components of the oscillator circuit. Although we speak of an oscillator as "generating" a frequency, it should be noted that it does not create energy, but merely acts as an energy converter. It receives d.c. energy and changes it into a.c. energy of desired frequency. The frequency of oscillations depends upon the constants of the device.

It may be mentioned here that although an alternator produces sinusoidal oscillations of 50Hz, it cannot be called an oscillator. Firstly, an alternator is a mechanical device having rotating parts whereas an oscillator is a non-rotating electronic device. Secondly, an alternator converts mechanical energy into a.c. energy while an oscillator converts d.c. energy into a.c. energy. Thirdly, an alternator cannot produce high frequency oscillations whereas an oscillator can produce oscillations ranging from a few Hz to several MHz.

Advantages

Although oscillations can be produced by mechanical devices (*e.g.* alternators), but electronic oscillators have the following advantages :

(*i*) An oscillator is a non-rotating device. Consequently, there is little wear and tear and hence longer life.

(*ii*) Due to the absence of moving parts, the operation of an oscillator is quite silent.

(*iii*) An oscillator can produce waves from small (20Hz) to extremely high frequencies (> 100 MHz).

(*iv*) The frequency of oscillations can be easily changed when desired.

(*v*) It has good frequency stability *i.e.* frequency once set remains constant for a considerable period of time.

(*vi*) It has very high efficiency.

12.2 Types of Sinusoidal Oscillations

Sinusoidal electrical oscillations can be of two types *viz.* *damped oscillations* and *undamped oscillations*.

(*i*) Damped oscillations. The electrical oscillations whose amplitude goes on decreasing with time are called *damped oscillations*. Fig. 12.1 (*i*) shows waveform of damped electrical oscillations. Obviously, the electrical system in which these oscillations are generated has losses and some energy is lost during each oscillation. Further, no means are provided to compensate for the losses and consequently the amplitude of the generated wave decreases gradually. It may be noted that frequency of oscillations remains unchanged since it depends upon the constants of the electrical system.

(*ii*) Undamped oscillations. The electrical oscillations whose amplitude remains constant with time are called *undamped oscillations*. Fig. 12.1 (*ii*) shows waveform of undamped electrical oscillations. Although the electrical system in which these oscillations are being

generated has also losses, but now right amount of energy is being supplied to overcome the losses. Consequently, the amplitude of the generated wave remains constant. It should be emphasised that an oscillator is required to produce undamped electrical oscillations for utilising in various electronics equipment.

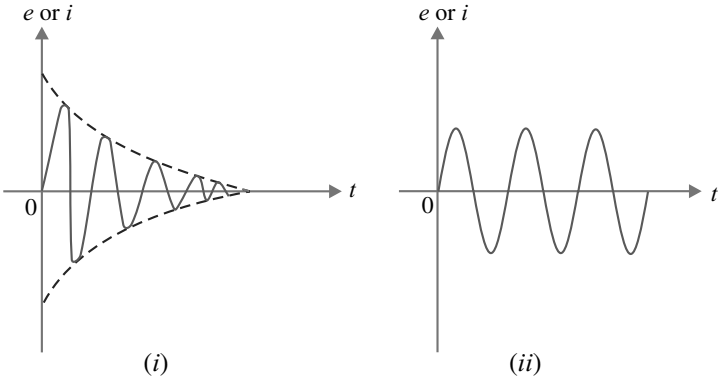


Fig. 12.1

12.3 Oscillatory Circuit

A circuit which produces electrical oscillations of any desired frequency is known as an **oscillatory circuit** *or* **tank circuit.**

A simple oscillatory circuit consists of a capacitor (C) and inductance coil (L) in parallel as shown in Fig. 12.2. This electrical system can produce electrical oscillations of frequency determined by the values of L and C. To understand how this comes about, suppose the capacitor is charged from a d.c. source with a polarity as shown in Fig. 12.2 (*i*).

(*i*) In the position shown in Fig. 12.2 (*i*), the upper plate of capacitor has deficit of electrons and the lower plate has excess of electrons. Therefore, there is a voltage across the capacitor and the capacitor has electrostatic energy.

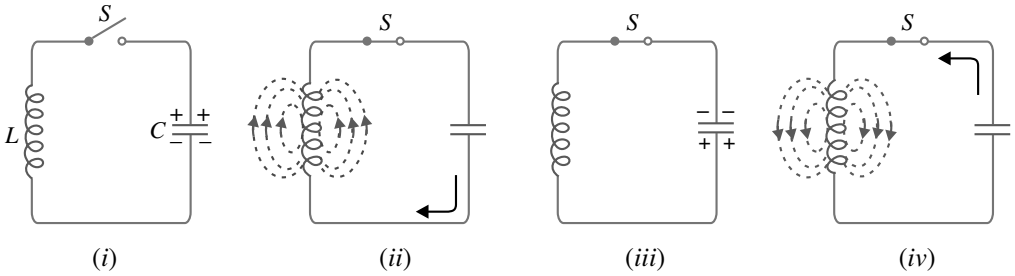


Fig. 12.2

(*ii*) When switch S is closed as shown in Fig. 12.2 (*ii*), the capacitor will discharge through inductance and the electron flow will be in the direction indicated by the arrow. This current flow sets up magnetic field around the coil. Due to the inductive effect, the current builds up slowly towards a maximum value. The circuit current will be maximum when the capacitor is fully discharged. At this instant, electrostatic energy is zero but because electron motion is greatest (*i.e.* maximum current), the magnetic field energy around the coil is maximum. This is shown in Fig. 12.2 (*ii*). Obviously, the electrostatic energy across the capacitor is completely converted into magnetic field energy around the coil.

(*iii*) Once the capacitor is discharged, the magnetic field will begin to collapse and produce a counter e.m.f. According to Lenz's law, the counter e.m.f. will keep the current flowing in the same direction. The result is that the capacitor is now charged with opposite polarity, making upper plate of capacitor negative and lower plate positive as shown in Fig. 12.2 (*iii*).

(*iv*) After the collapsing field has recharged the capacitor, the capacitor now begins to discharge ; current now flowing in the opposite direction. Fig. 12.2 (*iv*) shows capacitor fully discharged and maximum current flowing.

The sequence of charge and discharge results in alternating motion of electrons or an oscillating current. The energy is alternately stored in the electric field of the capacitor (C) and the magnetic field of the inductance coil (L). This interchange of energy between L and C is repeated over and again resulting in the production of oscillations.

Waveform. If there were no losses in the tank circuit to consume the energy, the interchange of energy between L and C would continue indefinitely. In a practical tank circuit, there are resistive and radiation losses in the coil and dielectric losses in the capacitor. During each cycle, a small part of the originally imparted energy is used up to overcome these losses. The result is that the amplitude of oscillating current decreases gradually and eventually it becomes zero when all the energy is consumed as losses. Therefore, the tank circuit by itself will produce *damped oscillations* as shown in Fig. 12.3.

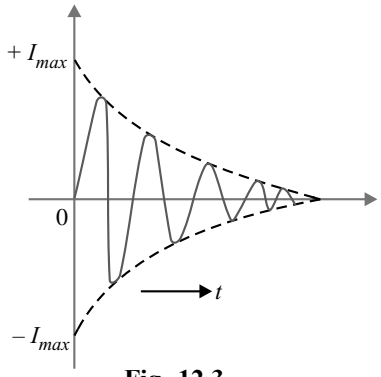


Fig. 12.3

Frequency of oscillations. The frequency of oscillations in the tank circuit is determined by the constants of the circuit *viz.* L and C. The actual frequency of oscillations is the resonant frequency (or natural frequency) of the tank circuit given by ;

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

It is clear that frequency of oscillations in the tank circuit is inversely proportional to L and C. This can be easily explained. If a large value of capacitor is used, it will take longer for the capacitor to charge fully and also longer to discharge. This will lengthen the period of oscillations in the tank circuit, or equivalently lower its frequency. With a large value of inductance, the opposition to change in current flow is greater and hence the time required to complete each cycle will be longer. Therefore, the greater the value of inductance, the longer is the period or the lower is the frequency of oscillations in the tank circuit.

12.4 Undamped Oscillations from Tank Circuit

As discussed before, a tank circuit produces damped oscillations. However, in practice, we need continuous undamped oscillations for the successful operation of electronics equipment. In order to make the oscillations in the tank circuit undamped, it is necessary to supply correct amount of energy to the tank circuit at the proper time intervals to meet the losses. Thus referring back to Fig. 12.2, any energy which would be applied to the circuit must have

a polarity conforming to the existing polarity at the instant of application of energy. If the applied energy is of opposite polarity, it would oppose the energy in the tank circuit, causing stoppage of oscillations. Therefore, in order to make the oscillations in the tank circuit undamped, the following conditions must be fulfilled :

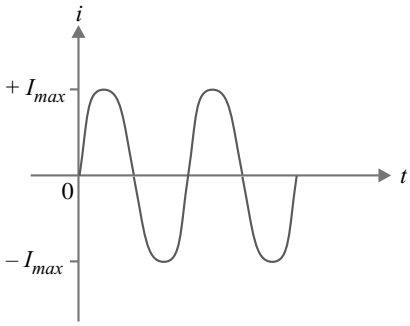


Fig. 12.4

(*i*) The amount of energy supplied should be such so as to meet the losses in the tank circuit and the a.c. energy removed from the circuit by the load. For instance, if losses in *LC* circuit amount to 5 mW and a.c. output being taken is 100 mW, then power of 105 mW should be continuously supplied to the circuit.

(*ii*) The applied energy should have the same frequency as that of the oscillations in the tank circuit.

(*iii*) The applied energy should be in phase with the oscillations set up in the tank circuit *i.e.* it should aid the tank circuit oscillations.

If these conditions are fulfilled, the circuit will produce continuous undamped output as shown in Fig. 12.4.

12.5 Positive Feedback Amplifier — Oscillator

A transistor amplifier with *proper* positive feedback can act as an oscillator *i.e.* it can generate oscillations without any external signal source. Fig. 12.5 shows a transistor amplifier with positive feedback. Remember that a positive feedback amplifier is one that produces a feedback voltage (V_f) that *is in phase* with the original input signal. As you can see, this condition is met in the circuit shown in Fig. 12.5. A phase shift of 180° is produced by the amplifier and a further phase shift of 180° is introduced by feedback network. Consequently, the signal is shifted by 360° and fed to the input *i.e.* feedback voltage is in phase with the input signal.

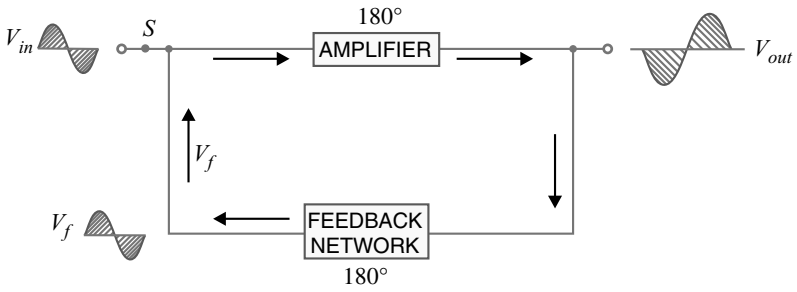


Fig. 12.5

(*i*) We note that the circuit shown in Fig. 12.5 is producing oscillations in the output. However, this circuit has an input signal. This is inconsistent with our definition of an oscillator *i.e.* an oscillator is a circuit that produces oscillations without any external signal source.

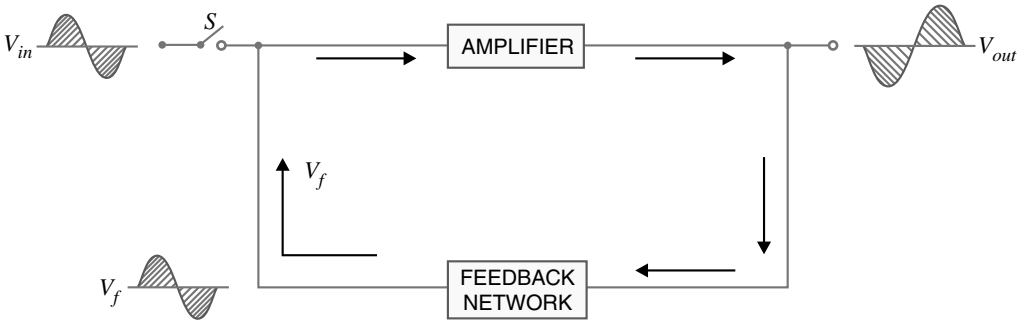


Fig. 12.6

(*ii*) When we open the switch *S* of Fig. 12.5, we get the circuit shown in Fig. 12.6. This means the input signal (V_{in}) is removed. However, V_f (which is in phase with the original signal) is still applied to the input signal. The amplifier will respond to this signal in the same way that it did to V_{in} *i.e.* V_f will be amplified and sent to the output. The feedback network sends a portion of the output back to the input. Therefore, the amplifier receives another input cycle and another output cycle is produced. This process will continue so long as the amplifier is turned on. Therefore, the amplifier will produce sinusoidal output without any external signal source. The only condition is that the amount of positive feedback should be such that the feedback voltage $V_f = V_{in}$. In that case, the circuit will provide its own input signal (V_{in}) *via* the feedback circuit and no externally applied input signal is required. The only input power to the oscillator is the d.c. power supply.

12.6 Condition for Proper Oscillator Operation

An oscillator is an amplifier that uses *positive feedback* for its operation. Therefore, it is a positive feedback amplifier. Proper operation of oscillator means that it should produce continuous oscillations of the desired frequency without any external signal source except the d.c. supply. This is possible only if the positive feedback amplifier provides its own input signal *via* the feedback circuit. Let us find the mathematical condition under which it can happen. Consider the block diagram of positive feedback amplifier shown in Fig. 12.7. Here A_v is the open-loop voltage gain of the amplifier and m_v is the feedback fraction. The output voltage is $V_o = A_v V_i$ where V_i is the input voltage. Clearly, the feedback voltage $V_f = m_v V_o = A_v m_v V_i$. If the link *L* were removed and the output of feedback circuit connected to the input circuit with +ve to +ve and –ve to –ve, the feedback would be positive. This means that the feedback voltage V_f ($= A_v m_v V_i$) is in phase with the input voltage. Since V_i is the required input voltage to produce the output voltage V_o, it means that for continuous output (*i.e.* oscillations), the feedback voltage (V_f) must be equal to the input voltage (V_i) *i.e.*

$$\text{Feedback voltage, } V_f = V_i$$

$$\text{or} \qquad A_v m_v V_i = V_i$$

$$\therefore \qquad A_v m_v = 1$$

Under this condition (*i.e.* $A_v m_v = 1$), even if these were no input generator (*i.e.* V_i), a continuous output will be obtained across R_L immediately after connecting the necessary d.c. power supply. Fig. 12.8 shows the general layout of an oscillator.

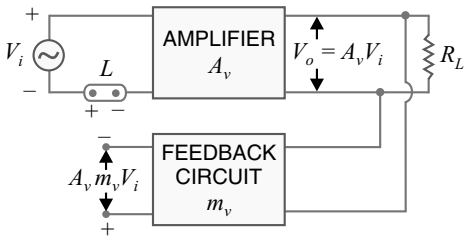


Fig. 12.7

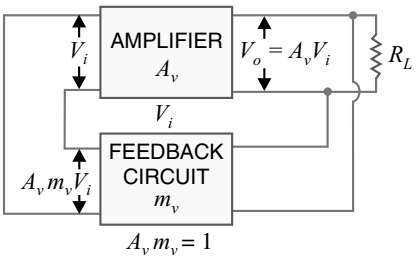


Fig. 12.8

It is clear from Fig. 12.8 that continuous output voltage (V_o) will be obtained with no externally applied input signal. The circuit provides its own input signal *via* the feedback circuit.Thus output voltage $V_o = A_v V_i$ and the feedback voltage, $V_f = A_v m_v V_i = V_i$ ($\because A_v m_v = 1$).

Therefore, for proper operation of oscillator (positive feedback amplifier), the product $A_v m_v$ should be equal to 1 i.e.

$$A_v m_v = 1$$

This relation is called **Barkhausen criterion** and is discussed in detail in the next article.

Note. Let us see what happens if the condition $A_v m_v = 1$ is not met in an oscillator.

(*i*) If $A_v m_v < 1$, the oscillations will die out within a few cycles. It is because the oscillations will lose amplitude on each progressive cycle and will eventually fade out. This loss of signal is called *damping*.

(*ii*) If $A_v m_v > 1$, the output will be distorted. It is because the output will increase from each cycle to the next. Eventually, the output will start to experience saturation and cut-off clipping.

12.7 Explanation of Barkhausen Criterion

Barkhausen criterion is that in order to produce continuous undamped oscillations at the output of an amplifier, the positive feedback should be such that :

$$m_v A_v = 1$$

Once this condition is set in the positive feedback amplifier, continuous undamped oscillations can be obtained at the output immediately after connecting the necessary power supplies.

(*i*) Mathematical explanation. The voltage gain of a positive feedback amplifier is given by ;

$$A_{vf} = \frac{A_v}{1 - m_v A_v}$$

$$\text{If } m_v A_v = 1, \text{ then, } A_{vf} \to \infty.$$

We know that we cannot achieve infinite gain in an amplifier. So what does this result infer in physical terms? It means that a vanishing small input voltage would give rise to finite (*i.e.* a definite amount of) output voltage even when the input signal is zero. Thus once the circuit receives the input trigger, it would become an oscillator, generating oscillations with no external signal source.

(*ii*) Graphical Explanation. Let us discuss the condition $m_v A_v = 1$ graphically. Suppose the voltage gain of the amplifier without positive feedback is 100. In order to produce continuous

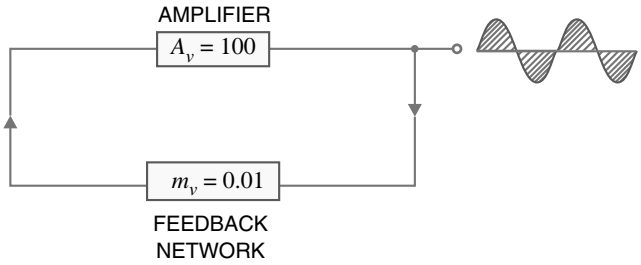


Fig. 12.9

undamped oscillations, $m_v A_v = 1$ or $m_v \times 100 = 1$ or $m_v = 0.01$. This is illustrated in Fig. 12.9. Since the condition $m_v A_v = 1$ is met in the circuit shown in Fig. 12.9, it will produce sustained oscillations.

Suppose the initial triggering voltage is 0.1V peak. Starting with this value, circuit ($A_v = 100$; $m_v = 0.01$) will progress as follows.

Cycle	V_{in}	V_{out}	V_f
1.	0.1Vpk	10Vpk	0.1Vpk
2.	0.1Vpk	10Vpk	0.1Vpk

The same thing will repeat for 3rd, 4th cycles and so on. Note that during each cycle, $V_f = 0.1$Vpk and $V_{out} = 10$ Vpk. Clearly, the oscillator is producing continuous undamped oscillations.

Note. The relation $m_v A_v = 1$ holds good for true ideal circuits. However, practical circuits need an $m_v A_v$ product that is slightly greater than 1. This is to compensate for power loss (*e.g.* in resistors) in the circuit.

12.8 Essential Parts of an Oscillator

An oscillator is a positive feedback amplifier. The amplifier may employ a transistor or any other electronic device capable of doing the job of amplification. Fig. 12.10 shows the block diagram of an oscillator. Its essential components are :

(*i*) Frequency-determining circuit. It consists of either parallel tuned circuit (*LC*) or *RC* circuit. The radio-frequency oscillators generally employ the former while the audio frequency oscillators use the latter. The frequency of oscillations in the circuit depends upon these circuit elements.

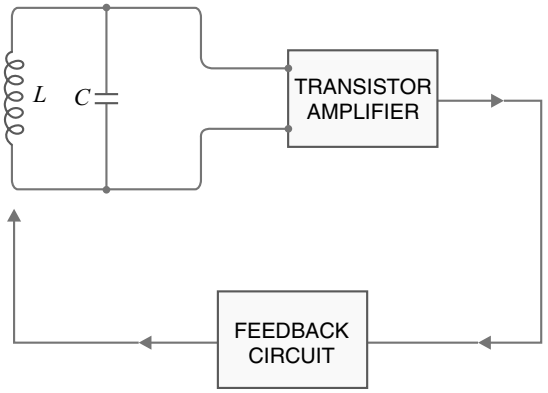


Fig. 12.10

(*ii*) Transistor Amplifier. The transistor amplifier receives d.c. power from the battery and changes it into a.c. power of desired frequency for supplying to the frequency-determining circuit. The oscillations occurring in the frequency-determining circuit are applied to the input of the transistor amplifier. Because of amplifying properties of the transistor, we get increased output of these oscillations. The output of the transistor is supplied to the frequency-determining circuit to meet the losses.

(*iii*) Feedback Circuit. The feedback circuit supplies a part of output energy to the frequency determining circuit satisfying the relation $A_v m_v = 1$. The entire circuit arrangement is so designed that energy fed back is in phase with oscillations in the frequency-determining circuit *i.e.* positive feedback is provided.

Discussion. The following points are worth noting about an oscillator :

(*i*) A transistor amplifier with *proper positive feedback* will work as an oscillator.

(*ii*) The circuit needs only a quick trigger to start the oscillations. This trigger may be provided when d.c. supply is applied to the oscillator. Once oscillations have started, no external signal source is needed.

(*iii*) The amount of positive feedback should be such that $A_v\, m_v = 1$.

12.9 Different Types of Oscillators

All oscillators are positive feedback amplifiers in which the amount of positive feedback is such that the condition $A_v\, m_v = 1$ is met. They differ only in the manner in which this objective is achieved. There are two basic types of oscillators *viz*.

(*i*) *LC* oscillators **(*ii*)** *RC* oscillators

(*i*) In ***LC* oscillators**, the *LC* circuit determines the frequency of oscillations as well as produces 180° phase shift. A further phase shift of 180° is introduced by the amplifier. In this way, positive feedback is ensured. The important types of *LC* oscillators are :

(*a*) Tuned collector oscillator

(*b*) Colpitt's oscillator

(*c*) Hartley oscillator

(*d*) Crystal oscillator

(*ii*) In ***RC* oscillators**, the *RC* network determines the operating frequency f_0. Of course, the *RC* circuit cannot produce oscillations as in *LC* circuit. The *RC* network may produce 180° phase shift or 0° phase shift, depending upon the design of *RC* network. The two important types of *RC* oscillators are :

(*a*) *RC* phase shift oscillator

(*b*) Wien bridge oscillator

12.10 Tuned Collector Oscillator

Fig. 12.11 shows the circuit of tuned collector oscillator. It contains tuned circuit L_1-C_1 in the collector and hence the name. The frequency of oscillations depends upon the values of L_1 and C_1 and is given by ;

$$f = \frac{1}{2\pi \sqrt{L_1\, C_1}} \qquad ...(i)$$

The feedback coil L_2 in the base circuit is magnetically coupled to the tank circuit coil L_1. In practice, L_1 and L_2 form the primary and secondary of the transformer respectively. The biasing is provided by potential divider arrangement. The capacitor C connected in the base circuit provides low reactance path to the oscillations.

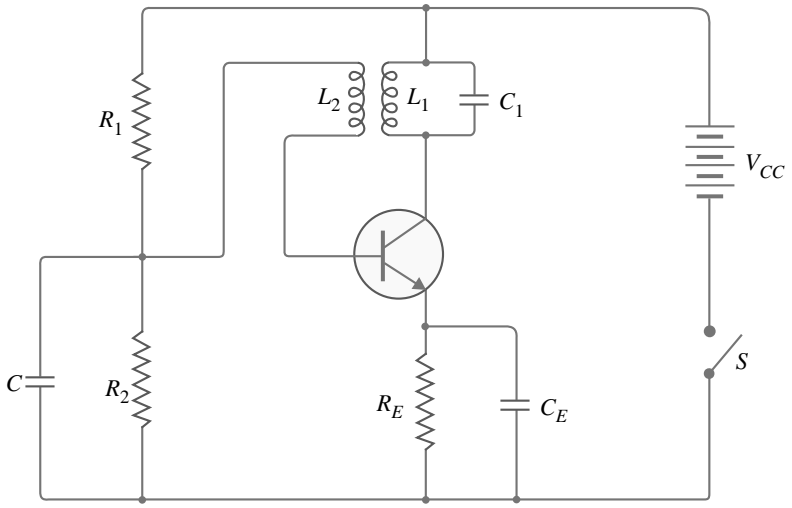


Fig. 12.11

Circuit operation. When switch S is closed, collector current starts increasing and charges the capacitor C_1. When this capacitor is fully charged, it discharges through coil L_1, setting up oscillations of frequency determined by exp. (*i*). These oscillations induce some voltage in coil L_2 by mutual induction. The frequency of voltage in coil L_2 is the same as that of tank circuit but its magnitude depends upon the number of turns of L_2 and coupling between L_1 and L_2. The voltage across L_2 is applied between base and emitter and appears in the amplified form in the collector circuit, thus overcoming the losses occurring in the tank circuit. The number of turns of L_2 and coupling between L_1 and L_2 are so adjusted that oscillations across L_2 are amplified to a level just sufficient to supply losses to the tank circuit.

It may be noted that the phase of feedback is correct *i.e.* energy supplied to the tank circuit is in phase with the generated oscillations. A phase shift of 180º is created between the voltages of L_1 and L_2 due to transformer *action. A further phase shift of 180º takes place between base-emitter and collector circuit due to transistor properties. As a result, the energy fed back to the tank circuit is in phase with the generated oscillations.

Example 12.1. *The tuned collector oscillator circuit used in the local oscillator of a radio receiver makes use of an LC tuned circuit with L_1 = 58.6 μH and C_1 = 300 pF. Calculate the frequency of oscillations.*

Solution.

$$L_1 = 58.6\ \mu\text{H} = 58.6 \times 10^{-6}\ \text{H}$$

$$C_1 = 300\ \text{pF} = 300 \times 10^{-12}\ \text{F}$$

$$\text{Frequency of oscillations, } f = \frac{1}{2\pi\sqrt{L_1 C_1}}$$

* All transformers introduce a phase shift of 180° between primary and secondary.

$$= \frac{1}{2\pi\sqrt{58.6\times10^{-6}\times300\times10^{-12}}}\text{ Hz}$$

$$= 1199\times10^{3}\text{ Hz} = \mathbf{1199\ kHz}$$

Example 12.2. *Find the capacitance of the capacitor required to build an LC oscillator that uses an inductance of L_1 = 1 mH to produce a sine wave of frequency 1 GHz (1 GHz = 1 × 10^{12} Hz).*

Solution.

Frequency of oscillations is given by ;

$$f = \frac{1}{2\pi\sqrt{L_1C_1}}$$

or
$$C_1 = \frac{1}{L_1(2\pi f)^2} = \frac{1}{(1\times10^{-3})(2\pi\times1\times10^{12})^2}$$

$$= 2{\cdot}53\times10^{-23}\text{ F} = \mathbf{2{\cdot}53\times10^{-11}\ pF}$$

The *LC* circuit is often called *tuned circuit* or *tank circuit.*

12.11 Colpitt's Oscillator

Fig. 12.12 shows a Colpitt's oscillator. It uses two capacitors and placed across a common inductor *L* and the centre of the two capacitors is tapped. The tank circuit is made up of C_1, C_2 and *L*. The frequency of oscillations is determined by the values of C_1, C_2 and *L* and is given by ;

$$f = \frac{1}{2\pi\sqrt{LC_T}} \qquad ...(i)$$

where
$$C_T = \frac{C_1C_2}{C_1+C_2}$$

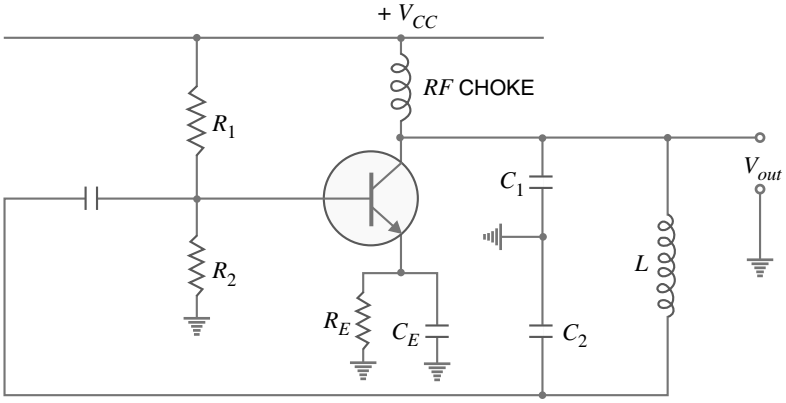


Fig. 12.12

*Note that $C_1 - C_2 - L$ is also the feedback circuit that produces a phase shift of 180°.

Circuit operation. When the circuit is turned on, the capacitors C_1 and C_2 are charged. The capacitors discharge through *L*, setting up oscillations of frequency determined by

* The *RF* choke decouples any a.c. signal on the power lines from affecting the output signal.

exp. *(*i*). The output voltage of the amplifier appears across C_1 and feedback voltage is developed across C_2. The voltage across it is 180° out of phase with the voltage developed across C_1 (V_{out}) as shown in Fig. 12.13. It is easy to see that voltage fed back (voltage across C_2) to the transistor provides positive feedback. A phase shift of 180° is produced by the transistor and a further phase shift of 180° is produced by $C_1 - C_2$ voltage divider. In this way, feedback is properly phased to produce continuous undamped oscillations.

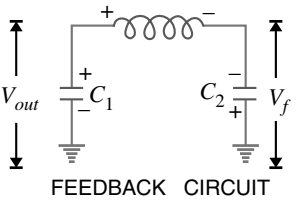


Fig. 12.13

Feedback fraction m_v. The amount of feedback voltage in Colpitt's oscillator depends upon feedback fraction m_v of the circuit. For this circuit,

$$\text{Feedback fraction, } m_v = \frac{V_f}{V_{out}} = \frac{X_{C2}}{X_{C1}} = \frac{C_1}{C_2} \quad **$$

or
$$m_v = \frac{C_1}{C_2}$$

We know that the condition for continuous oscillations in an oscillator is $A_v\, m_v = 1$.

$$\therefore \quad \text{Amplifier voltage gain, } A_v = \frac{1}{m_v} = \frac{C_2}{C_1}$$

Actually, the product $A_v\, m_v$ should be slightly greater than 1 to prevent decrease of oscillations due to power loss within the amplifier.

Example 12.3. *Determine the (i) operating frequency and (ii) feedback fraction for Colpitt's oscillator shown in Fig. 12.14.*

Solution.

(*i*) Operating Frequency. The operating frequency of the circuit is always equal to the resonant frequency of the feedback network. As noted previously, the capacitors C_1 and C_2 are in series.

$$\therefore \quad C_T = \frac{C_1\, C_2}{C_1 + C_2} = \frac{0.001 \times 0.01}{0.001 + 0.01} = 9.09 \times 10^{-4}\ \mu\text{F}$$

$$= 909 \times 10^{-12}\ \text{F}$$

$$L = 15\ \mu\text{H} = 15 \times 10^{-6}\ \text{H}$$

$$\therefore \quad \text{Operating frequency, } f = \frac{1}{2\pi\sqrt{LC_T}}$$

$$= \frac{1}{2\pi\sqrt{15 \times 10^{-6} \times 909 \times 10^{-12}}}\ \text{Hz}$$

$$= 1361 \times 10^3\ \text{Hz} = \mathbf{1361\ kHz}$$

---

* Referring to Fig. 12.13, it is clear that C_1 and C_2 are in series. Therefore, total capacitance C_T is given by ; $C_T = \frac{C_1 C_2}{C_1 + C_2}$.

** Referring to Fig. 12.13, the circulating current for the two capacitors is the same. Further, capacitive reactance is inversely proportional to capacitance.

(*ii*) Feedback fraction

$$m_v = \frac{C_1}{C_2} = \frac{0.001}{0.01} = \mathbf{0.1}$$

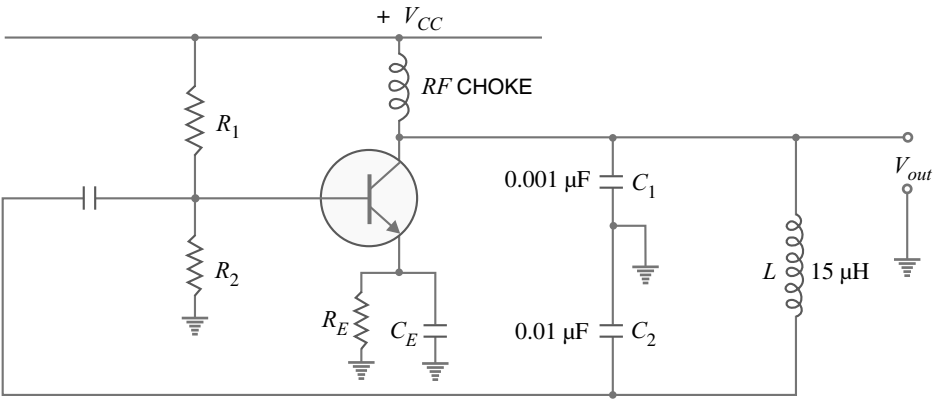


Fig. 12.14

Example 12.4. *A 1 mH inductor is available. Choose the capacitor values in a Colpitt's oscillator so that f = 1 MHz and m_v = 0.25.*

Solution.

Feedback fraction, $m_v = \dfrac{C_1}{C_2}$

or $0.25 = \dfrac{C_1}{C_2} \qquad \therefore\ C_2 = 4C_1$

Now, $f = \dfrac{1}{2\pi\sqrt{LC_T}}$

or $C_T = \dfrac{1}{L(2\pi f)^2} = \dfrac{1}{(1\times10^{-3})\ (2\pi\times1\times10^{6})^2}$

$= 25.3 \times 10^{-12}$ F

$= 25.3$ pF

or $\dfrac{C_1 C_2}{C_1 + C_2} = 25.3$ pF $\qquad \left[\because C_T = \dfrac{C_1 C_2}{C_1 + C_2}\right]$

or $\dfrac{C_2}{1 + \dfrac{C_2}{C_1}} = 25.3$

or $\dfrac{C_2}{1+4} = 25.3 \qquad \therefore\ C_2 = 25.3 \times 5 = \mathbf{126.5\ pF}$

and $C_1 = C_2/4 = 126.5/4 = \mathbf{31.6\ pF}$

Example 12.5. *For the Colpitt's oscillator shown in Fig. 12.15 determine the (i) d.c. emitter current (ii) d.c. collector-emitter voltage (iii) frequency of oscillations (iv) feedback fraction and (v) minimum value of voltage gain to start oscillations.*

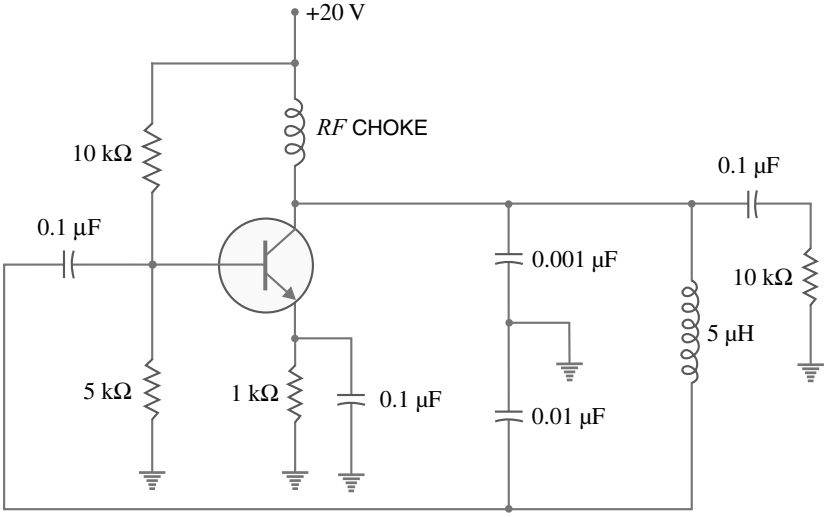


Fig. 12.15

Solution. $R_1 = 10\ k\Omega$; $R_2 = 5\ k\Omega$; $C_1 = 0.001\ \mu F$; $C_2 = 0.01\ \mu F$; $R_E = 1\ k\Omega$

(*i*) The d.c. emitter current I_E is found as under :

$$\text{D.C. base voltage, } V_B = \frac{V_{CC}}{R_1 + R_2} \times R_2 = \frac{20\ V}{(10+5)\ k\Omega} \times 5\ k\Omega = 6.67\ V$$

$$\text{D.C. emitter voltage, } V_E = V_B - V_{BE} = 6.67 - 0.7 = 5.97\ V$$

$$\therefore \quad \text{D.C. emitter current, } I_E = \frac{V_E}{R_E} = \frac{5.97\ V}{1\ k\Omega} = \mathbf{5.97\ mA}$$

(*ii*) Since *RF* choke is short to d.c. ($f = 0$; $X_L = 0$), the collector voltage is $V_C = 20$ V.

$$\therefore \quad V_{CE} = V_C - V_E = 20 - 5.97 = \mathbf{14.03\ V}$$

(*iii*)
$$C_T = \frac{C_1 C_2}{C_1 + C_2} = \frac{(0.001 \times 0.01)}{(0.001 + 0.01)} = 909 \times 10^{-12}\ F$$

$\therefore$ Frequency of oscillations of the oscillator is

$$f = \frac{1}{2\pi\sqrt{LC_T}} = \frac{1}{2\pi\sqrt{(5 \times 10^{-6}) \times 909 \times 10^{-12}}}$$

$$= 2.36 \times 10^6\ Hz = \mathbf{2.36\ MHz}$$

(*iv*) Feedback fraction of this type of oscillator is

$$m_v = \frac{C_1}{C_2} = \frac{0.001\ \mu F}{0.01\ \mu F} = \mathbf{0.1}$$

(*v*) For the oscillations to start, the minimum value of voltage gain (A_v) is

$$A_v = \frac{1}{m_v} = \frac{C_2}{C_1} = \frac{1}{0.1} = \mathbf{10}$$

12.12 Hartley Oscillator

The Hartley oscillator is similar to Colpitt's oscillator with minor modifications. Instead of using tapped capacitors, two inductors L_1 and L_2 are placed across a common capacitor C and the centre of the inductors is tapped as shown in Fig. 12.16. The tank circuit is made up of L_1, L_2 and C. The frequency of oscillations is determined by the values of L_1, L_2 and C and is given by ;

$$f = \frac{1}{2\pi\sqrt{CL_T}} \qquad ...(i)$$

where $L_T = L_1 + L_2 + 2M$

Here M = Mutual inductance between L_1 and L_2

Note that $L_1 - L_2 - C$ is also the feedback network that produces a phase shift of 180°.

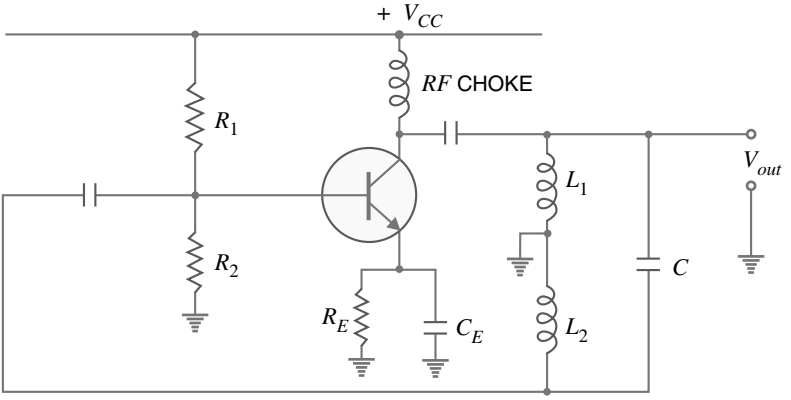


Fig. 12.16

Circuit operation. When the circuit is turned on, the capacitor is charged. When this capacitor is fully charged, it discharges through coils L_1 and L_2 setting up oscillations of frequency determined by *exp. (i). The output voltage of the amplifier appears across L_1 and feedback voltage across L_2. The voltage across L_2 is 180° out of phase with the voltage developed across L_1 (V_{out}) as shown in Fig. 12.17. It is easy to see that voltage fed back (*i.e.* voltage across L_2) to the transistor provides positive feedback. A phase shift of 180° is produced by the transistor and a further phase shift of 180° is produced by $L_1 - L_2$ voltage divider. In this way, feedback is properly phased to produce continuous undamped oscillations.

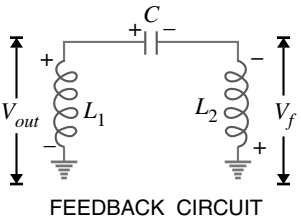


Fig. 12.17

Feedback fraction m_v. In Hartley oscillator, the feedback voltage is across L_2 and output voltage is across L_1.

$$\therefore \quad \text{Feedback fraction, } m_v = \frac{V_f}{V_{out}} = \frac{X_{L_2}}{X_{L_1}} = {}^{**}\frac{L_2}{L_1}$$

---

* Referring to Fig. 12.17, it is clear that L_1 and L_2 are in series. Therefore, total inductance L_T is given by ; $L_T = L_1 + L_2 + 2M$.

** Referring to Fig. 12.17, the circulating current for the two inductors is the same. Further, inductive reactance is directly proportional to inductance.

or $$m_v = \frac{L_2}{L_1}$$

We know that condition for continuous oscillations in an oscillator is $A_v\, m_v = 1$.

$\therefore$ Amplifier voltage gain, $A_v = \dfrac{1}{m_v} = \dfrac{L_1}{L_2}$

Actually, the product $A_v\, m_v$ should be slightly greater than 1 to prevent the loss of oscillations due to power loss within the amplifier.

Example 12.6. *Calculate the (i) operating frequency and (ii) feedback fraction for Hartley oscillator shown in Fig. 12.18. The mutual inductance between the coils, M = 20 μH.*

Solution.

(*i*) $L_1 = 1000\ \mu\text{H}\ ;\quad L_2 = 100\ \mu\text{H}\ ;\quad M = 20\ \mu\text{H}$

$\therefore$ Total inductance, $L_T = L_1 + L_2 + 2M$

$= 1000 + 100 + 2 \times 20$

$= 1140\ \mu\text{H} = 1140 \times 10^{-6}\ \text{H}$

Capacitance, $C = 20\ \text{pF} = 20 \times 10^{-12}\ \text{F}$

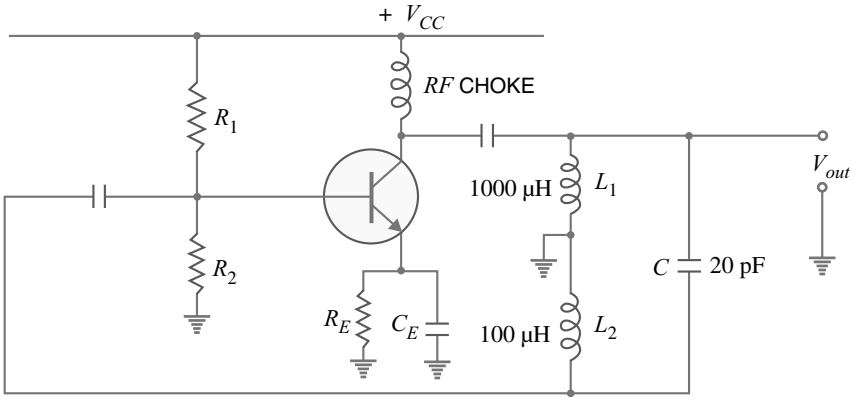


Fig. 12.18

$\therefore$ Operating frequency, $f = \dfrac{1}{2\pi\sqrt{L_T C}} = \dfrac{1}{2\pi\sqrt{1140 \times 10^{-6} \times 20 \times 10^{-12}}}$ Hz

$= 1052 \times 10^3$ Hz = **1052 kHz**

(*ii*) Feedback fraction, $m_v = \dfrac{L_2}{L_1} = \dfrac{100\ \mu\text{H}}{1000\ \mu\text{H}} = \mathbf{0.1}$

Example 12.7. *A 1 pF capacitor is available. Choose the inductor values in a Hartley oscillator so that f = 1 MHz and m_v = 0.2.*

Solution.

Feedback fraction, $m_v = \dfrac{L_2}{L_1}$

or $0.2 = \frac{L_2}{L_1} \quad \therefore L_1 = 5L_2$

Now, $f = \frac{1}{2\pi\sqrt{L_T C}}$

or $L_T = \frac{1}{C(2\pi f)^2} = \frac{1}{(1\times10^{-12})(2\pi\times1\times10^6)^2}$

$= 25.3 \times 10^{-3}$ H $= 25.3$ mH

or $L_1 + L_2 = 25.3$ mH $\quad (\because L_T = L_1 + L_2)$

or $5L_2 + L_2 = 25.3 \quad \therefore L_2 = 25.3/6 =$ **4.22 mH**

and $L_1 = 5L_2 = 5 \times 4.22 =$ **21.1 mH**

Example 12.8. *The Hartley oscillator shown in Fig. 12.19 has the values as indicated. Find the feedback fraction and the minimum voltage gain needed to start the oscillations. Also find the frequency of oscillations.*

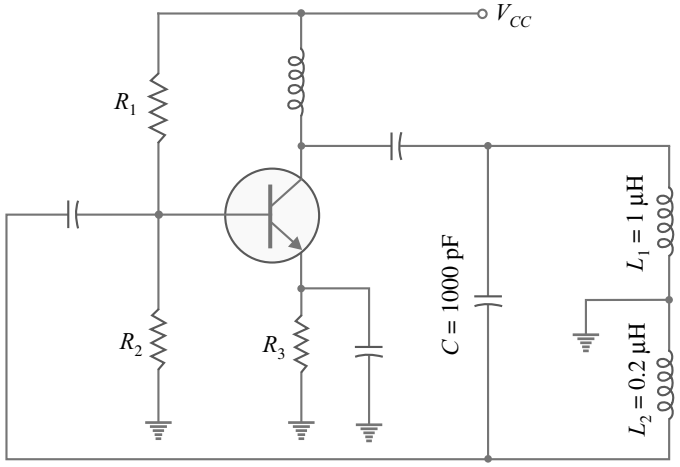


Fig. 12.19

Solution.

$L_1 = 1$ μH $= 1 \times 10^{-6}$ H ; $L_2 = 0.2$ μH $= 0.2 \times 10^{-6}$ H ; $C = 1000$ pF

$L_T = L_1 + L_2 = (1 + 0.2)$ μH $= 1.2 \times 10^{-6}$ H

$C = 1000$ pF $= 1000 \times 10^{-12}$ F $= 10^{-9}$ F

The frequency of oscillations is given by ;

$$f = \frac{1}{2\pi\sqrt{L_T C}} = \frac{1}{2\pi\sqrt{(1.2\times10^{-6})\times10^{-9}}}$$

$= 4.59 \times 10^6$ Hz $=$ **4.59 MHz**

Feedback fraction, $m_v = \frac{L_2}{L_1} = \frac{0.2\,\mu\text{H}}{1\,\mu\text{H}} =$ **0.2**

Minimum voltage gain, $A_v = \frac{1}{m_v} = \frac{L_1}{L_2} = \frac{1}{0.2} =$ **5**

12.13 Principle of RC Phase Shift Oscillators

One desirable feature of an oscillator is that it should feed back energy of correct phase to the tank circuit to overcome the losses occurring in it. In the oscillator circuits discussed so far, the tank circuit employed inductive (L) and capacitive (C) elements. In such circuits, a phase shift of 180º was obtained due to inductive or capacitive coupling and a further phase shift of 180º was obtained due to transistor properties. In this way, energy supplied to the tank circuit was in phase with the generated oscillations. The oscillator circuits employing L-C elements have two general drawbacks. Firstly, they suffer from frequency instability and poor waveform. Secondly, they cannot be used for very low frequencies because they become too much bulky and expensive.

Good frequency stability and waveform can be obtained from oscillators employing resistive and capacitive elements. Such amplifiers are called *R-C phase shift oscillators* and have the additional advantage that they can be used for very low frequencies. In a phase shift oscillator, a phase shift of 180º is obtained with a phase shift circuit instead of inductive or capacitive coupling. A further phase shift of 180º is introduced due to the transistor properties. Thus, energy supplied back to the tank circuit is assured of correct phase.

Phase shift circuit. A phase-shift circuit essentially consists of an R-C network. Fig. 12.20 (*i*) shows a single section of RC network. From the elementary theory of electrical engineering, it can be shown that alternating voltage V_1' across R leads the applied voltage V_1 by ϕº. The value of ϕ depends upon the values of R and C. If resistance R is varied, the value of ϕ also changes. If R were reduced to zero, V_1' will lead V_1 by 90º *i.e.* $\phi = 90º$. However, adjusting R to zero would be impracticable because it would lead to no voltage across R. Therefore, in practice, R is varied to such a value that makes V_1' to lead V_1 by 60º.

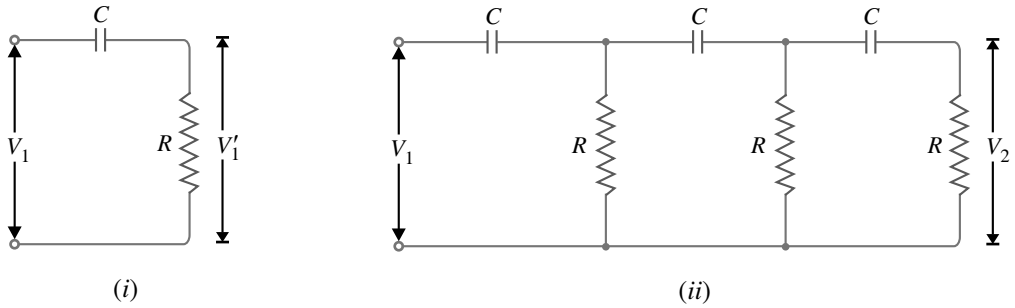


Fig. 12.20

Fig. 12.20 (*ii*) shows the three sections of RC network. Each section produces a phase shift of 60º. Consequently, a total phase shift of 180º is produced *i.e.* voltage V_2 leads the voltage V_1 by 180º.

12.14 RC Phase Shift Oscillator

In RC phase shift oscillator, the RC network (consisting of 3 identical RC sections) determines the frequency of oscillations as well as produces 180° phase shift. There is one frequency f_0 at which the RC network produces a phase shift of 180° and at the same time the condition $A_v\, m_v = 1$ is met by the oscillator circuit. The value of f_0 depends upon the values of R and C. By properly changing the values of R and C, the desired value of f_0 can be obtained.

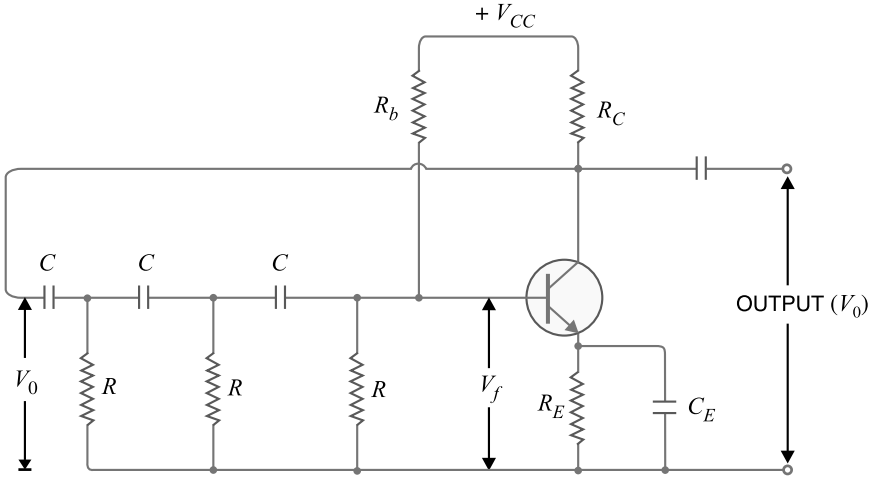


Fig. 12.21

Fig. 12.21 shows the circuit of *RC* phase shift oscillator. It consists of conventional single transistor amplifier and a *RC* phase shift network. The phase shift network consists of three identical *RC* sections (*i.e.* three equal-value resistors *R* and three equal-value capacitors *C*). At some frequency f_0, the phase shift in each *RC* section is 60° so that the total phase shift produced by the *RC* network is 180°. It can be proved mathematically that the frequency of oscillations f_0 is given by ;

$$f_0 = \frac{1}{2\pi RC\sqrt{6}} \qquad \text{... (}i\text{) [See Example 12.11]}$$

By changing the values of *R* or *C* or both, the desired frequency f_0 can be obtained.

Feedback fraction (m_v). The *RC* network produces phase shift as well as reduces the output voltage (V_0). It can be shown mathematically that at frequency f_0, the *RC* network produces a phase shift of 180° as well as decreases the output voltage by 29 *i.e.* the feedback voltage $V_f = V_0/29$. This feedback voltage V_f is applied to the input of the amplifier as shown in Fig. 12.21.

$$\therefore \quad \text{Feedback fraction, } m_v = \frac{V_f}{V_0} = \frac{V_0/29}{V_0} = \frac{1}{29}$$

We know that the condition for continuous oscillations in an oscillator is $A_v m_v = 1$.

$$\therefore \quad \text{Amplifier voltage gain, } A_v = \frac{1}{m_v} = 29$$

Therefore, the amplifier must provide a minimum voltage gain (A_v) of 29 in order to maintain oscillations. If A_v is less than 29, the circuit will not oscillate. However, if A_v is much greater than 29, the output waveform will be distorted. In practice, the voltage gain of the amplifier is made slightly more than 29 to ensure that the circuit will oscillate.

Circuit operation. When the circuit is switched on, it produces oscillations of frequency determined by exp. (*i*). The output V_0 of the amplifier is fed back to *RC* feedback network.

This network produces a phase shift of 180° and a voltage V_f appears at its output which is applied to the transistor amplifier.

Obviously, the feedback fraction $m_v = V_f/V_0$. The feedback phase is correct. A phase shift of 180° is produced by the transistor amplifier. A further phase shift of 180° is produced by the *RC* network. As a result, the phase shift around the entire loop is 360°.

Advantages

(*i*) It does not require transformers or inductors.

(*ii*) It can be used to produce very low frequencies.

(*iii*) The circuit provides good frequency stability.

Disadvantages

(*i*) It is difficult for the circuit to start oscillations as the feedback is generally small.

(*ii*) The circuit gives small output.

Example 12.9. *In the phase shift oscillator shown in Fig. 12.21, R = 1 MΩ and C = 68 pF. At what frequency does the circuit oscillate?*

Solution. $R = 1\ \text{M}\Omega = 10^6\ \Omega$; $C = 68\ \text{pF} = 68 \times 10^{-12}\ \text{F}$

Frequency of oscillations is given by ;

$$f_0 = \frac{1}{2\pi\, RC\sqrt{6}}$$

$$= \frac{1}{2\pi \times 10^6 \times 68 \times 10^{-12}\sqrt{6}}\ \text{Hz}$$

$$= \mathbf{954\ Hz}$$

Example 12.10. *A phase shift oscillator uses 5 pF capacitors. Find the value of R to produce a frequency of 800 kHz.*

Solution. Here, $C = 5\ \text{pF} = 5 \times 10^{-12}\ \text{F}$; $R = ?$

$$f_0 = \frac{1}{2\pi\, RC\sqrt{6}}$$

or

$$R = \frac{1}{2\pi f_0 C\sqrt{6}} = \frac{1}{2\pi \times 800 \times 10^3 \times 5 \times 10^{-12} \times \sqrt{6}}$$

$$= 16.2 \times 10^3\ \Omega = \mathbf{16.2\ k\Omega}$$

Example 12.11. *Draw the frequency determining circuit of RC phase shift oscillator and find the expression for the frequency (called operating frequency) at which the phase shift produced by this circuit is exactly 180°. Also find the feedback fraction at this frequency.*

Solution. Fig. 12.22 shows the frequency-determining circuit of *RC* phase shift oscillator. It consists of three identical *RC* sections. The output voltage V_0 of the amplifier is applied to the input of *RC* phase shift circuit and reduced feedback voltage V_f appears across the output terminals of the *RC* network. The frequency f_0 at which the phase shift produced by the *RC* network is exactly 180° and the feedback fraction $m_v (= V_f/V_0)$ at this frequency can be found as follows.

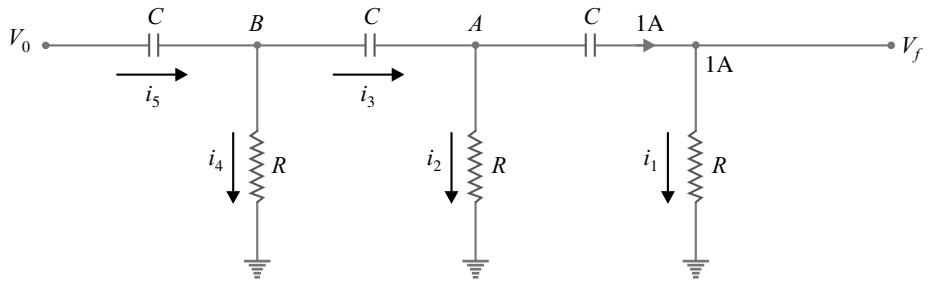


Fig. 12.22

Suppose current i_1 through R in Fig. 12.22 is 1 A *i.e.* $i_1 = 1$ A. Then,

Feedback voltage, $V_f = i_1 R = 1 \text{ A} \times R = R$ volt

Voltage at point A, $\boldsymbol{V_A} = V_f + i_1 (-j X_C) = V_f + 1(-j X_C)$ ($\because i_1 = 1$ A)

or $\boldsymbol{V_A} = R - j X_C$ ($\because V_f = R$ volt)

$$\text{Current } \boldsymbol{i_2} = \frac{\boldsymbol{V_A}}{R} = \frac{R - j X_C}{R} = 1 - \frac{j X_C}{R}$$

$$\text{Current } \boldsymbol{i_3} = i_1 + \boldsymbol{i_2} = 1 + \left(1 - \frac{j X_C}{R}\right) = 2 - \frac{j X_C}{R}$$

Voltage at point B, $\boldsymbol{V_B} = \boldsymbol{V_A} + \boldsymbol{i_3} (-j X_C)$

$$= (R - j X_C) + \left(2 - \frac{j X_C}{R}\right)(-j X_C)$$

$\therefore$ $$\boldsymbol{V_B} = R - \frac{X_C^2}{R} - j(3X_C)$$

$$\text{Current } \boldsymbol{i_4} = \frac{\boldsymbol{V_B}}{R} = \frac{1}{R}\left[R - \frac{X_C^2}{R} - j(3X_C)\right]$$

or $$\boldsymbol{i_4} = 1 - \frac{X_C^2}{R^2} - j\left(\frac{3X_C}{R}\right)$$

$$\text{Current } \boldsymbol{i_5} = \boldsymbol{i_3} + \boldsymbol{i_4} = 3 - \frac{X_C^2}{R^2} - j\left(\frac{4X_C}{R}\right)$$

$\therefore$ Voltage $\boldsymbol{V_0} = \boldsymbol{V_B} + \boldsymbol{i_5} (-j X_C)$

Putting the values of $\boldsymbol{V_B}$ and $\boldsymbol{i_5}$ in the above equation, we have,

$$\boldsymbol{V_0} = R - \frac{5X_C^2}{R} - j\left(6X_C - \frac{X_C^3}{R^2}\right)$$

or $$\boldsymbol{V_0} = \frac{(R^3 - 5RX_C^2) + j(X_C^3 - 6R^2 X_C)}{R^2}$$

Feedback fraction, $\boldsymbol{m_v} = \frac{V_f}{\boldsymbol{V_0}} = \frac{R}{\boldsymbol{V_0}}$ ($\because V_f = R$ volt)

Putting the value of $\boldsymbol{V_0}$, we get,

$$m_v = \frac{R^3}{(R^3 - 5RX_C^2) + j(X_C^3 - 6R^2X_C)} \quad \ldots (i)$$

At some frequency f_0, the phase shift produced by the *RC* network is exactly 180° and m_v is a pure real number. This means that *j* term in the denominator in eq. (*i*) must be zero *i.e.*

$$X_C^3 - 6R^2X_C = 0$$

or $$X_C = R\sqrt{6}$$

or $$\frac{1}{2\pi f_0 C} = R\sqrt{6}$$

$\therefore$ $$\text{Frequency, } f_0 = \frac{1}{2\pi RC\sqrt{6}}$$

This is the expression for the frequency of oscillations for the *RC* phase shift oscillator.

At frequency f_0, the *j* term in the denominator in eq. (*i*) is zero so that m_v is given by ;

$$m_v = \frac{R^3}{R^3 - 5RX_C^2}$$

Putting $X_C = R\sqrt{6}$ in the above expression, we have,

$$m_v = \frac{R^3}{R^3 - 5R(R\sqrt{6})^2} = \frac{R^3}{R^3 - 30R^3} = -\frac{1}{29}$$

The negative sign shows that *RC* network produces a phase shift of 180°.

12.15 Wien Bridge Oscillator

It is an *RC* oscillator that uses a 2-stage amplifier and the Wien bridge. The 2-stage amplifier produces 360° phase shift required for positive feedback. The Wien bridge has *RC* circuit in the two adjacent branches (the other two adjacent branches have pure resistors) and determines the frequency of oscillations as well as acts as a feedback circuit. At one particular frequency f_0, the Wien bridge is balanced and produces 0° phase shift. Therefore, at the balance frequency f_0, the feedback is positive. At all other frequencies, the bridge is not balanced and does not produce 0° phase shift. Therefore, positive feedback is not ensured.

The Wien-bridge oscillator is the standard oscillator circuit for all frequencies in the range of 10 Hz to about 1 MHz. It is the most frequently used type of audio oscillator as the output is free from circuit fluctuations and ambient temperature. Fig. 12.23 shows the circuit of Wien bridge oscillator. It is essentially a two-stage amplifier with *R-C* bridge circuit. The bridge circuit has the arms R_1C_1, R_3, R_2C_2 and tungsten lamp L_P. Resistances R_3 and L_P are used to stabilise the amplitude of the output. The transistor T_1 serves as an oscillator and amplifier while the other transistor T_2 serves as an inverter (*i.e.* to produce a phase shift of 180°). The circuit uses positive and negative feedbacks. The positive feedback is through R_1C_1, C_2R_2 to the transistor T_1. The negative feedback is through the voltage divider to the input of transistor T_2. The frequency of oscillations is determined by the series element R_1C_1 and parallel element R_2C_2 of the bridge. At one particular frequency f_0, the Wien bridge is balanced and produces 0° phase shift. It can be proved mathematically that the balance frequency f_0 is given by ;

Answers to Multiple-Choice Questions

1. (*ii*)	2. (*iii*)	3. (*i*)	4. (*iv*)	5. (*ii*)
6. (*i*)	7. (*iii*)	8. (*i*)	9. (*ii*)	10. (*iii*)
11. (*iii*)	12. (*i*)	13. (*i*)	14. (*iv*)	15. (*ii*)
16. (*ii*)	17. (*iii*)	18. (*i*)	19. (*iv*)	20. (*ii*)
21. (*iii*)	22. (*iii*)	23. (*iv*)	24. (*i*)	25. (*ii*)
26. (*iii*)	27. (*iv*)	28. (*iv*)	29. (*i*)	30. (*i*)

Chapter Review Topics

1. What is an oscillator? What is its need? Discuss the advantages of oscillators.
2. What do you understand by damped and undamped electrical oscillations? Illustrate your answer with examples.
3. Explain the operation of a tank circuit with neat diagrams.
4. What is the nature of oscillations produced by a tank circuit?
5. How will you get undamped oscillations from a tank circuit?
6. Discuss the essentials of an oscillator.
7. Discuss the circuit operation of tuned collector oscillator.
8. With a neat diagram, explain the action of Hartley and Colpitt's oscillators.
9. What are the drawbacks of *LC* oscillators?
10. Write short notes on the following:

    (*i*) *RC* oscillators (*ii*) Wien bridge oscillator (*iii*) Crystal oscillator

Problems

1. Figure 12.30 shows the Colpitt's oscillator. Determine the (*i*) operating frequency and (*ii*) feedback fraction. **[(*i*) 24.5 kHz (*ii*) 0.1]**

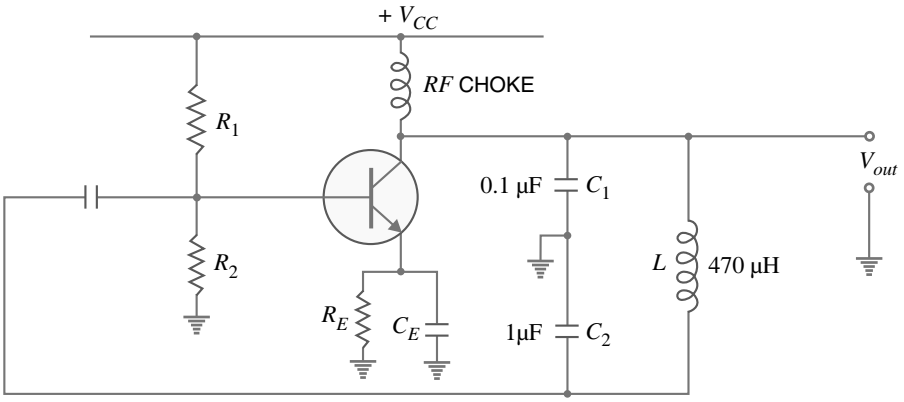


Fig. 12.30

2. Figure 12.31 shows the Hartley oscillator. If L_1 = 1000 μH, L_2 = 100 μH and C = 20 pF, find the (*i*) operating frequency and (*ii*) feedback fraction. **[(*i*) 1052 kHz (*ii*) 0.1]**
3. For the Colpitt's oscillator shown in Fig. 12.30, C_1 = 750 pF, C_2 = 2500 pF and L = 40 μH. Determine (*i*) the operating frequency and (*ii*) feedback fraction. **[(*i*) 1050 kHz (*ii*) 0.3]**
4. For the Hartley oscillator shown in Fig. 12.31, C = 250 pF, L_1 = 1.5 mH, L_2 = 1.5 mH and M = 0.58 mH. Determine the operating frequency. **[159.2 kHz]**

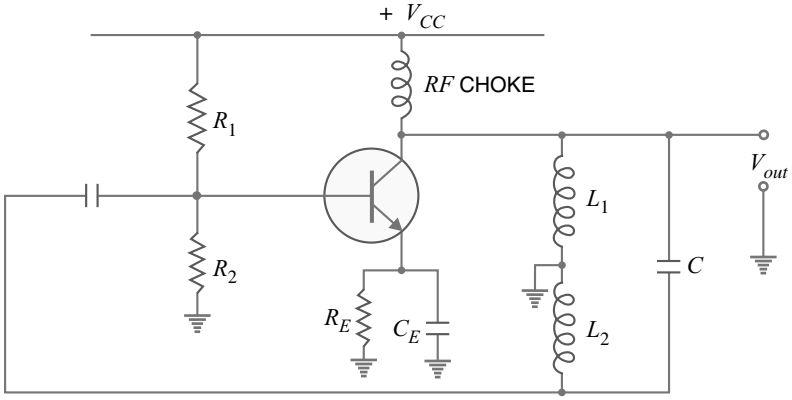


Fig. 12.31

5. A crystal has L = 3H, C = 0.05 pF, R = 2 kΩ and C_m = 10 pF. Calculate the series-resonant and parallel resonant frequencies of the crystal. **[411 kHz; 412 kHz]**